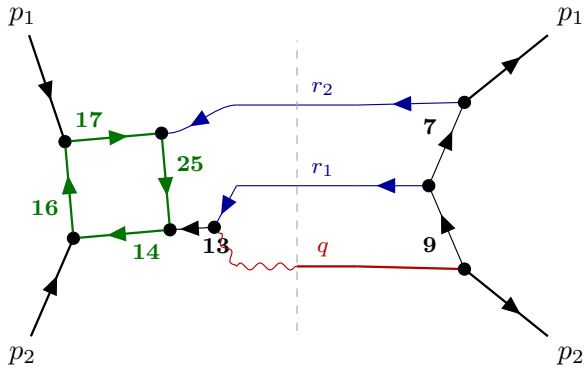
RRV top sectors as cut diagrams

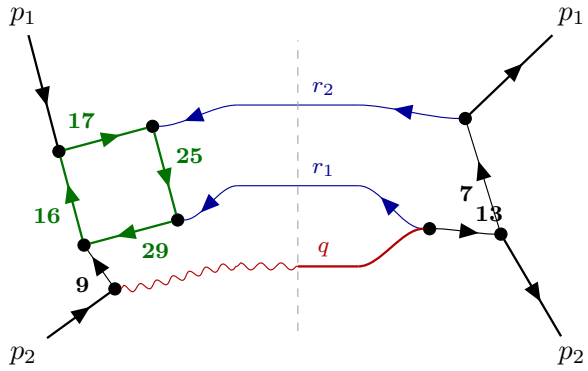
One-loop amplitude (left) × tree conjugate (right) for $p_1 p_2 \to V(q)\, r_1 r_2$, glued along the cut (dashed): r_1, r_2 (blue), q (red). The loop lines (green) are the propagators $(k_1 + S)^2$ of the family present in the drawing; the legs entering the loop between two of them meet at one loop vertex, through a tree when an RR propagator of the family separates them. The rapidity condition $-q \cdot p_1 + u\, q \cdot p_2 = 0$ is not a line: the q line is drawn wavy up to the cut ($q^2 = m^2$) and as a double line after it. Arrows: momentum flow with all legs incoming ($p_1 + p_2 + r_1 + r_2 + q = 0$, k_1 incoming); the conjugate is the mirror image. **Propagators** (DD numbers, all legs incoming): $5 = (p_1 + r_1)^2$; $6 = (p_2 + r_1)^2$; $7 = (p_1 + r_2)^2$; $8 = (p_2 + r_2)^2$; $9 = (p_1 + r_1 + r_2)^2$; $10 = (p_2 + r_1 + r_2)^2$; $11 = (r_1 + r_2)^2$; $12 = (p_1 + p_2 + r_1)^2$; $13 = (p_1 + p_2 + r_2)^2$; $14 = (k_1)^2$; $15 = (k_1 + p_1)^2$; $16 = (k_1 + p_2)^2$; $17 = (k_1 + p_1 + p_2)^2$; $18 = (k_1 + r_1)^2$; $19 = (k_1 + p_1 + r_1)^2$; $20 = (k_1 + p_2 + r_1)^2$; $21 = (k_1 + p_1 + p_2 + r_1)^2$; $22 = (k_1 + r_2)^2$; $23 = (k_1 + p_1 + r_2)^2$; $24 = (k_1 + p_2 + r_2)^2$; $25 = (k_1 + p_1 + p_2 + r_2)^2$; $26 = (k_1 + r_1 + r_2)^2$; $27 = (k_1 + p_1 + r_1 + r_2)^2$; $28 = (k_1 + p_2 + r_1 + r_2)^2$; $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$.



topo1 (RR family 33, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
7	$(p_1 + r_2)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate

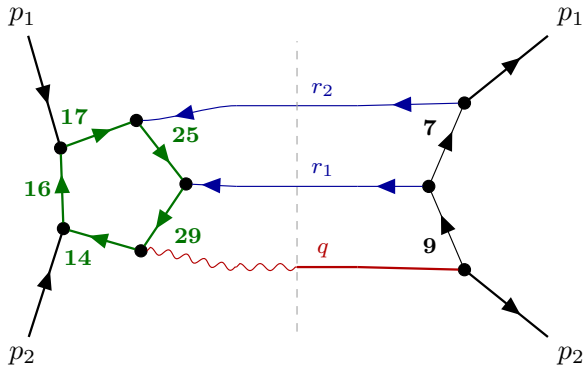
pinched: $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo1 (RR family 33, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
9	$(p_1 + r_1 + r_2)^2$	amplitude (tree on the loop)
7	$(p_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	conjugate

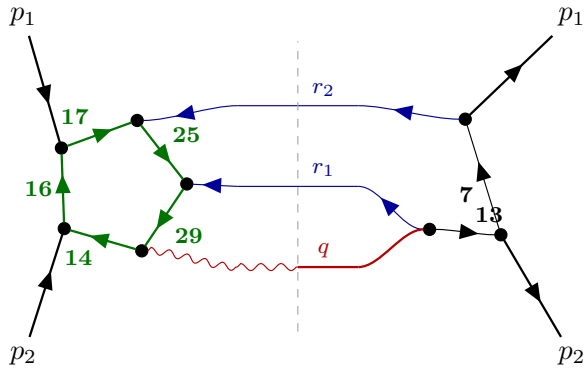
pinched: $14 = (k_1)^2$



topo1 (RR family 33, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
7	$(p_1 + r_2)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate

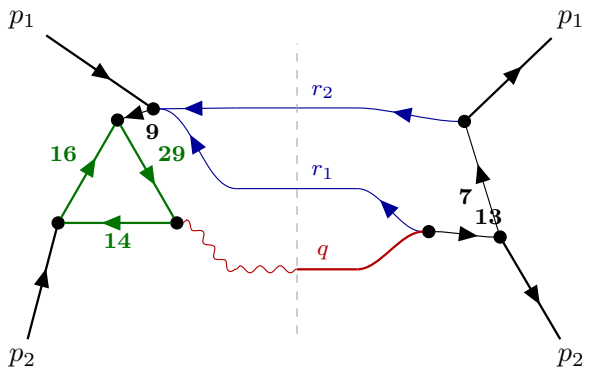
pinched: $13 = (p_1 + p_2 + r_2)^2$



topo1 (RR family 33, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
7	$(p_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	conjugate

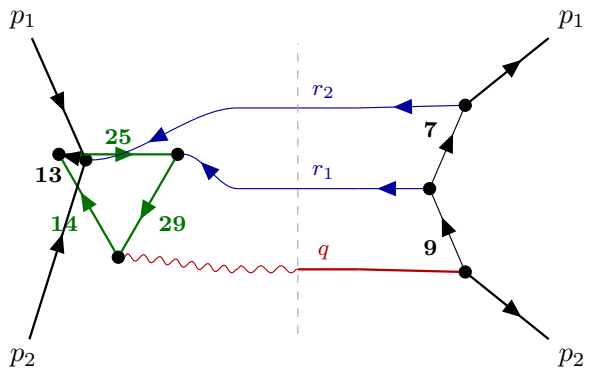
pinched: $9 = (p_1 + r_1 + r_2)^2$



topo1 (RR family 33, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

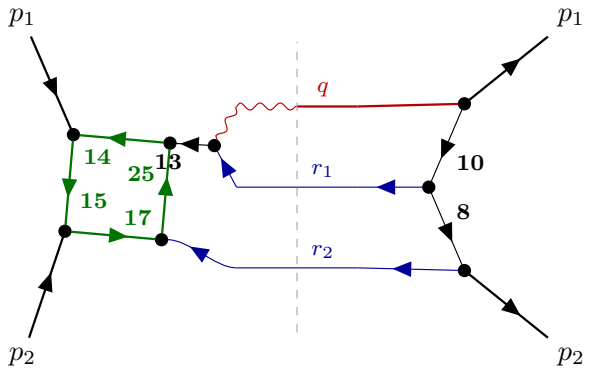
pinched: $17=(k_1+p_1+p_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo1 (RR family 33, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

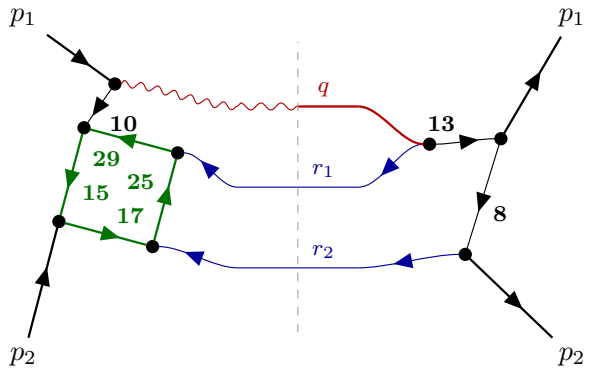
pinched: $16=(k_1+p_2)^2$, $17=(k_1+p_1+p_2)^2$



topo2 (RR family 29, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

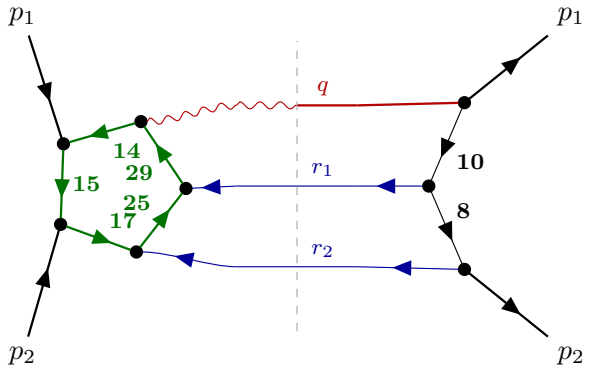
pinched: $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo2 (RR family 29, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

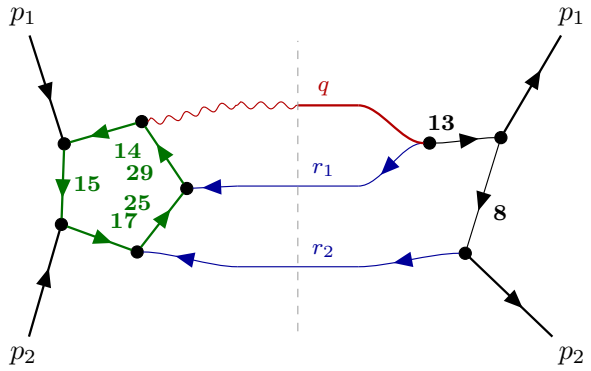
pinched: $14=(k_1)^2$



topo2 (RR family 29, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

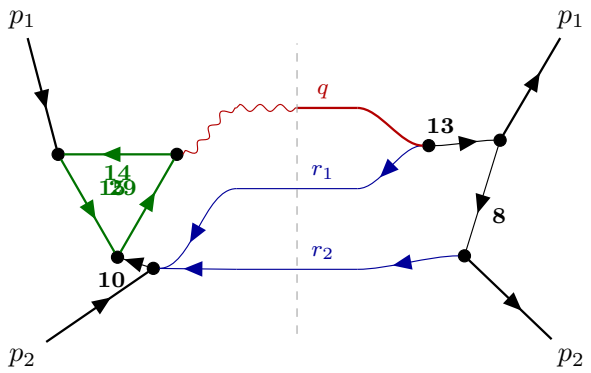
pinched: $13=(p_1+p_2+r_2)^2$



topo2 (RR family 29, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

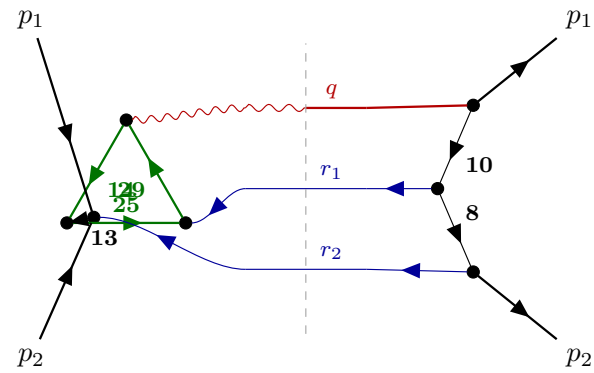
pinched: $10=(p_2+r_1+r_2)^2$



topo2 (RR family 29, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

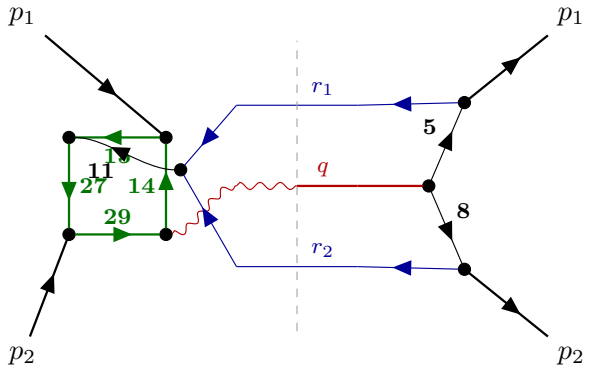
pinched: $17=(k_1+p_1+p_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo2 (RR family 29, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

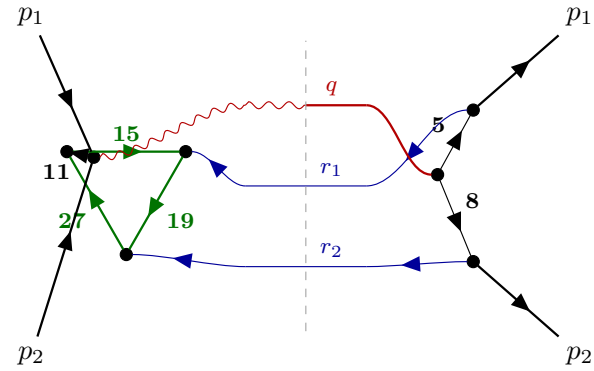
pinched: $15=(k_1+p_1)^2$, $17=(k_1+p_1+p_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

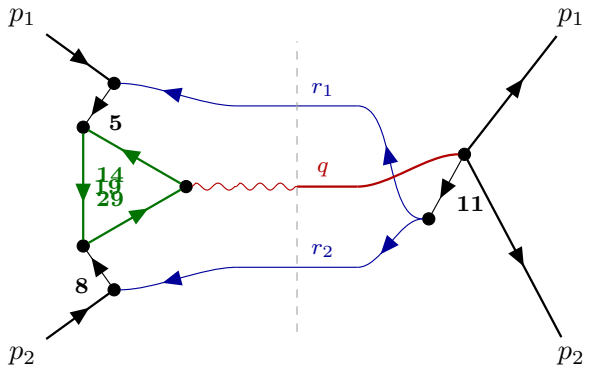
pinched: $19=(k_1+p_1+r_1)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

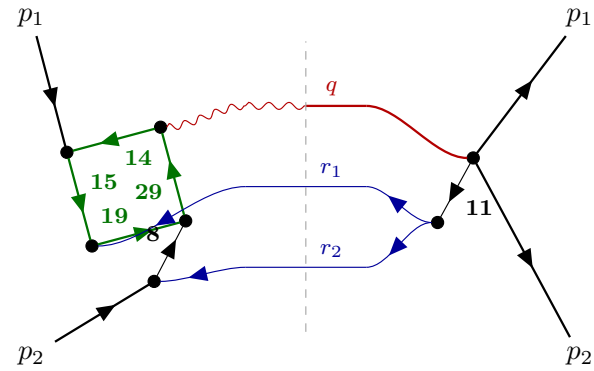
pinched: $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

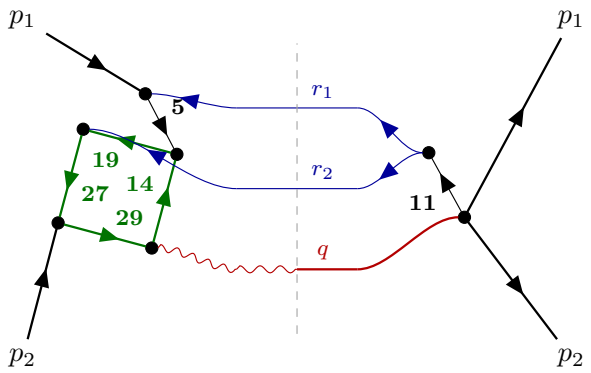
pinched: $15=(k_1+p_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

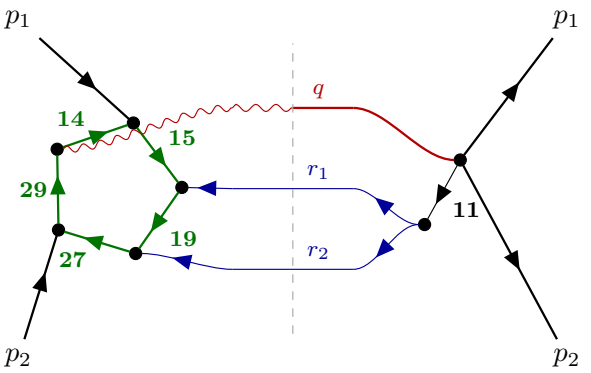
pinched: $5=(p_1+r_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

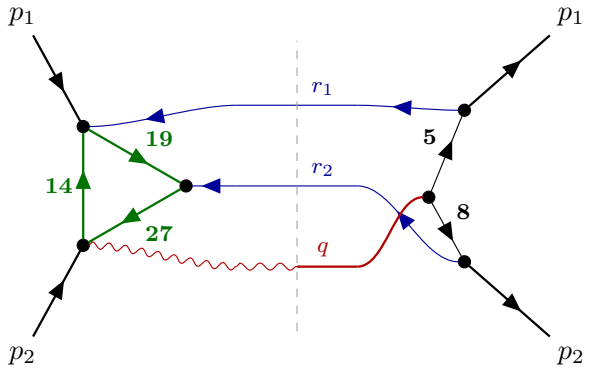
pinched: $8=(p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

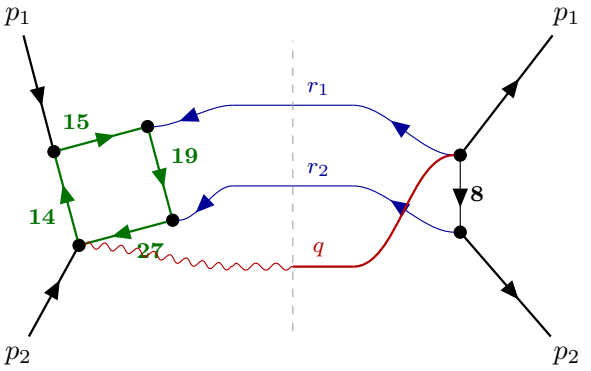
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

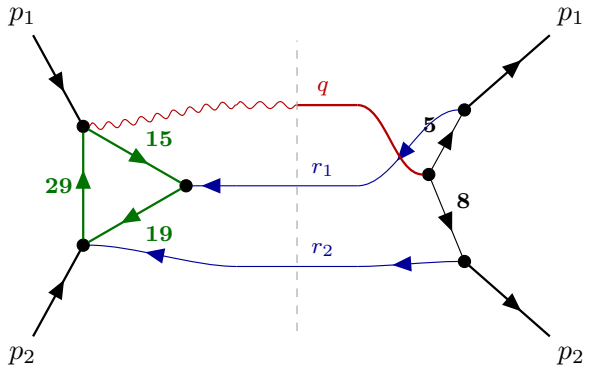
pinched: $11=(r_1+r_2)^2$, $15=(k_1+p_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate

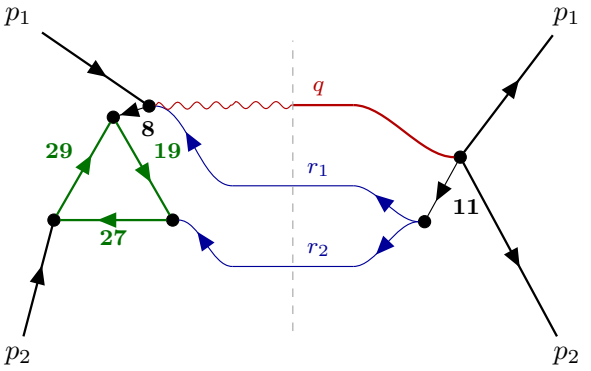
pinched: $5=(p_1+r_1)^2$, $11=(r_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

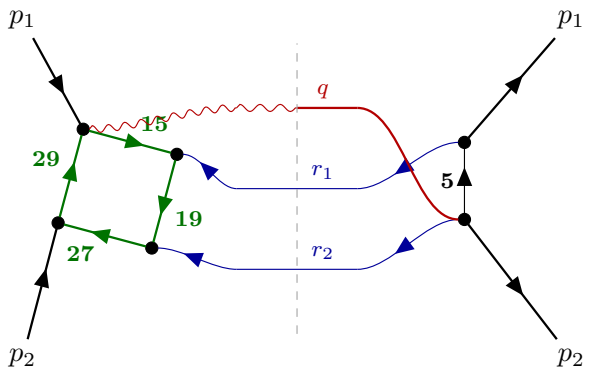
pinched: $11=(r_1+r_2)^2$, $14=(k_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 5 of 8 lines

19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

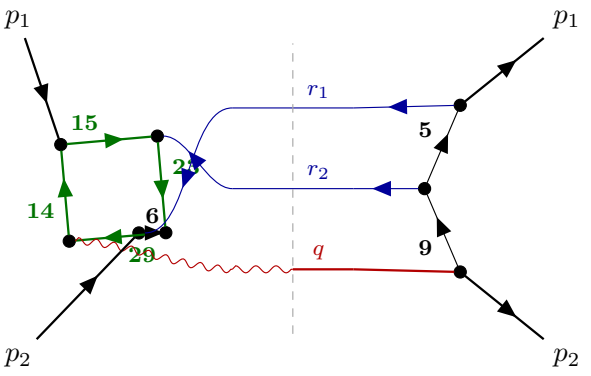
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo3 (RR family 8, pentagon q-p1-r1-r2-p2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate

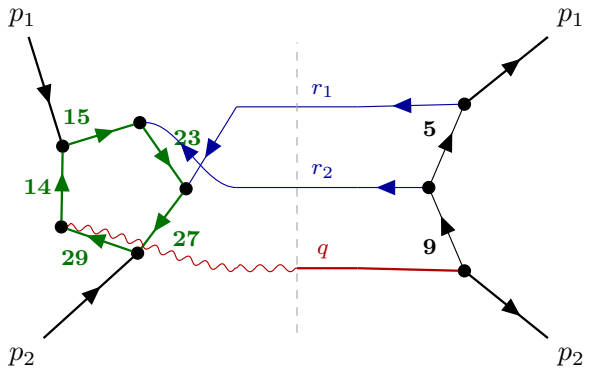
pinched: $8=(p_2+r_2)^2$, $11=(r_1+r_2)^2$, $14=(k_1)^2$



topo4 (RR family 3, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

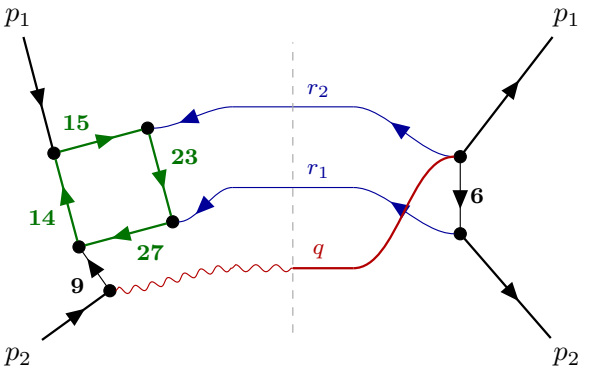
pinched: $27=(k_1+p_1+r_1+r_2)^2$



topo4 (RR family 3, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

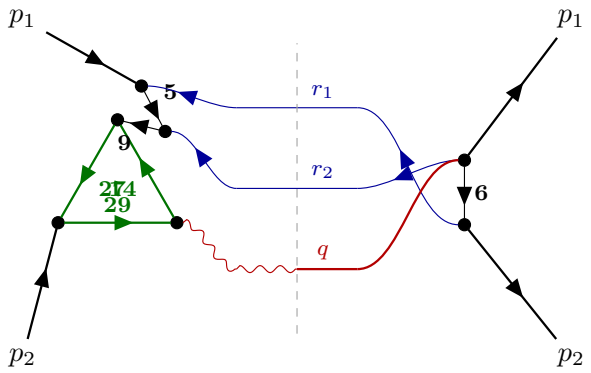
pinched: $6=(p_2+r_1)^2$



topo4 (RR family 3, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

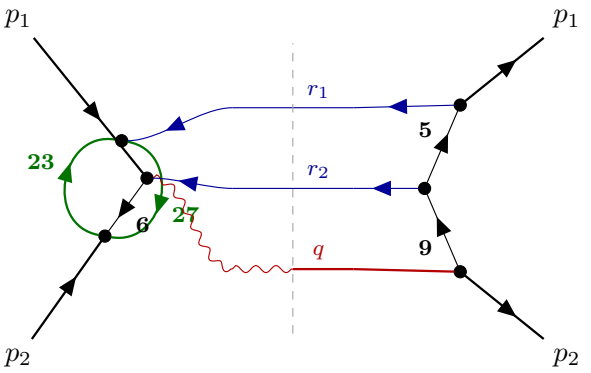
pinched: $5=(p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo4 (RR family 3, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

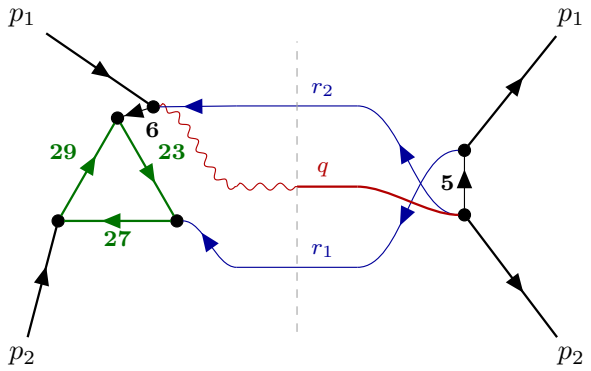
pinched: $15=(k_1+p_1)^2$, $23=(k_1+p_1+r_2)^2$



topo4 (RR family 3, pentagon q-p1-r2-r1-p2), sector with 5 of 8 lines

23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

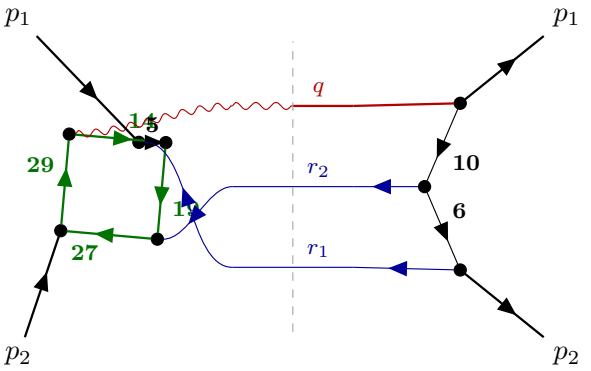
pinched: $14=(k_1)^2$, $15=(k_1+p_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo4 (RR family 3, pentagon q-p1-r2-r1-p2), sector with 5 of 8 lines

23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

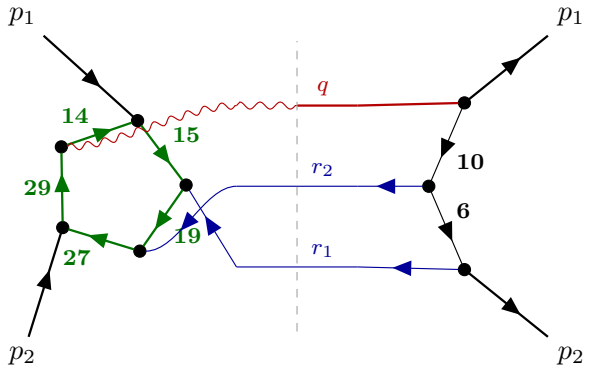
pinched: $9=(p_1+r_1+r_2)^2$, $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo5 (RR family 4, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

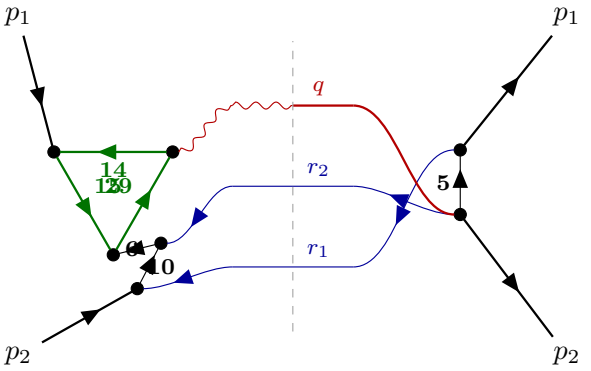
pinched: $15=(k_1+p_1)^2$



topo5 (RR family 4, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

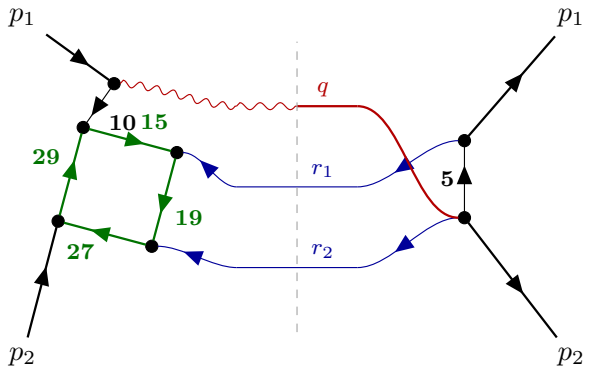
pinched: $5=(p_1+r_1)^2$



topo5 (RR family 4, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

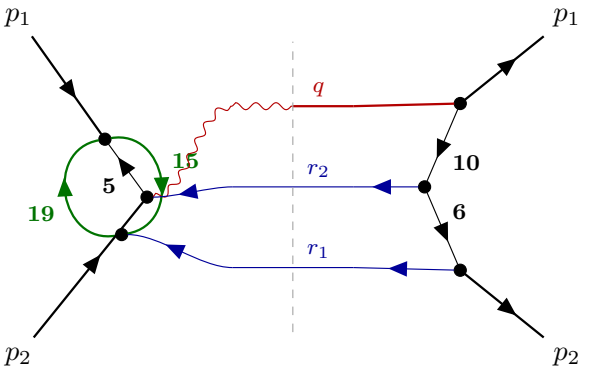
pinched: $19=(k_1+p_1+r_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo5 (RR family 4, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

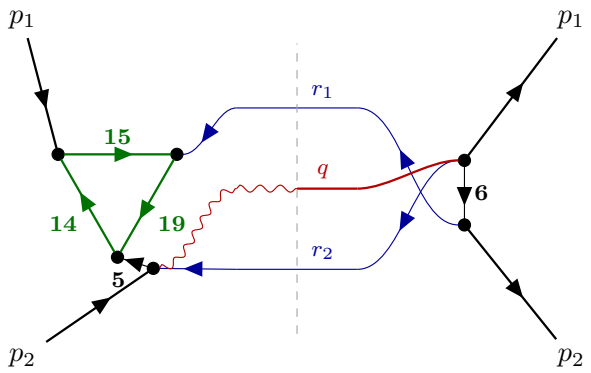
pinched: $6=(p_2+r_1)^2$, $14=(k_1)^2$



topo5 (RR family 4, pentagon q-p1-r1-r2-p2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

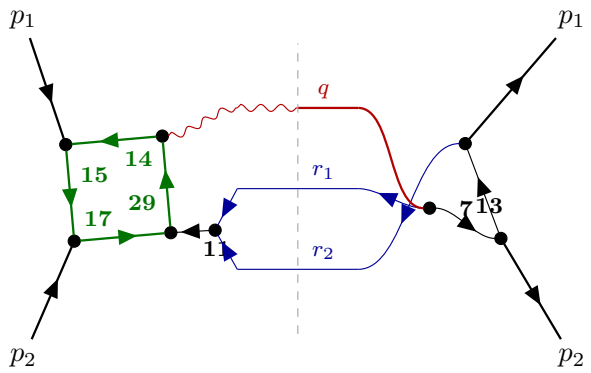
pinched: $14=(k_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo5 (RR family 4, pentagon q-p1-r1-r2-p2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

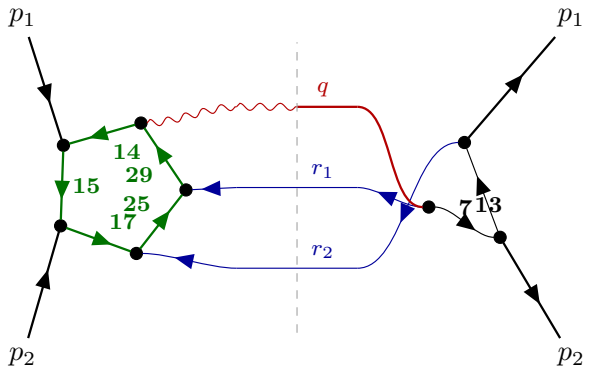
pinched: $10=(p_2+r_1+r_2)^2$, $27=(k_1+p_1+r_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo6 (RR family 30, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

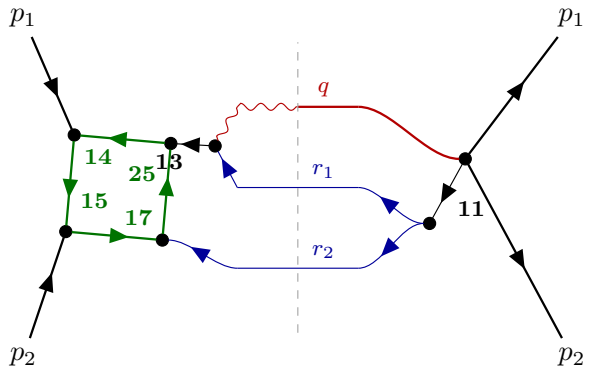
pinched: $25=(k_1+p_1+p_2+r_2)^2$



topo6 (RR family 30, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

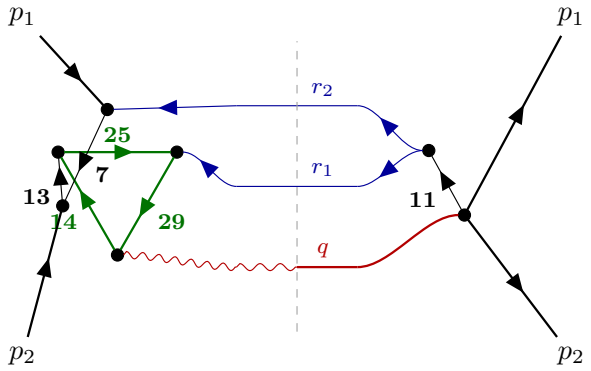
pinched: $11=(r_1+r_2)^2$



topo6 (RR family 30, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

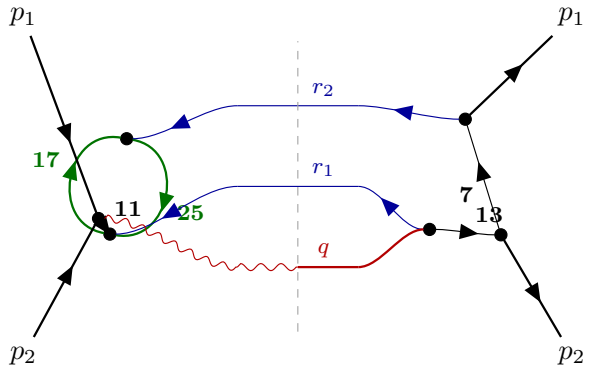
pinched: $7=(p_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo6 (RR family 30, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

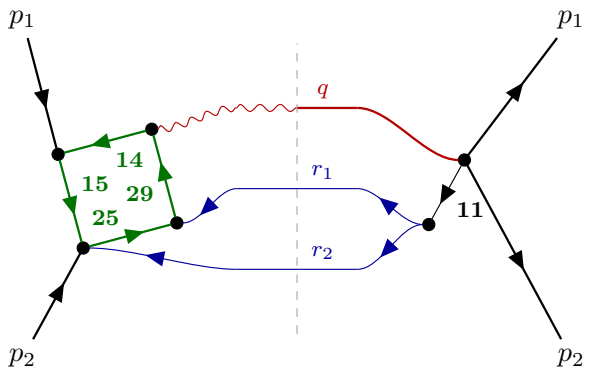
pinched: $15=(k_1+p_1)^2$, $17=(k_1+p_1+p_2)^2$



topo6 (RR family 30, pentagon q-p1-p2-r2-r1), sector with 5 of 8 lines

17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

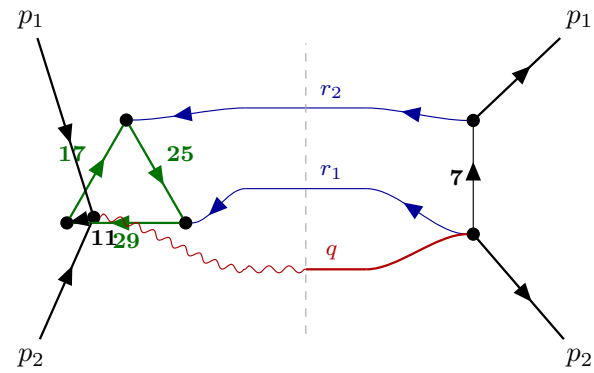
pinched: $14=(k_1)^2$, $15=(k_1+p_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo6 (RR family 30, pentagon q-p1-p2-r2-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

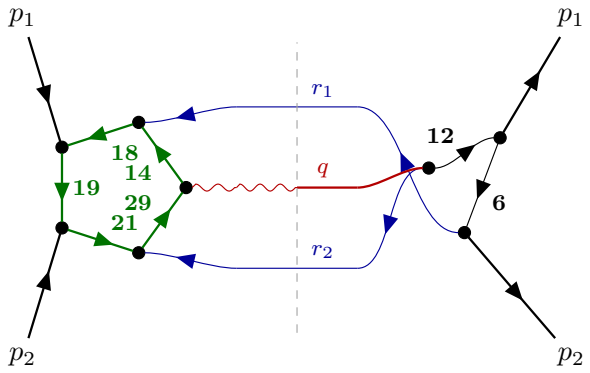
pinched: $7=(p_1+r_2)^2$, $13=(p_1+p_2+r_2)^2$, $17=(k_1+p_1+p_2)^2$



topo6 (RR family 30, pentagon q-p1-p2-r2-r1), sector with 5 of 8 lines

17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate

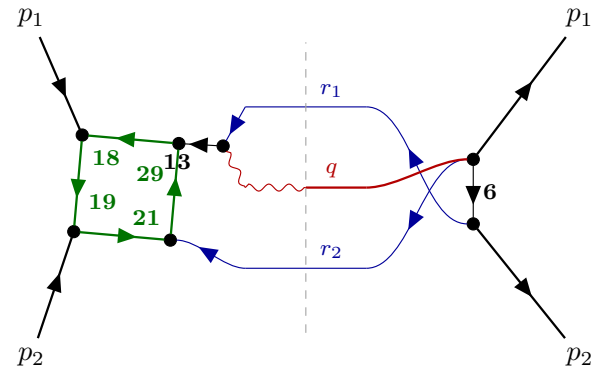
pinched: $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

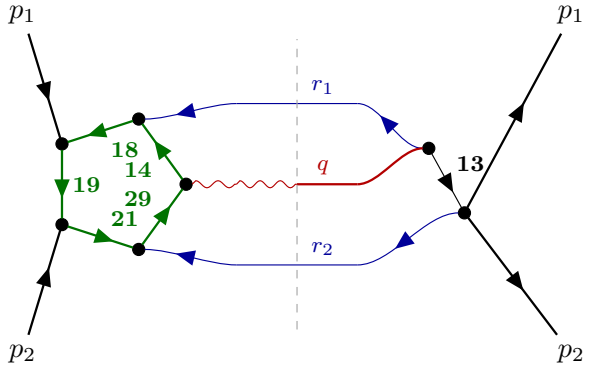
pinched: $13=(p_1+p_2+r_2)^2$



topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

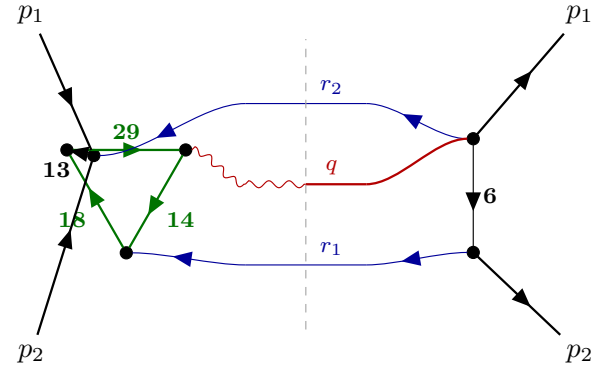
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$



topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	conjugate

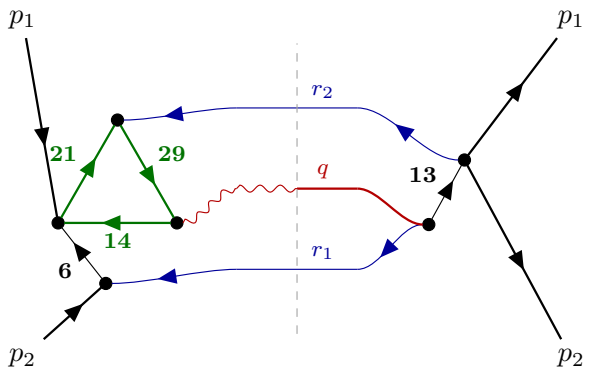
pinched: $6=(p_2+r_1)^2$, $12=(p_1+p_2+r_1)^2$



topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

pinched: $12=(p_1+p_2+r_1)^2$, $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

pinched: $12=(p_1+p_2+r_1)^2$, $18=(k_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$

topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

18	$(k_1+r_1)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

pinched: $14=(k_1)^2$, $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$

topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

pinched: $14=(k_1)^2$, $18=(k_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$

topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

14	$(k_1)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

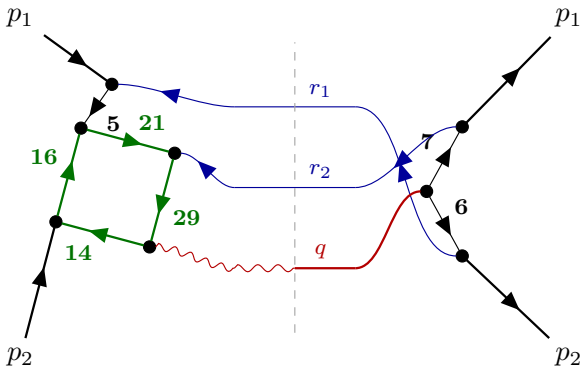
pinched: $18=(k_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$

topo7 (RR family 24, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

21	$(k_1+p_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

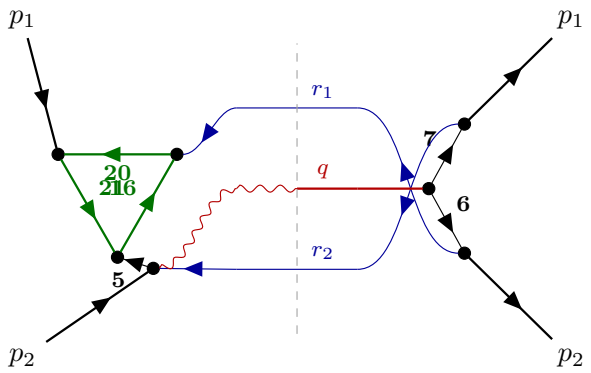
pinched: $14=(k_1)^2$, $18=(k_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

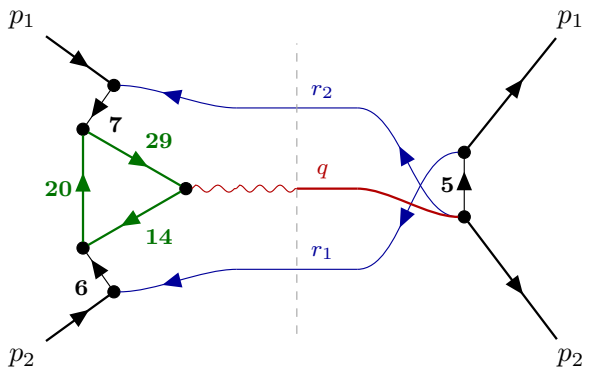
pinched: $20=(k_1+p_2+r_1)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

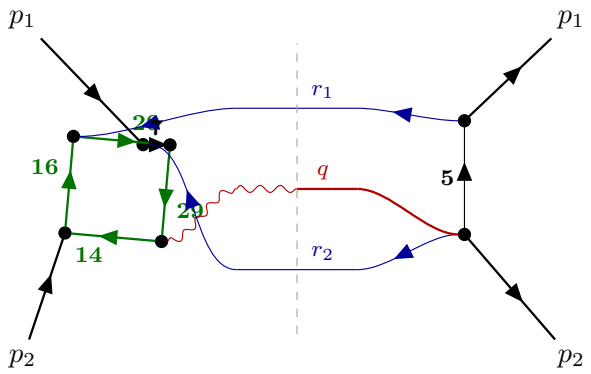
pinched: $14 = (k_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

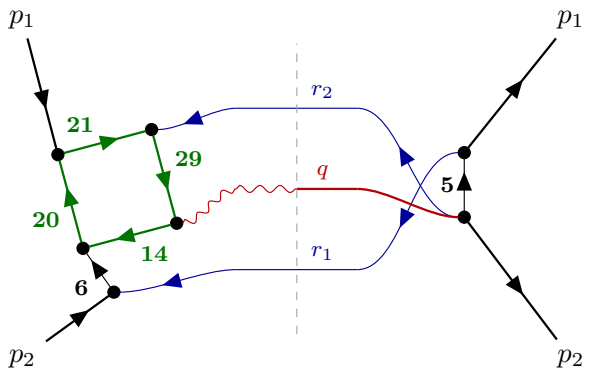
pinched: $16 = (k_1+p_2)^2$, $21 = (k_1+p_1+p_2+r_1)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

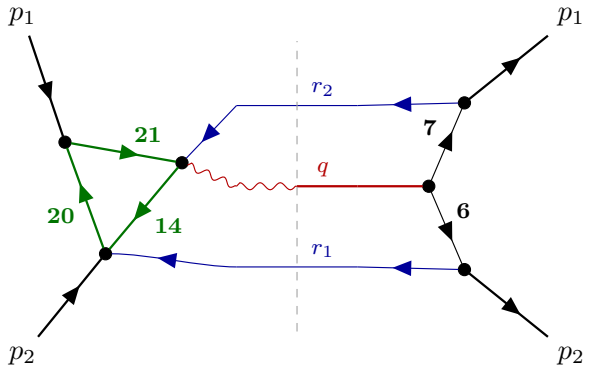
pinched: $6 = (p_2+r_1)^2$, $21 = (k_1+p_1+p_2+r_1)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

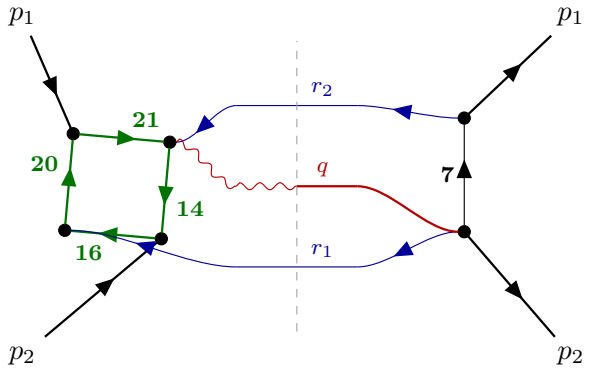
pinched: $7 = (p_1+r_2)^2$, $16 = (k_1+p_2)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

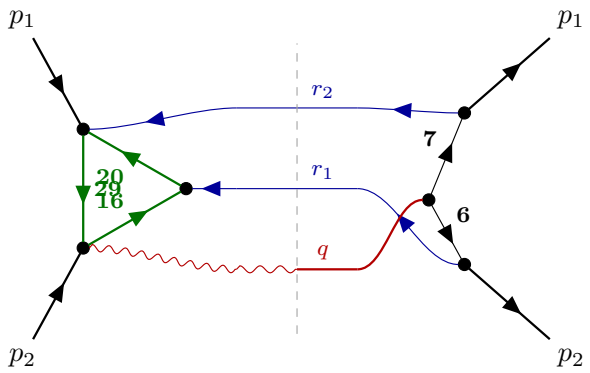
pinched: $5 = (p_1+r_1)^2$, $16 = (k_1+p_2)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
7	$(p_1+r_2)^2$	conjugate

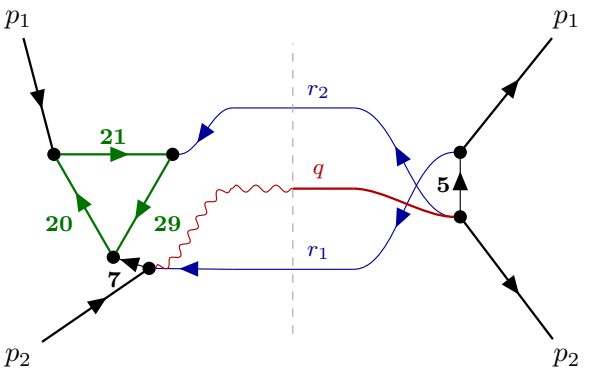
pinched: $5 = (p_1+r_1)^2$, $6 = (p_2+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

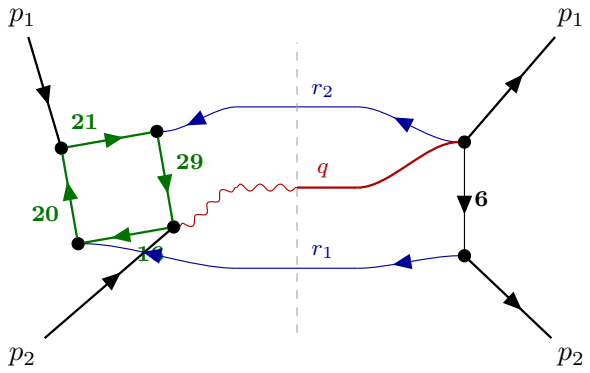
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

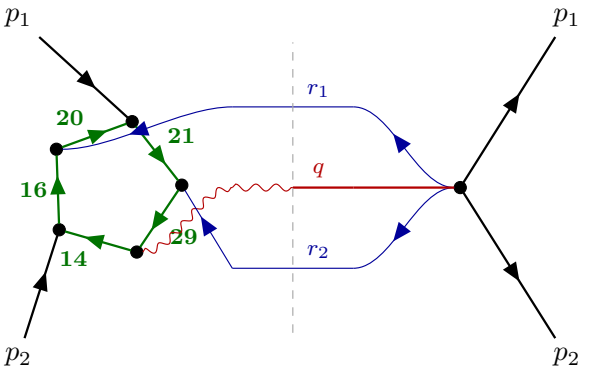
pinched: $6=(p_2+r_1)^2$, $14=(k_1)^2$, $16=(k_1+p_2)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate

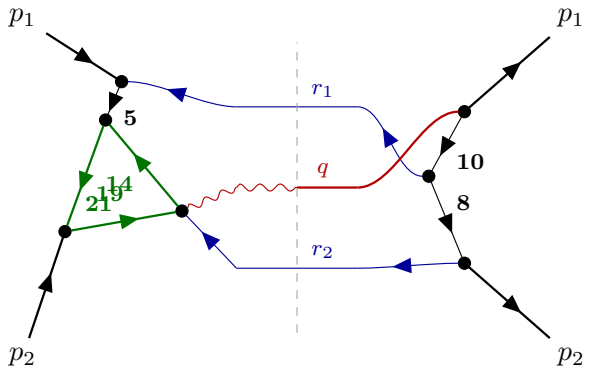
pinched: $5=(p_1+r_1)^2$, $7=(p_1+r_2)^2$, $14=(k_1)^2$



topo8 (RR family 1, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop

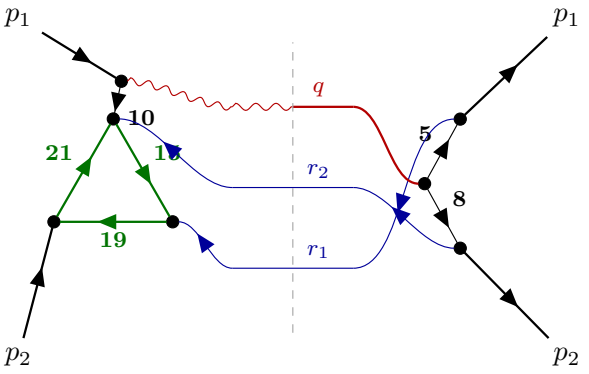
pinched: $5=(p_1+r_1)^2$, $6=(p_2+r_1)^2$, $7=(p_1+r_2)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

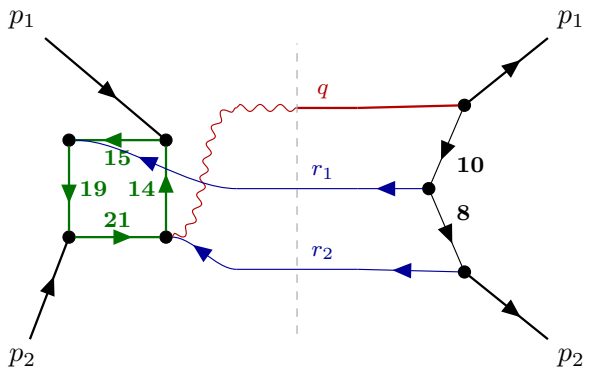
pinched: $15=(k_1+p_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

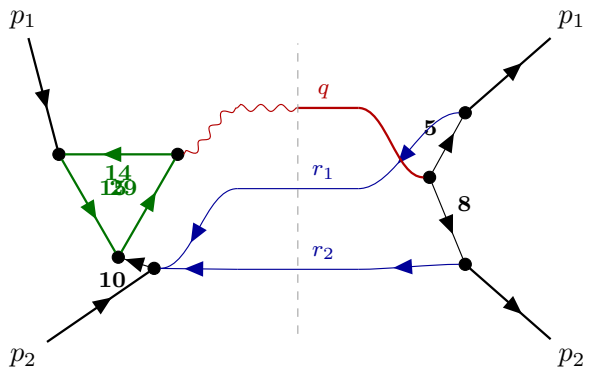
pinched: $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

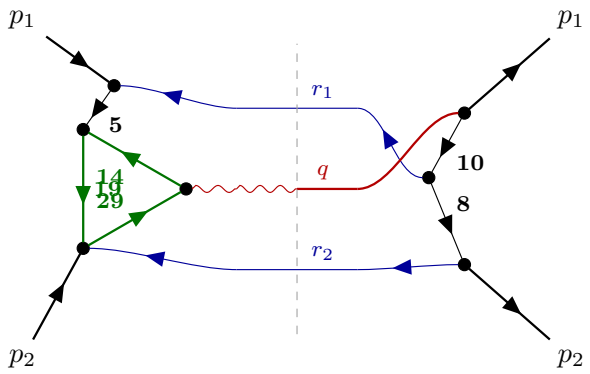
pinched: $5=(p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

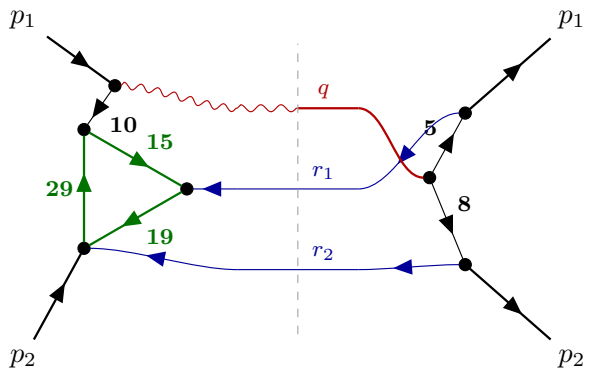
pinched: $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

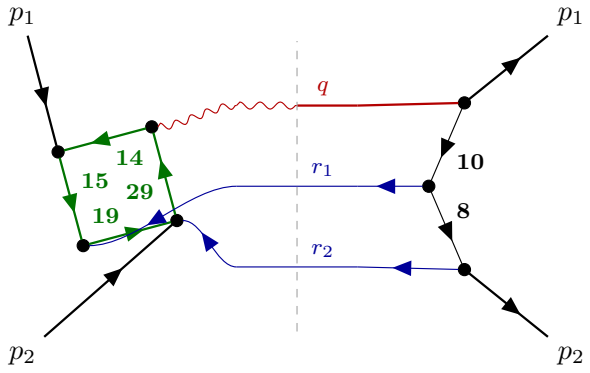
pinched: $15=(k_1+p_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

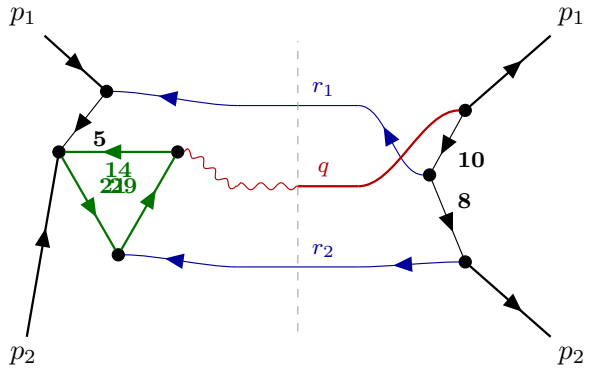
pinched: $14=(k_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

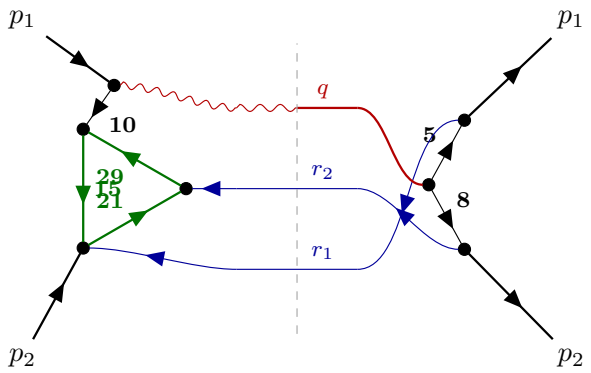
pinched: $5=(p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

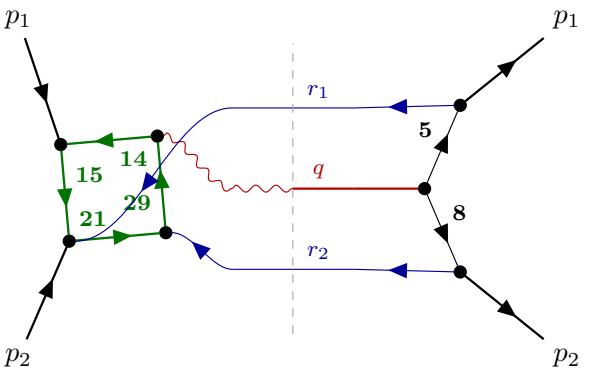
pinched: $15=(k_1+p_1)^2$, $19=(k_1+p_1+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

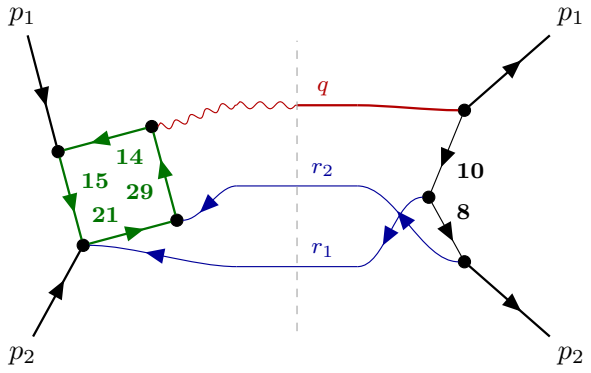
pinched: $14=(k_1)^2$, $19=(k_1+p_1+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

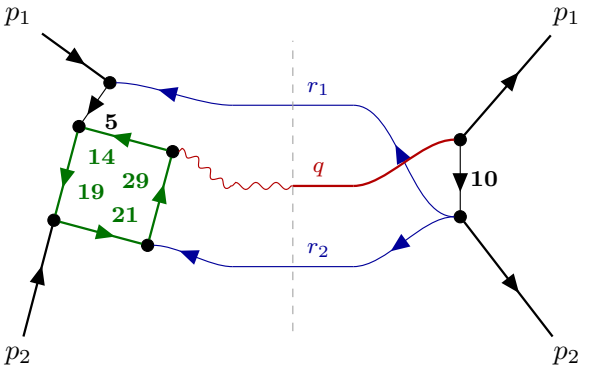
pinched: $10=(p_2+r_1+r_2)^2$, $19=(k_1+p_1+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

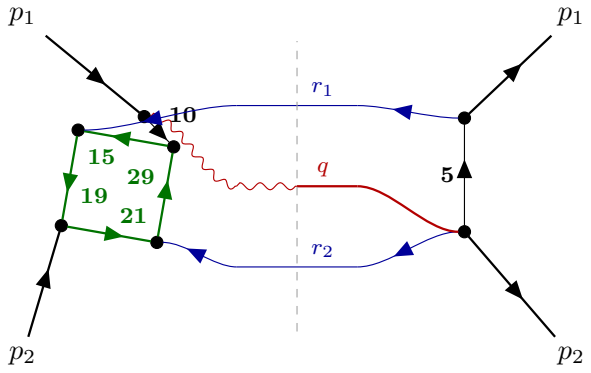
pinched: $5=(p_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	conjugate

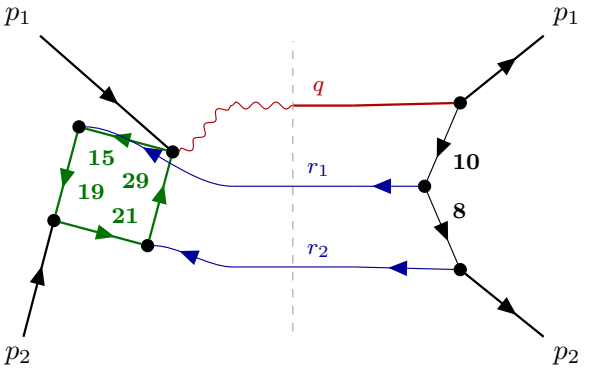
pinched: $8=(p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

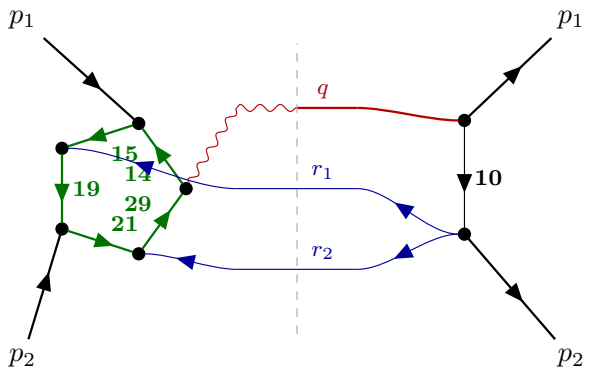
pinched: $8=(p_2+r_2)^2$, $14=(k_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

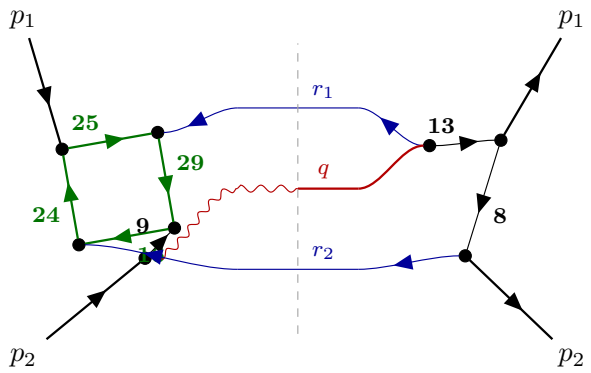
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$



topo9 (RR family 7, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	conjugate

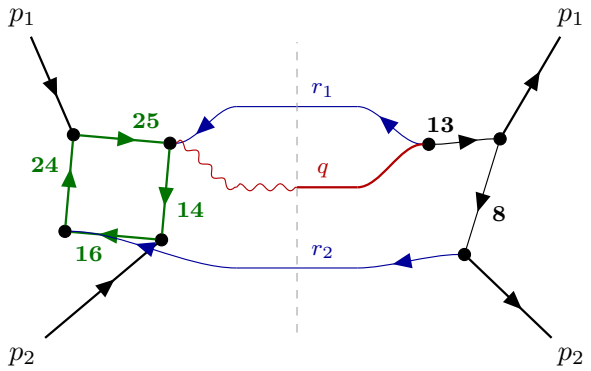
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

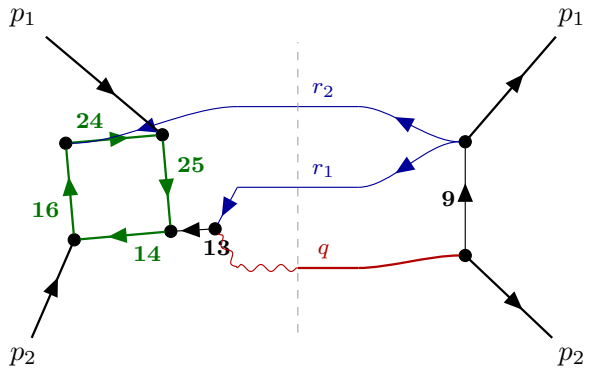
pinched: $14=(k_1)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

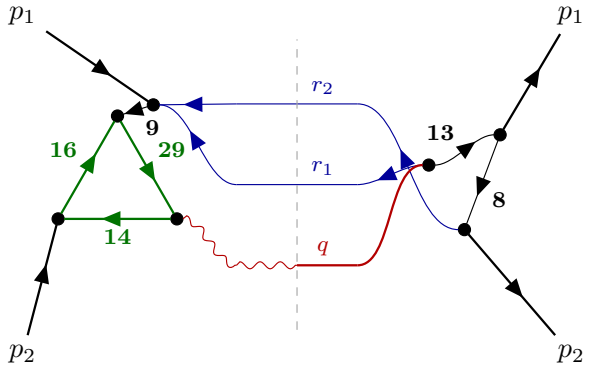
pinched: $9=(p_1+r_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	conjugate

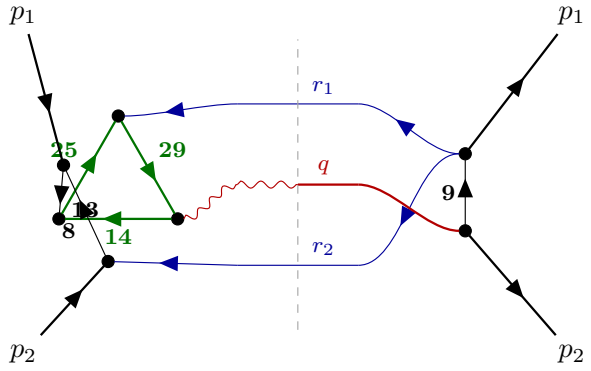
pinched: $8=(p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

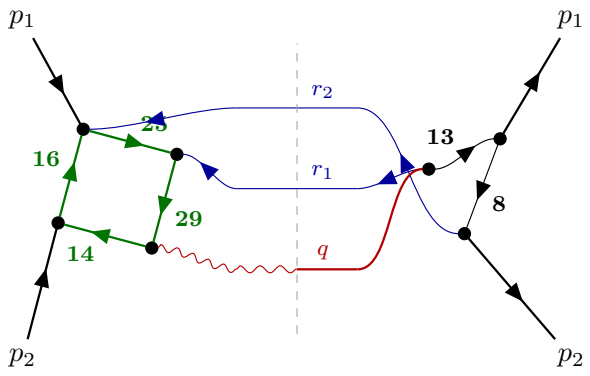
pinched: $24=(k_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	conjugate

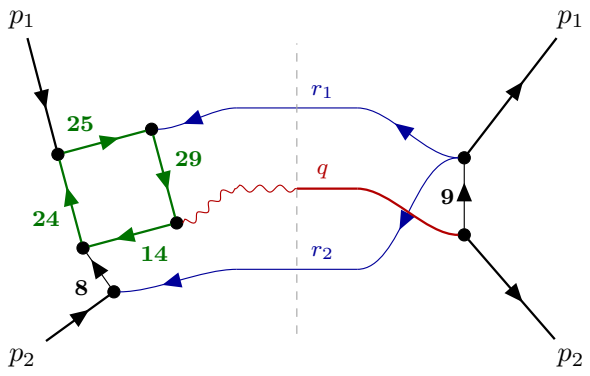
pinched: $16=(k_1+p_2)^2$, $24=(k_1+p_2+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

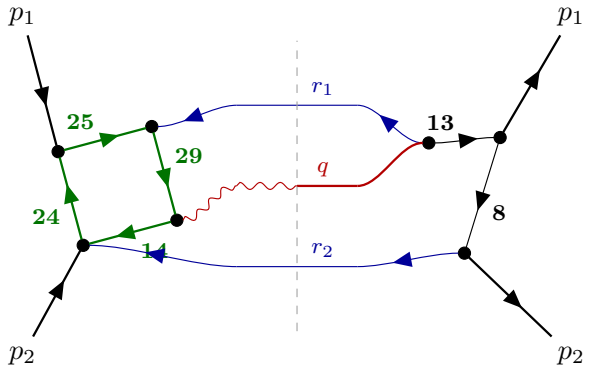
pinched: $9=(p_1+r_1+r_2)^2$, $24=(k_1+p_2+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	conjugate

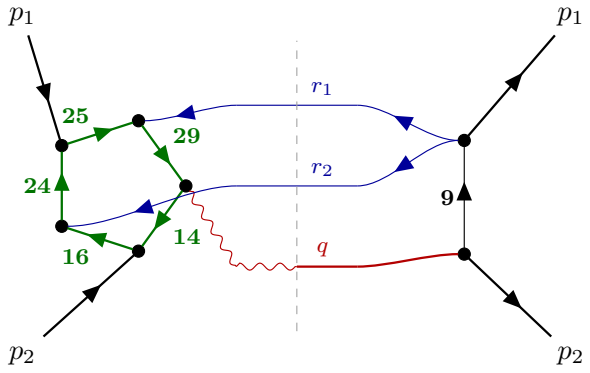
pinched: $13=(p_1+p_2+r_2)^2$, $16=(k_1+p_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

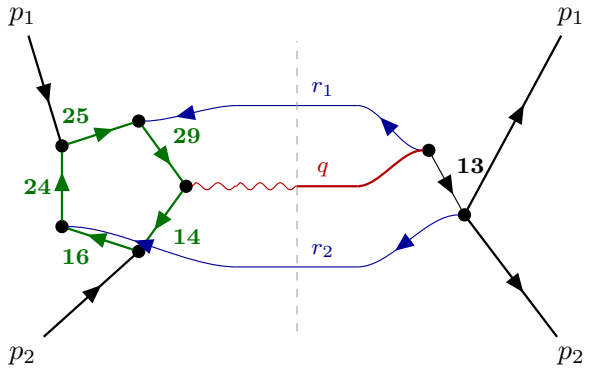
pinched: $9=(p_1+r_1+r_2)^2$, $16=(k_1+p_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	conjugate

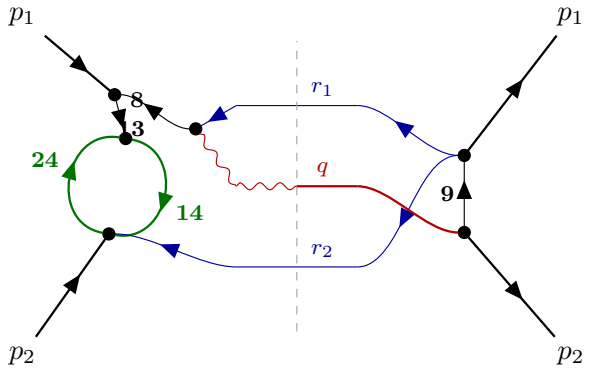
pinched: $8=(p_2+r_2)^2$, $13=(p_1+p_2+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	conjugate

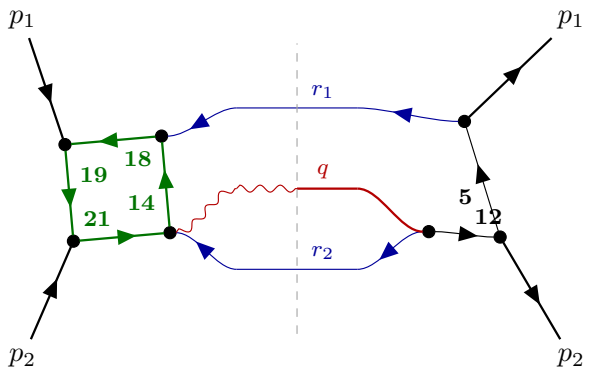
pinched: $8=(p_2+r_2)^2$, $9=(p_1+r_1+r_2)^2$



topo10 (RR family 34, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	conjugate

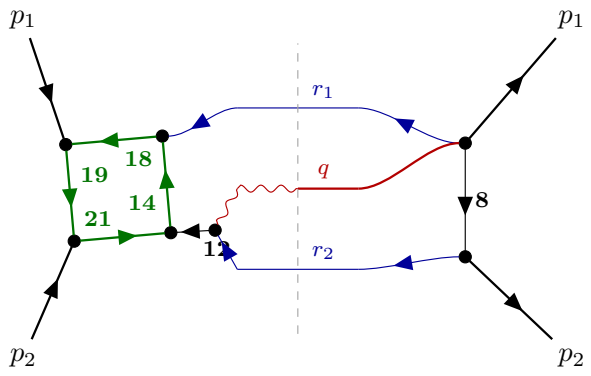
pinched: $16=(k_1+p_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

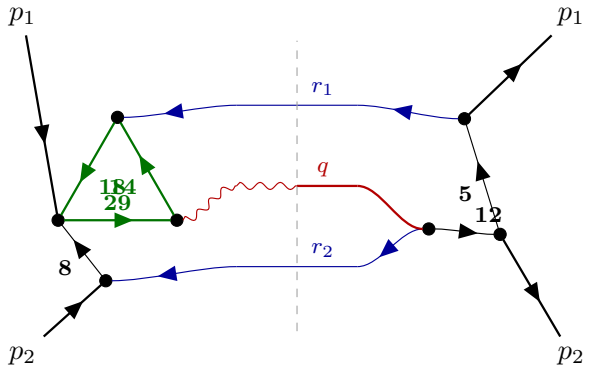
pinched: $8=(p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

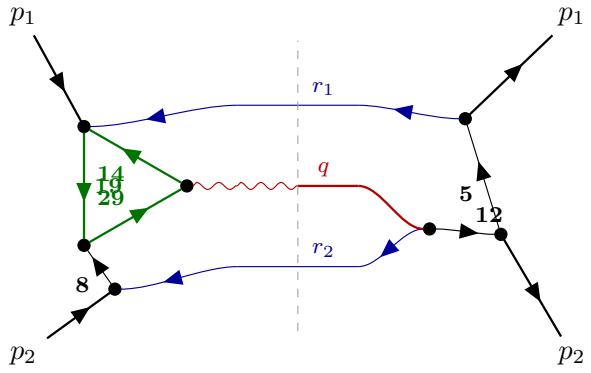
pinched: $5=(p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

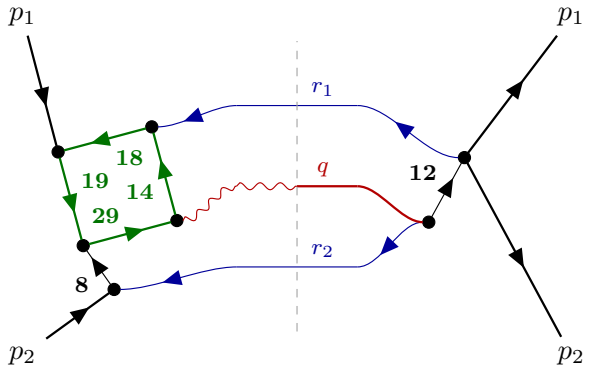
pinched: $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

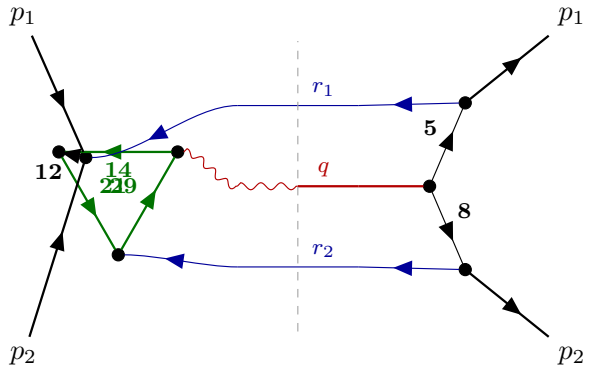
pinched: $18=(k_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	conjugate

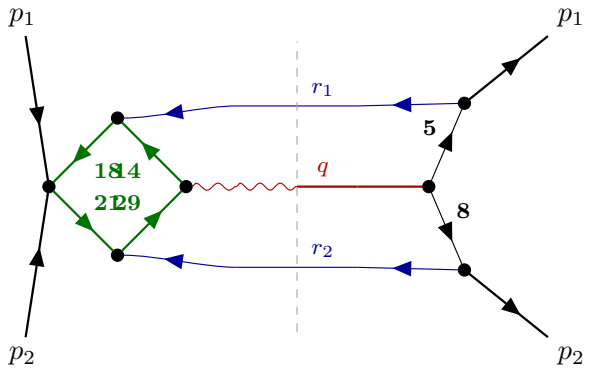
pinched: $5=(p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

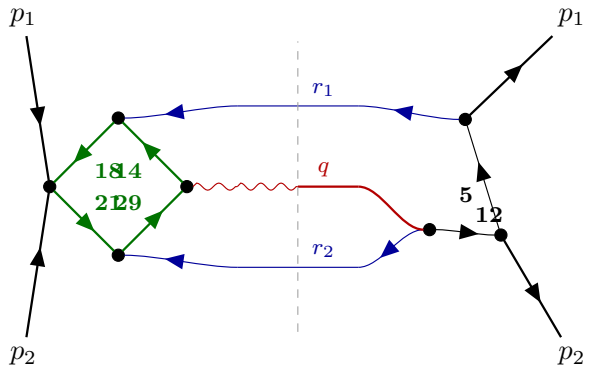
pinched: $18=(k_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

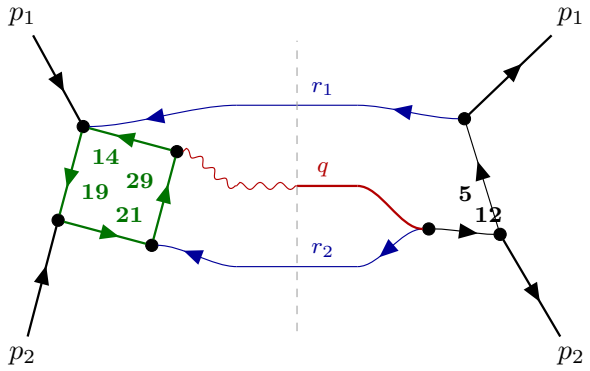
pinched: $12=(p_1+p_2+r_1)^2$, $19=(k_1+p_1+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

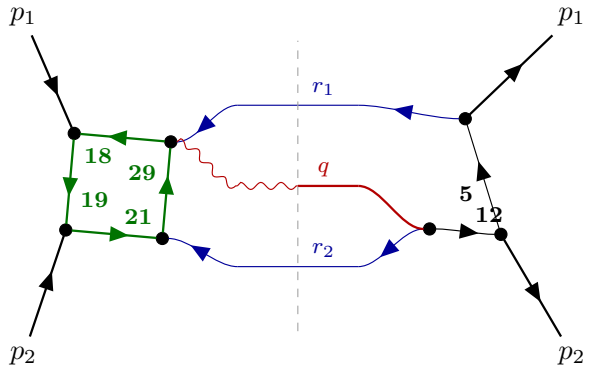
pinched: $8=(p_2+r_2)^2$, $19=(k_1+p_1+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

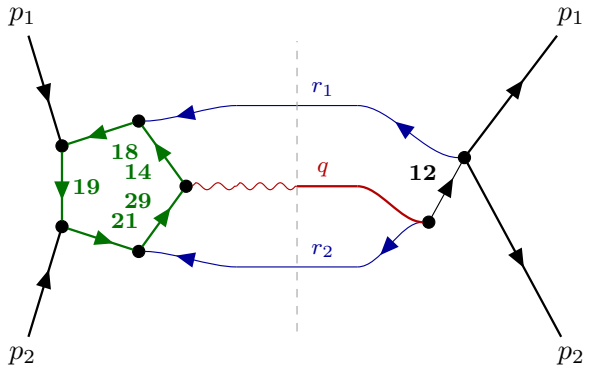
pinched: $8=(p_2+r_2)^2$, $18=(k_1+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

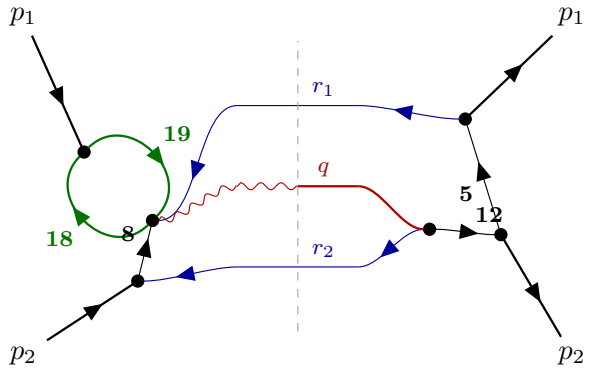
pinched: $8=(p_2+r_2)^2$, $14=(k_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
12	$(p_1+p_2+r_1)^2$	conjugate

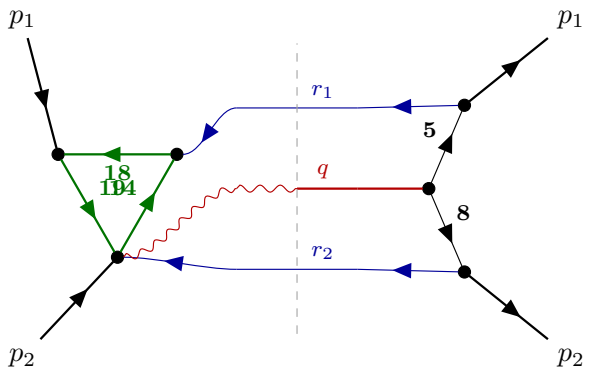
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

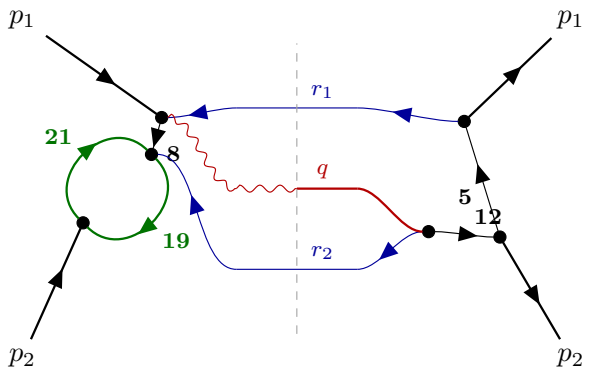
pinched: $14=(k_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

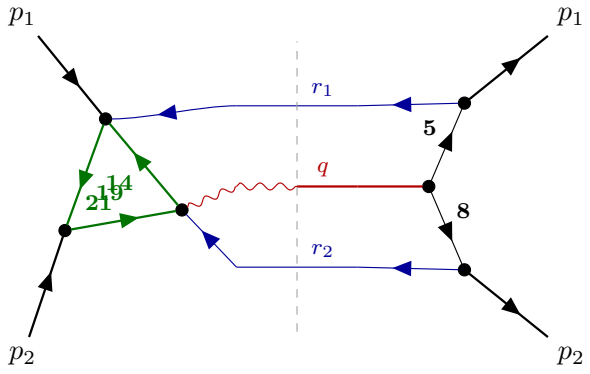
pinched: $12=(p_1+p_2+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

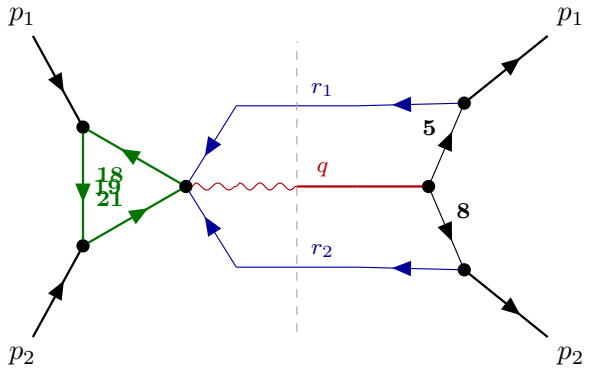
pinched: $14=(k_1)^2$, $18=(k_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

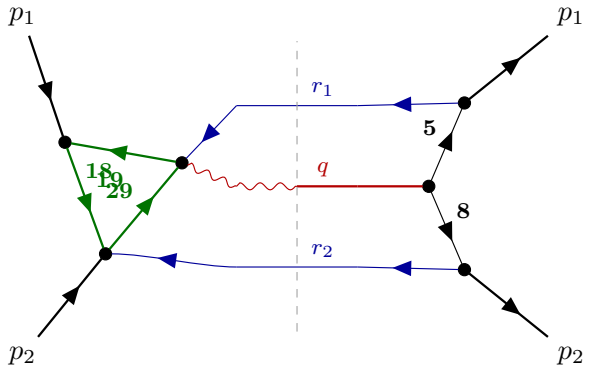
pinched: $12=(p_1+p_2+r_1)^2$, $18=(k_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

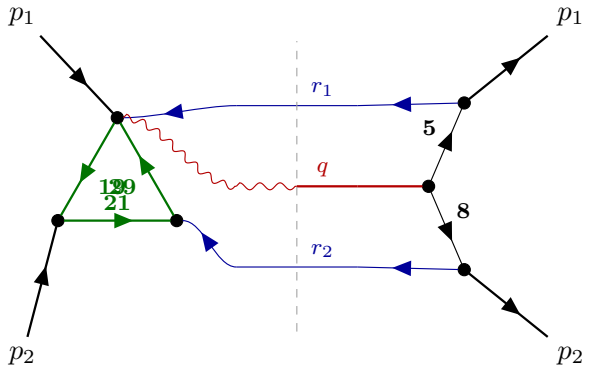
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

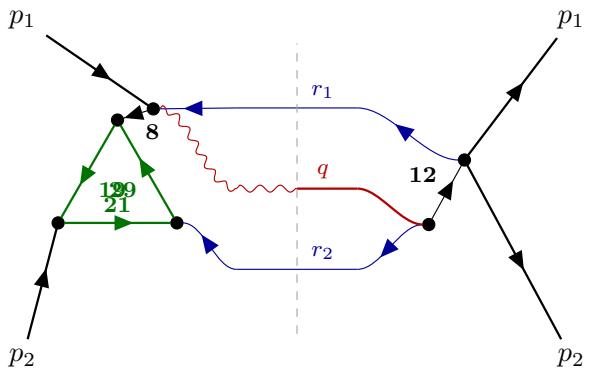
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

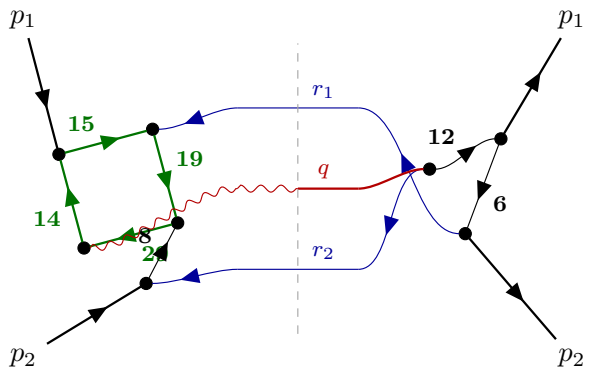
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$, $18=(k_1+r_1)^2$



topo11 (RR family 17, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	conjugate

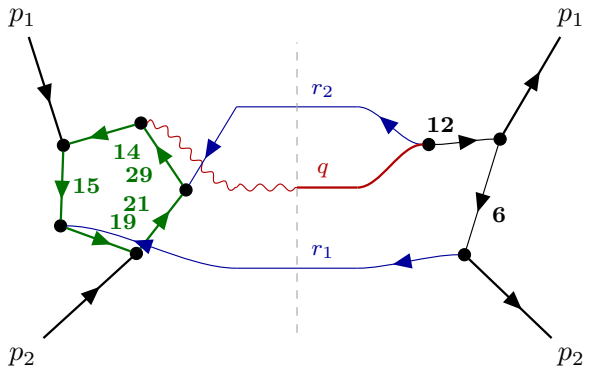
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $18=(k_1+r_1)^2$



topo12 (RR family 21, pentagon q-p1-r1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

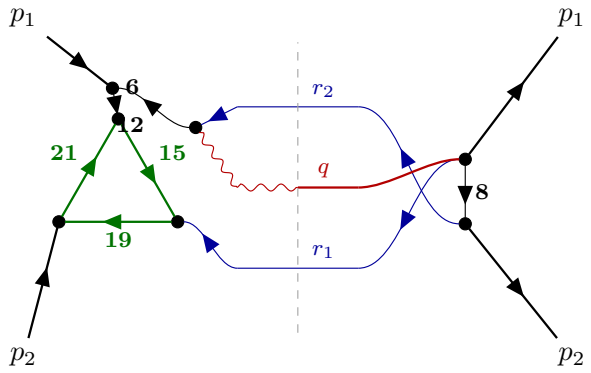
pinched: $21=(k_1+p_1+p_2+r_1)^2$



topo12 (RR family 21, pentagon q-p1-r1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

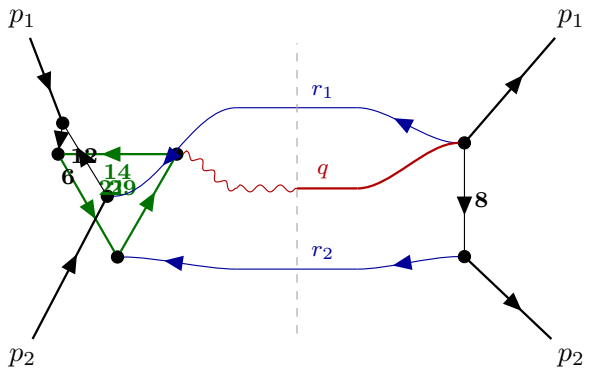
pinched: $8=(p_2+r_2)^2$



topo12 (RR family 21, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

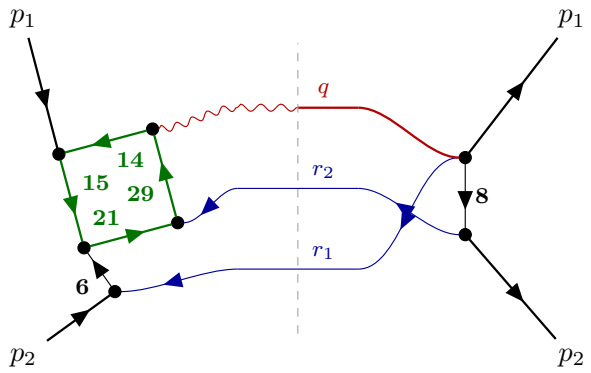
pinched: $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo12 (RR family 21, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

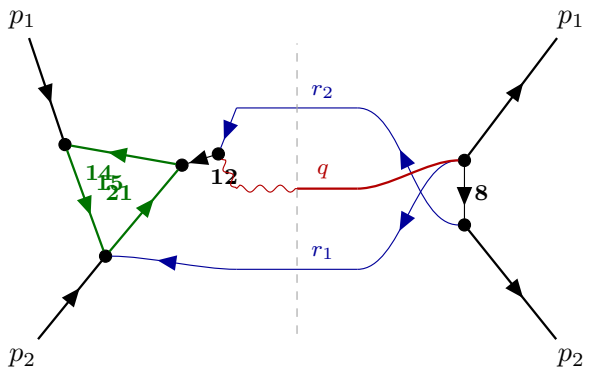
pinched: $15=(k_1+p_1)^2$, $19=(k_1+p_1+r_1)^2$



topo12 (RR family 21, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

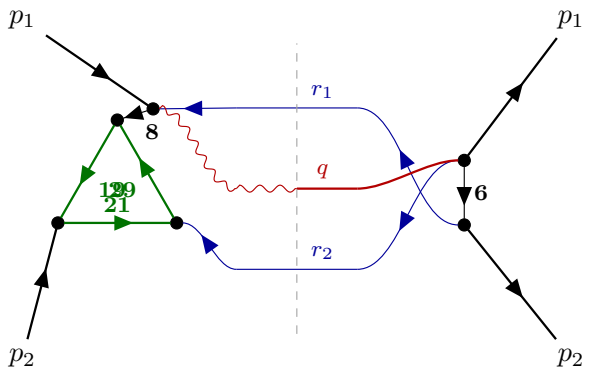
pinched: $12=(p_1+p_2+r_1)^2$, $19=(k_1+p_1+r_1)^2$



topo12 (RR family 21, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

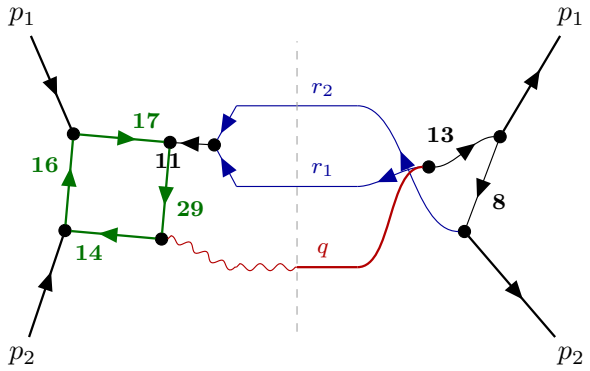
pinched: $6=(p_2+r_1)^2$, $19=(k_1+p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo12 (RR family 21, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

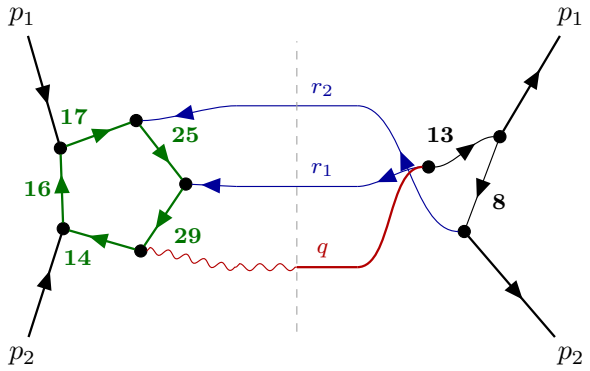
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo13 (RR family 31, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

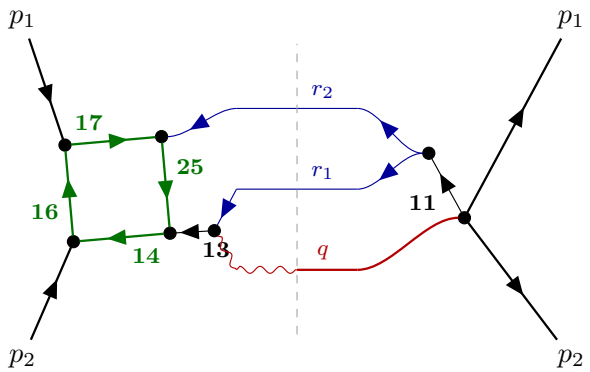
pinched: $25=(k_1+p_1+p_2+r_2)^2$



topo13 (RR family 31, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

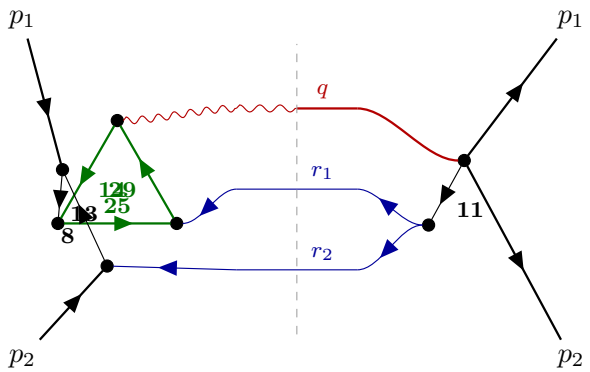
pinched: $11=(r_1+r_2)^2$



topo13 (RR family 31, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

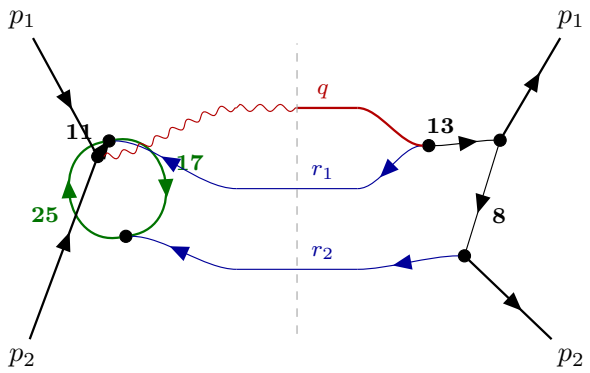
pinched: $8=(p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo13 (RR family 31, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

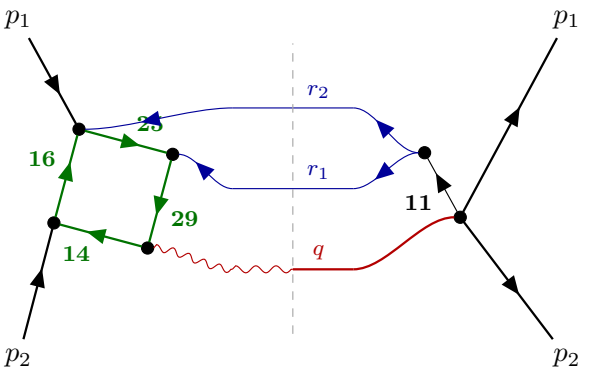
pinched: $16=(k_1+p_2)^2$, $17=(k_1+p_1+p_2)^2$



topo13 (RR family 31, pentagon q-p2-p1-r2-r1), sector with 5 of 8 lines

17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

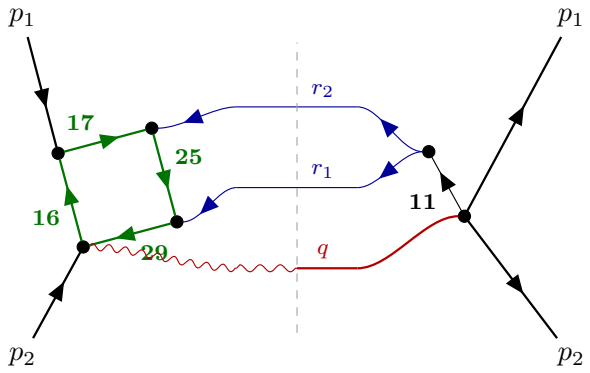
pinched: $14=(k_1)^2$, $16=(k_1+p_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo13 (RR family 31, pentagon q-p2-p1-r2-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

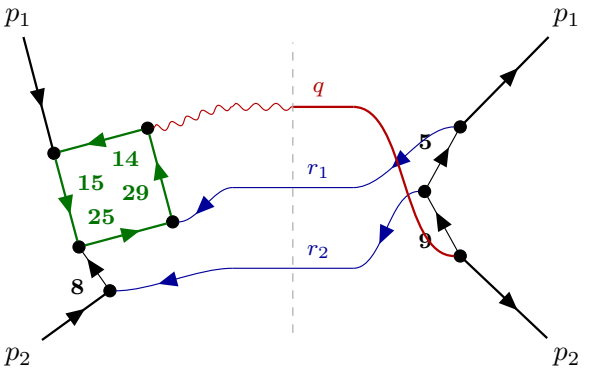
pinched: $8=(p_2+r_2)^2$, $13=(p_1+p_2+r_2)^2$, $17=(k_1+p_1+p_2)^2$



topo13 (RR family 31, pentagon q-p2-p1-r2-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

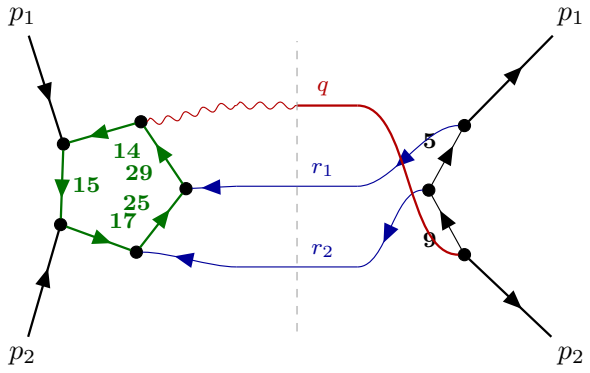
pinched: $8=(p_2+r_2)^2$, $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$



topo14 (RR family 6, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

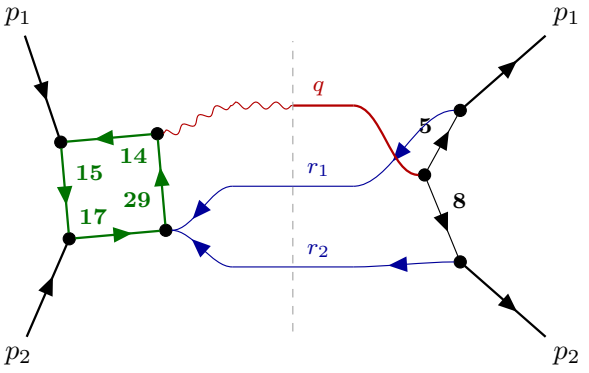
pinched: $17=(k_1+p_1+p_2)^2$



topo14 (RR family 6, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

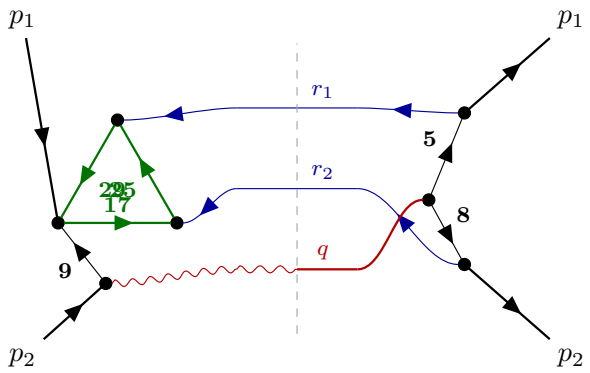
pinched: $8=(p_2+r_2)^2$



topo14 (RR family 6, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

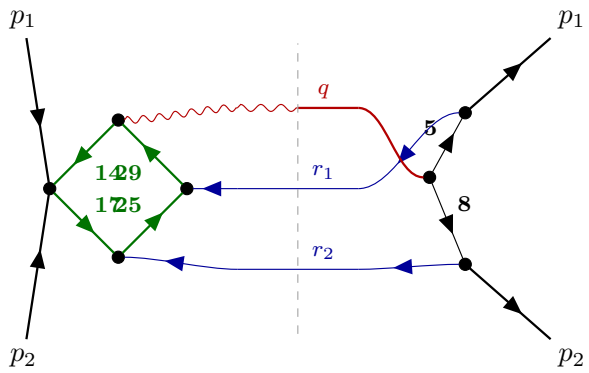
pinched: $9=(p_1+r_1+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo14 (RR family 6, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

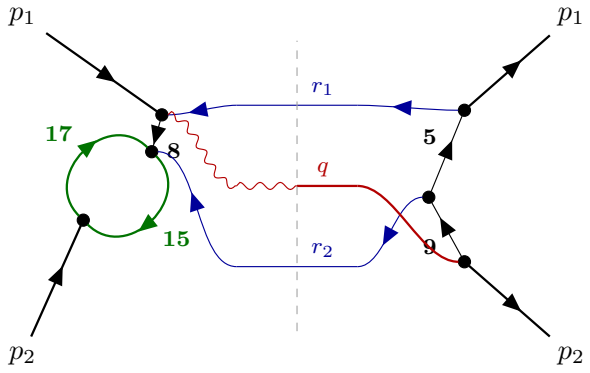
pinched: $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo14 (RR family 6, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

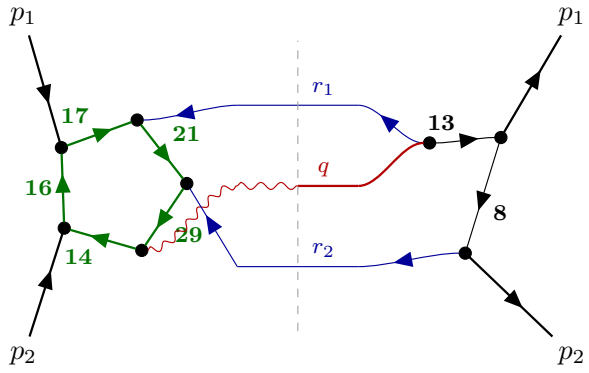
pinched: $9=(p_1+r_1+r_2)^2$, $15=(k_1+p_1)^2$



topo14 (RR family 6, pentagon q-p1-p2-r2-r1), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

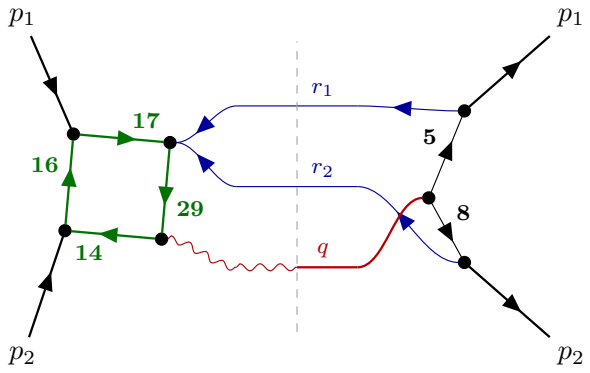
pinched: $14=(k_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

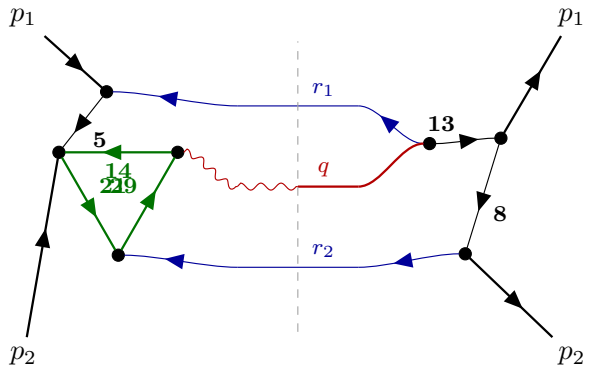
pinched: $5=(p_1+r_1)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

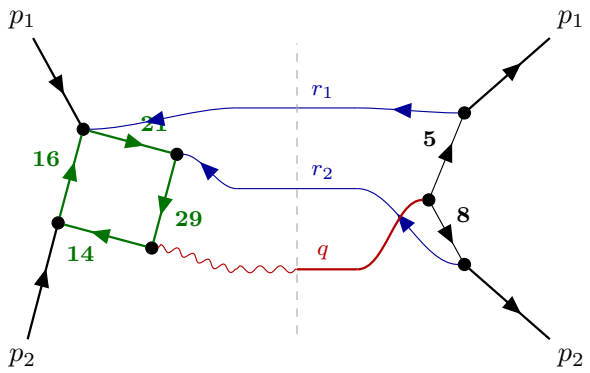
pinched: $13=(p_1+p_2+r_2)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

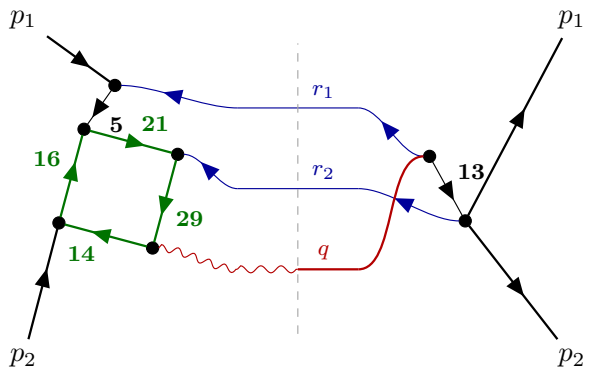
pinched: $16=(k_1+p_2)^2$, $17=(k_1+p_1+p_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

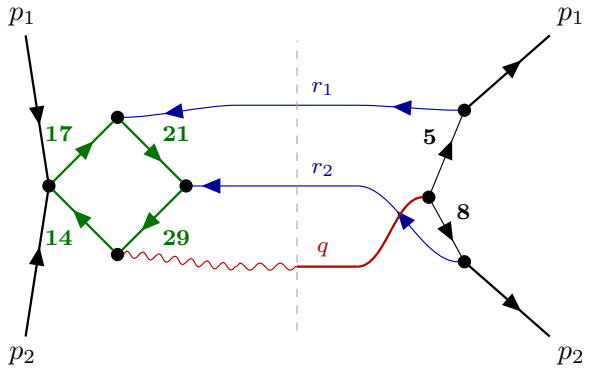
pinched: $13=(p_1+p_2+r_2)^2$, $17=(k_1+p_1+p_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

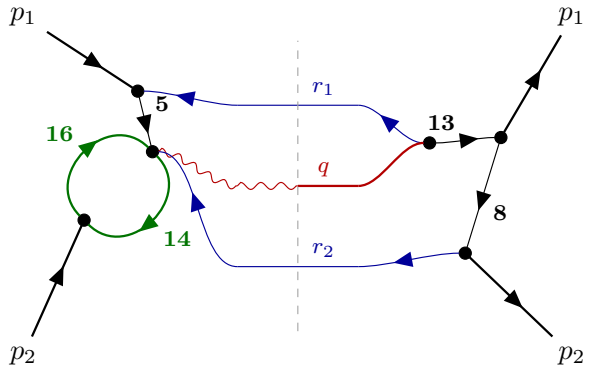
pinched: $8=(p_2+r_2)^2$, $17=(k_1+p_1+p_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

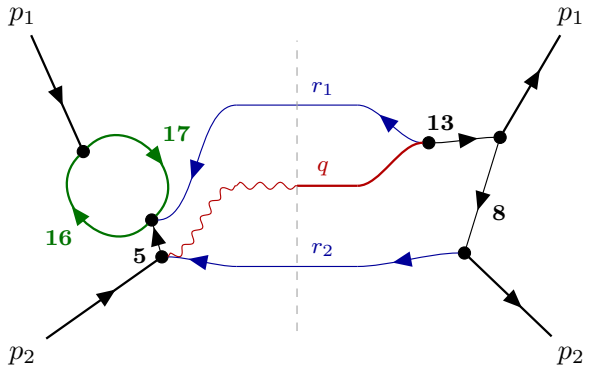
pinched: $13=(p_1+p_2+r_2)^2$, $16=(k_1+p_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

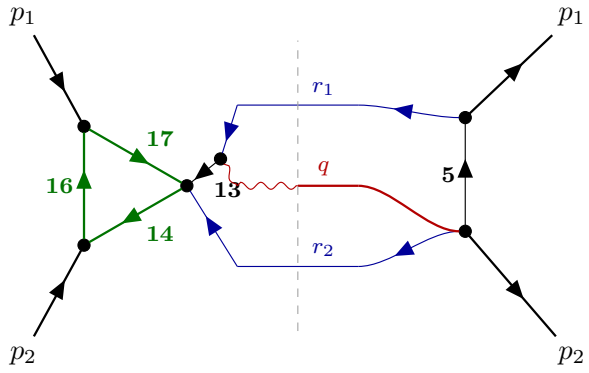
pinched: $17=(k_1+p_1+p_2)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

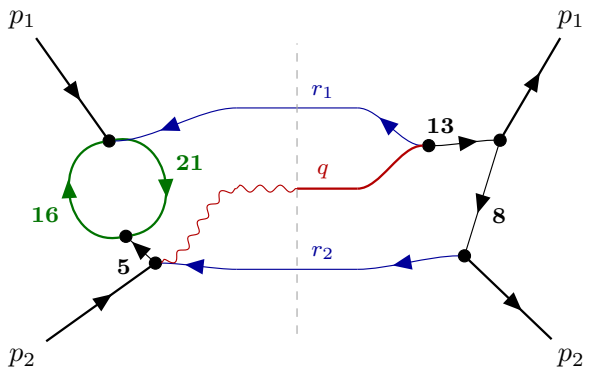
pinched: $14=(k_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

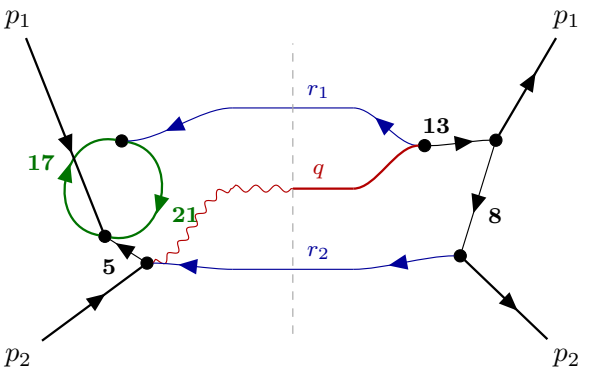
pinched: $8=(p_2+r_2)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

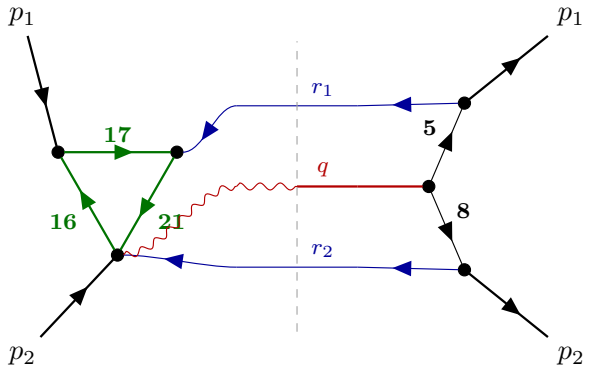
pinched: $14=(k_1)^2$, $17=(k_1+p_1+p_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

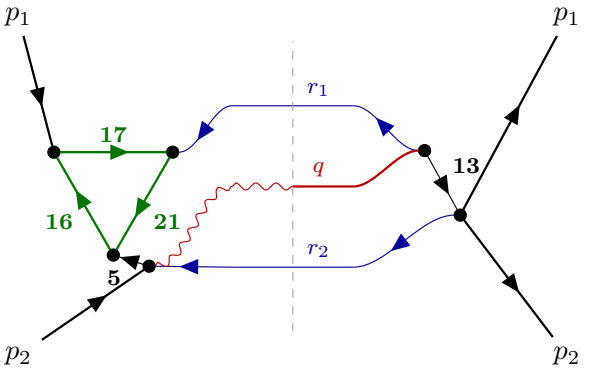
pinched: $14=(k_1)^2$, $16=(k_1+p_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

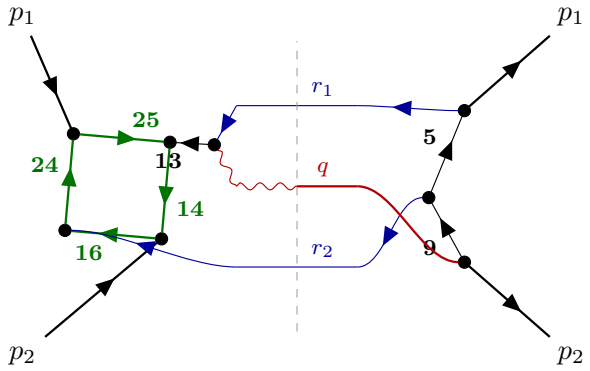
pinched: $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo15 (RR family 18, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

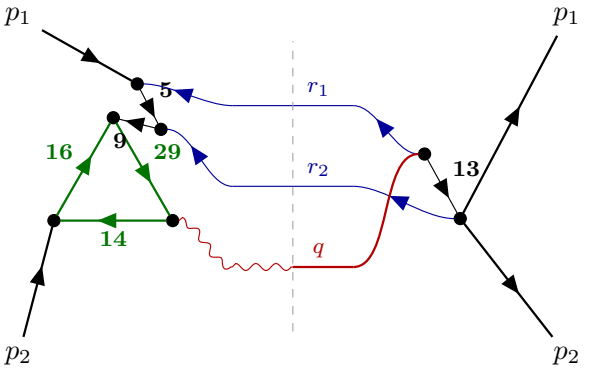
pinched: $8=(p_2+r_2)^2$, $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

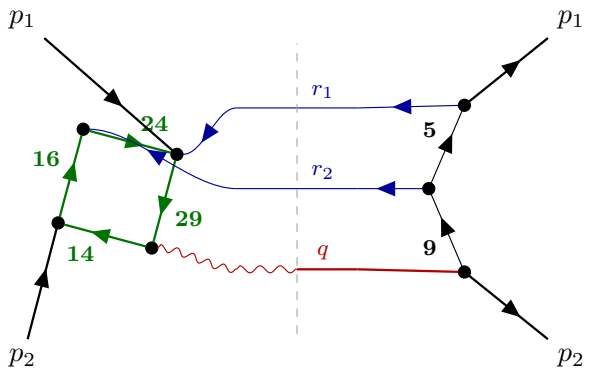
pinched: $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

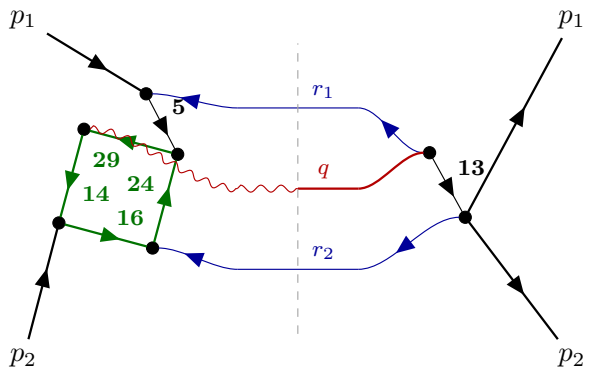
pinched: $24=(k_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

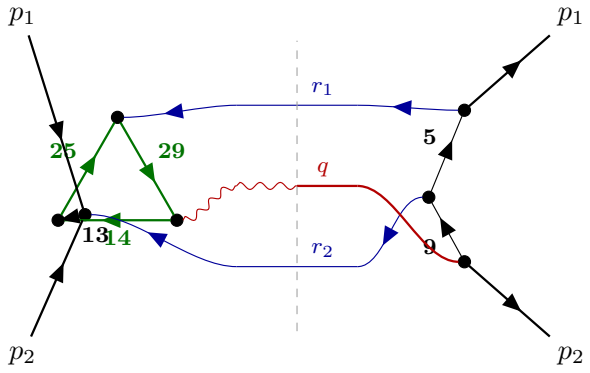
pinched: $13=(p_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

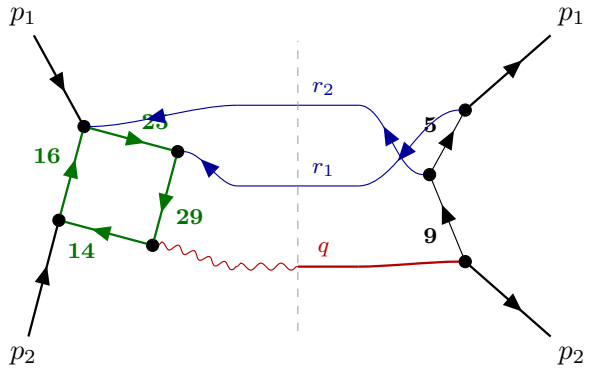
pinched: $9=(p_1+r_1+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

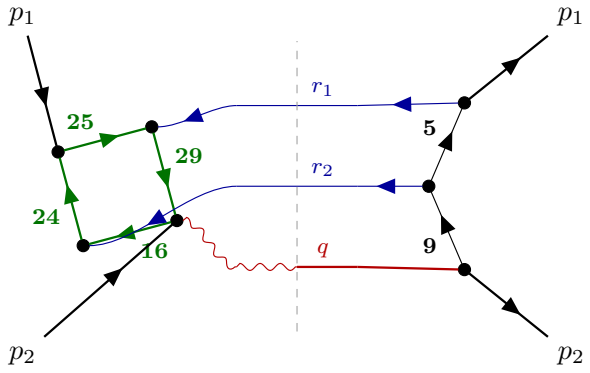
pinched: $16=(k_1+p_2)^2$, $24=(k_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

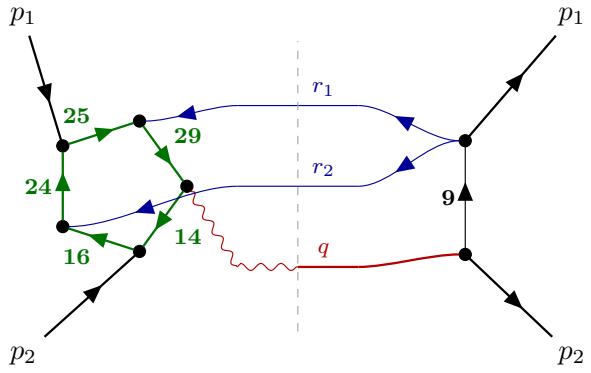
pinched: $13=(p_1+p_2+r_2)^2$, $24=(k_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

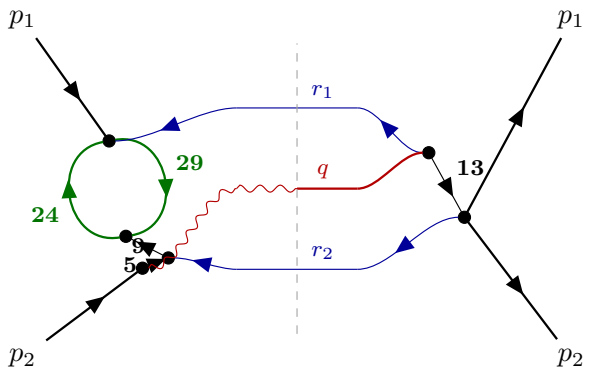
pinched: $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	conjugate

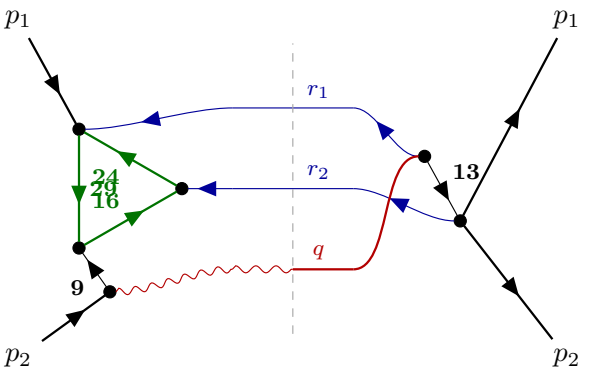
pinched: $5=(p_1+r_1)^2$, $13=(p_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

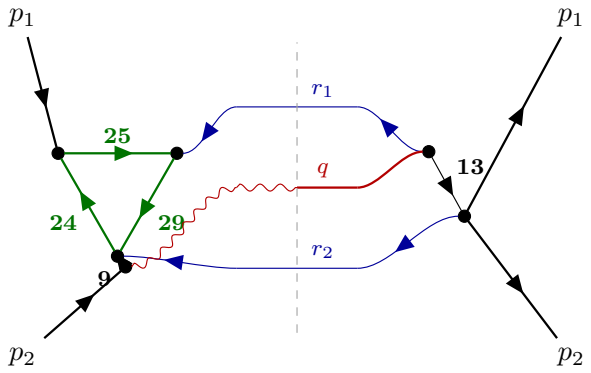
pinched: $14=(k_1)^2$, $16=(k_1+p_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

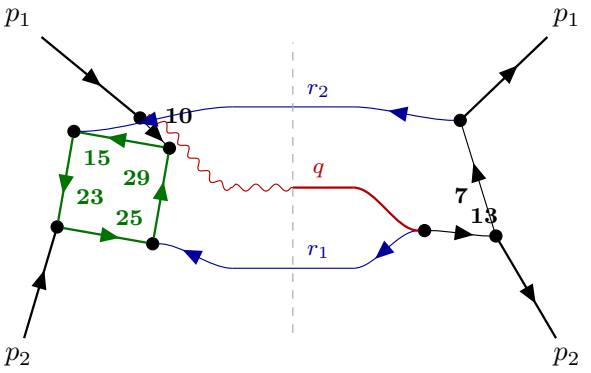
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo16 (RR family 32, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

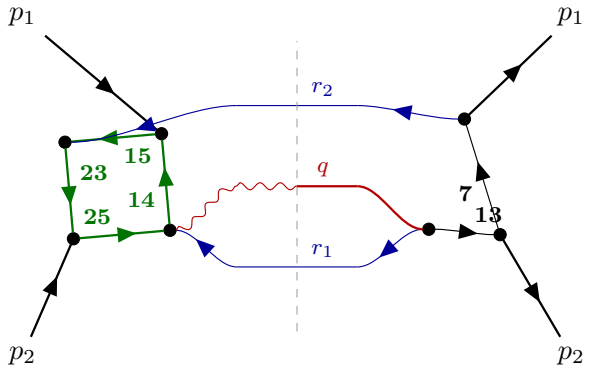
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $16=(k_1+p_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

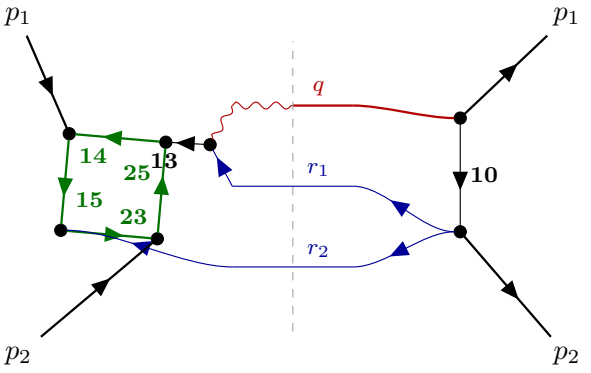
pinched: $14=(k_1)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

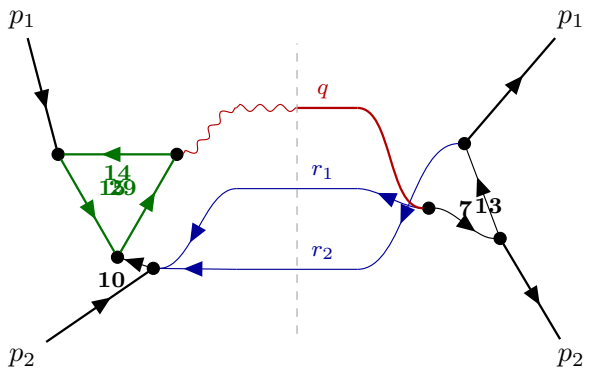
pinched: $10=(p_2+r_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	conjugate

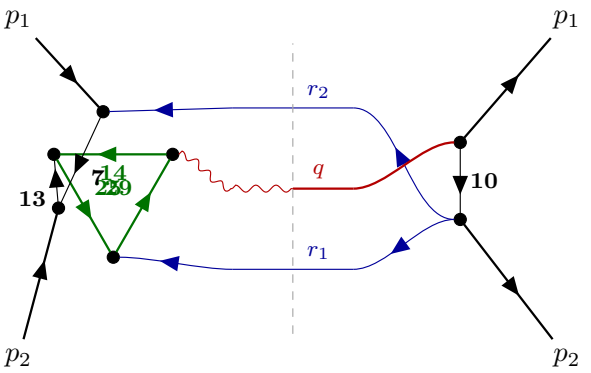
pinched: $7=(p_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

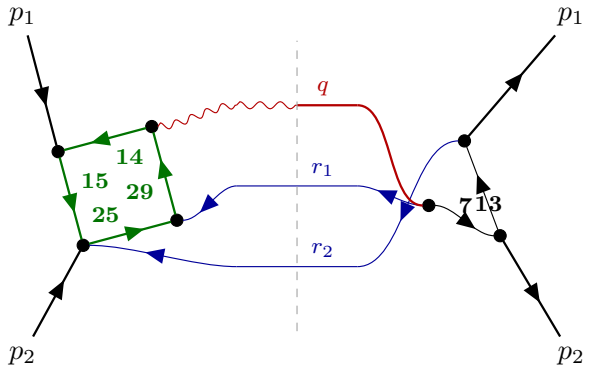
pinched: $23=(k_1+p_1+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	conjugate

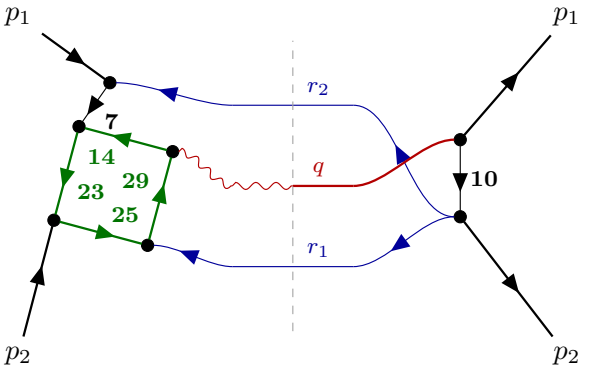
pinched: $15=(k_1+p_1)^2$, $23=(k_1+p_1+r_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

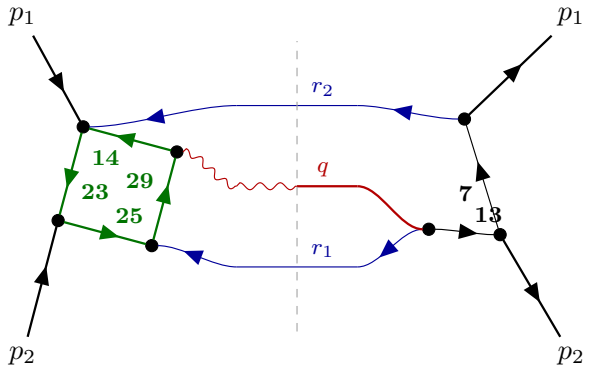
pinched: $10=(p_2+r_1+r_2)^2$, $23=(k_1+p_1+r_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	conjugate

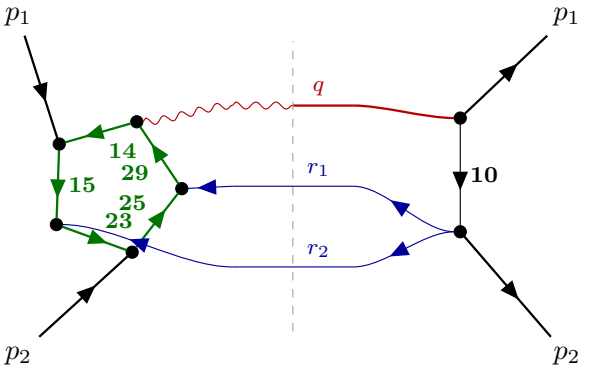
pinched: $13=(p_1+p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

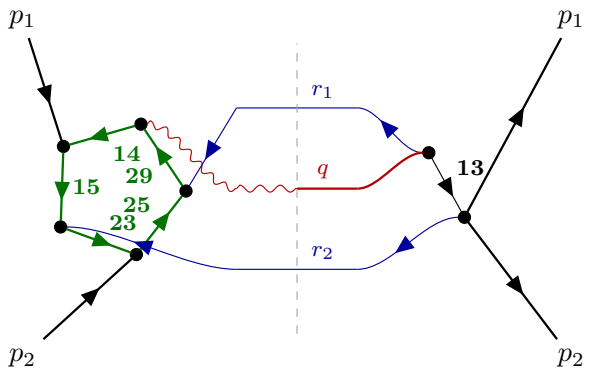
pinched: $10=(p_2+r_1+r_2)^2$, $15=(k_1+p_1)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	conjugate

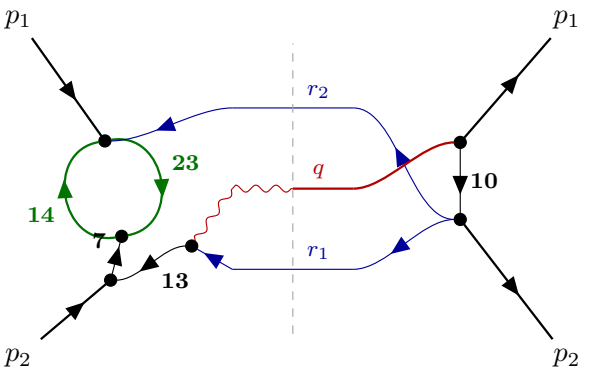
pinched: $7=(p_1+r_2)^2$, $13=(p_1+p_2+r_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	conjugate

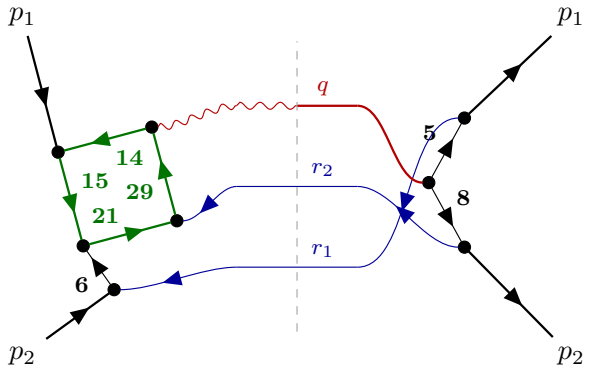
pinched: $7=(p_1+r_2)^2$, $10=(p_2+r_1+r_2)^2$



topo17 (RR family 28, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	conjugate

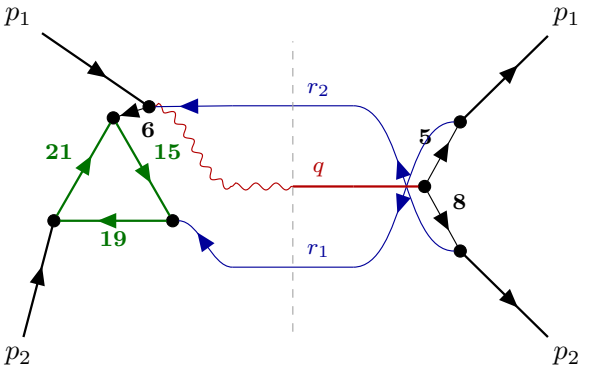
pinched: $15=(k_1+p_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo18 (RR family 2, pentagon q-p1-r1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

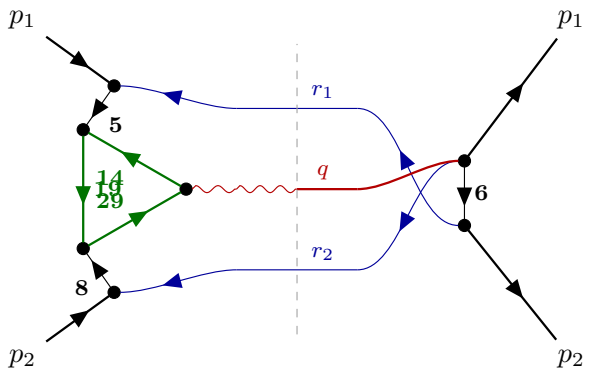
pinched: $19=(k_1+p_1+r_1)^2$



topo18 (RR family 2, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

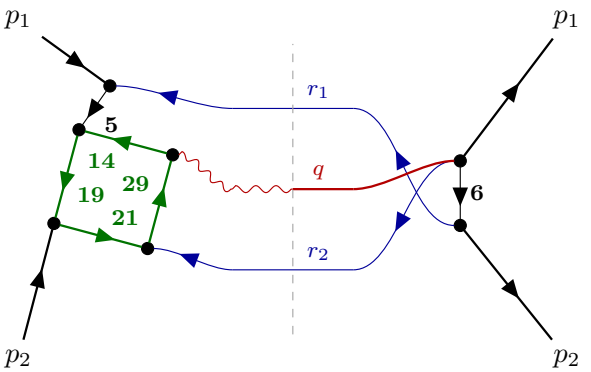
pinched: $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo18 (RR family 2, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

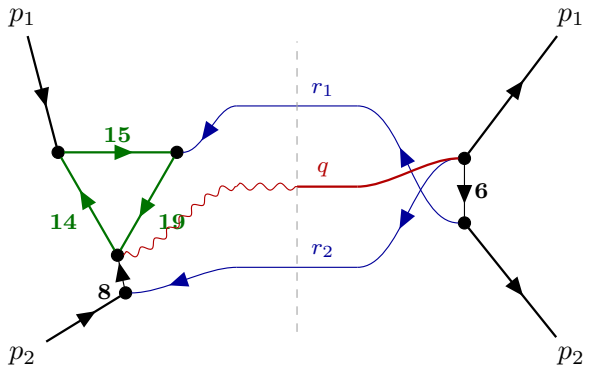
pinched: $15=(k_1+p_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo18 (RR family 2, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

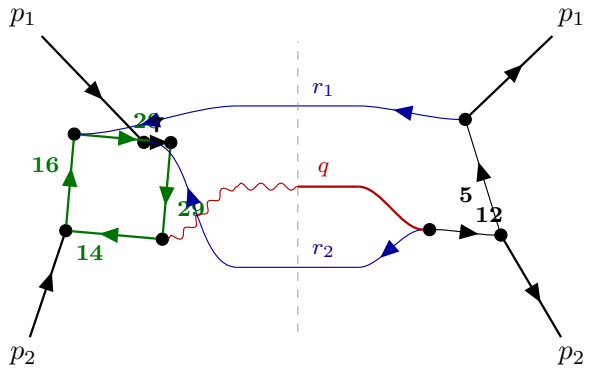
pinched: $8=(p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo18 (RR family 2, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

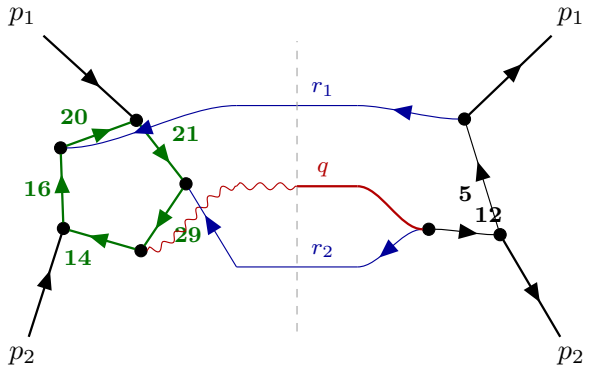
pinched: $5=(p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo19 (RR family 15, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

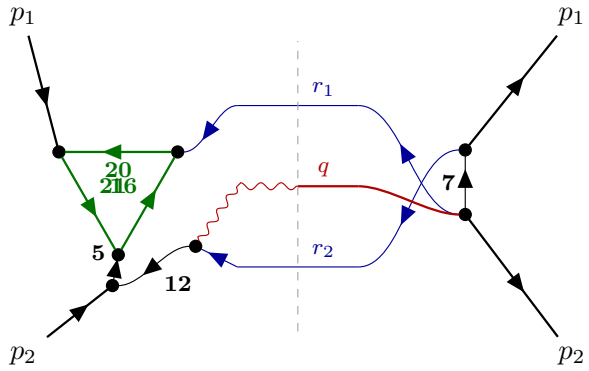
pinched: $21=(k_1+p_1+p_2+r_1)^2$



topo19 (RR family 15, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

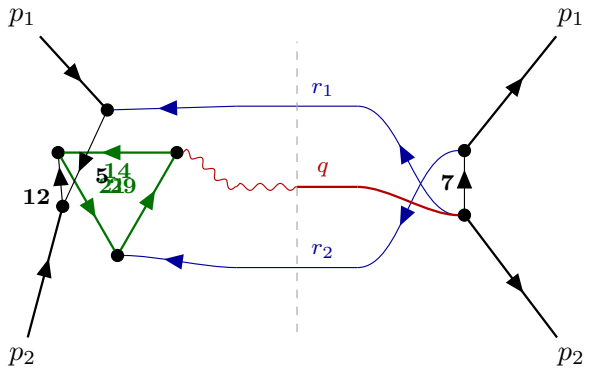
pinched: $7=(p_1+r_2)^2$



topo19 (RR family 15, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate

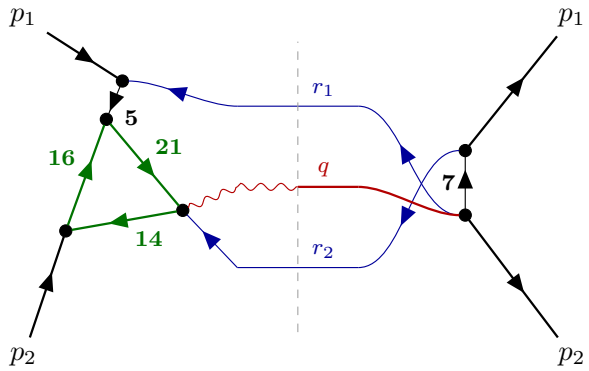
pinched: $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo19 (RR family 15, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate

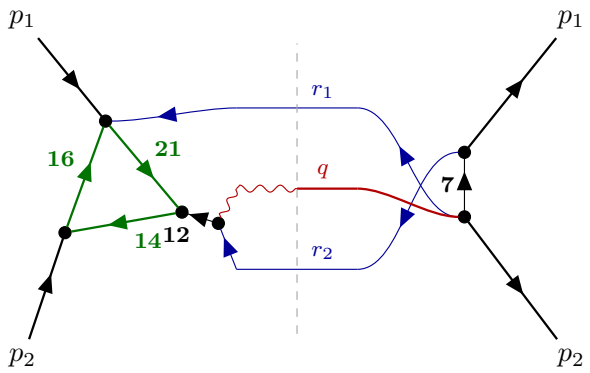
pinched: $16=(k_1+p_2)^2$, $20=(k_1+p_2+r_1)^2$



topo19 (RR family 15, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate

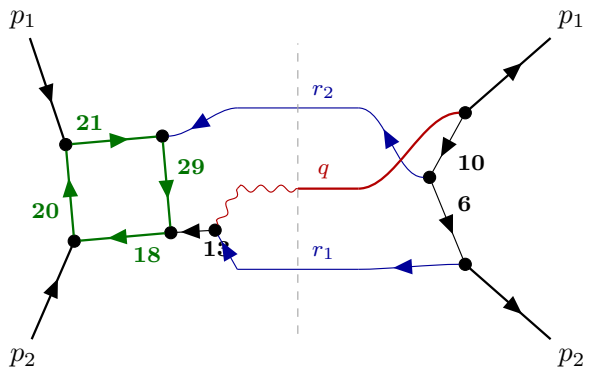
pinched: $12=(p_1+p_2+r_1)^2$, $20=(k_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo19 (RR family 15, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate

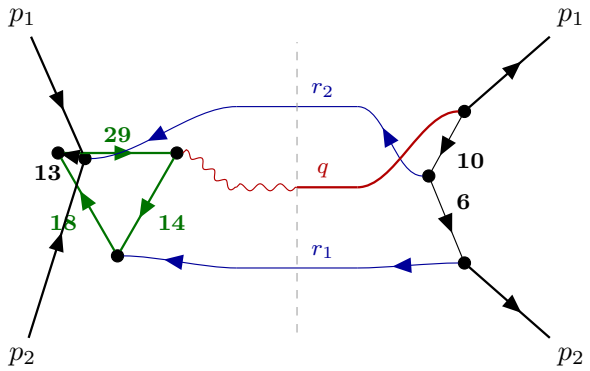
pinched: $5=(p_1+r_1)^2$, $20=(k_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 7 of 8 lines

18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

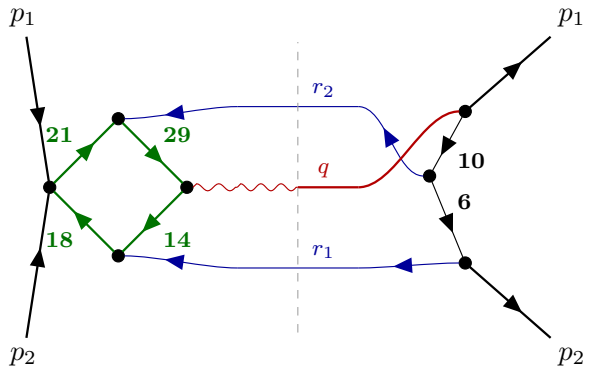
pinched: $14=(k_1)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

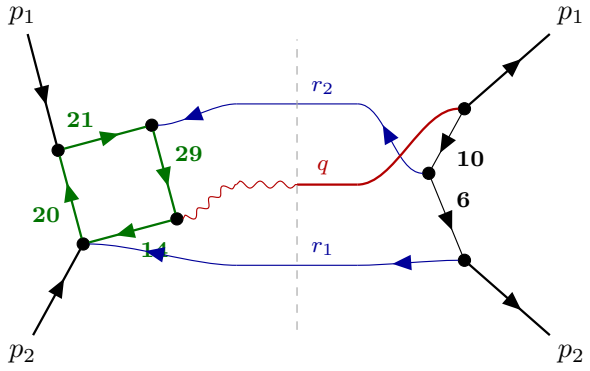
pinched: $20=(k_1+p_2+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

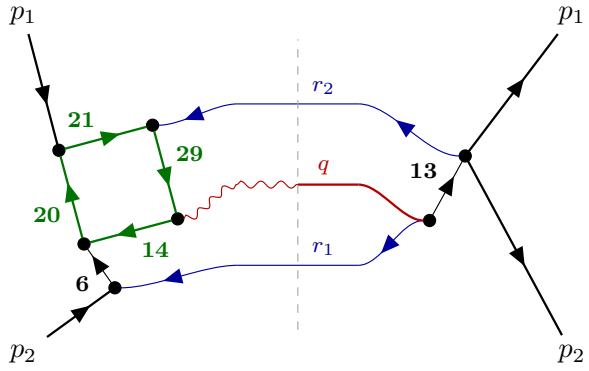
pinched: $13=(p_1+p_2+r_2)^2$, $20=(k_1+p_2+r_1)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

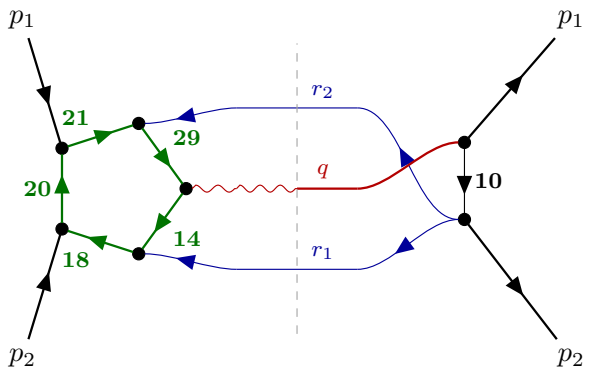
pinched: $13=(p_1+p_2+r_2)^2$, $18=(k_1+r_1)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

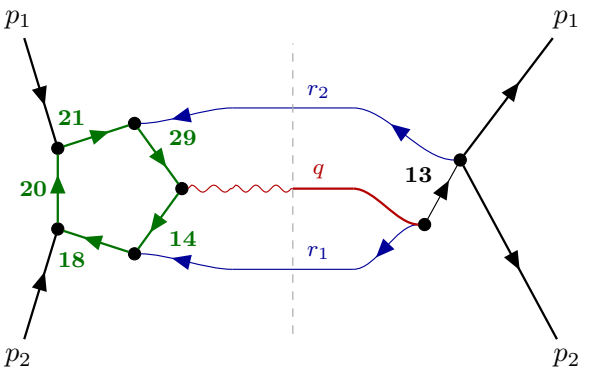
pinched: $10=(p_2+r_1+r_2)^2$, $18=(k_1+r_1)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	conjugate

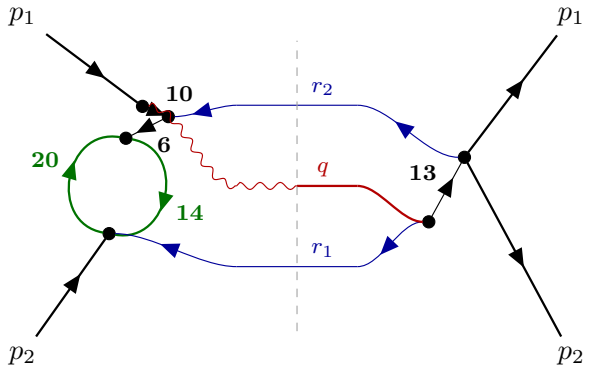
pinched: $6=(p_2+r_1)^2$, $13=(p_1+p_2+r_2)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	conjugate

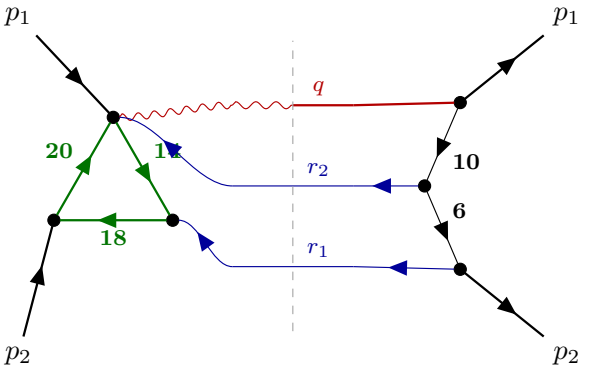
pinched: $6=(p_2+r_1)^2$, $10=(p_2+r_1+r_2)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

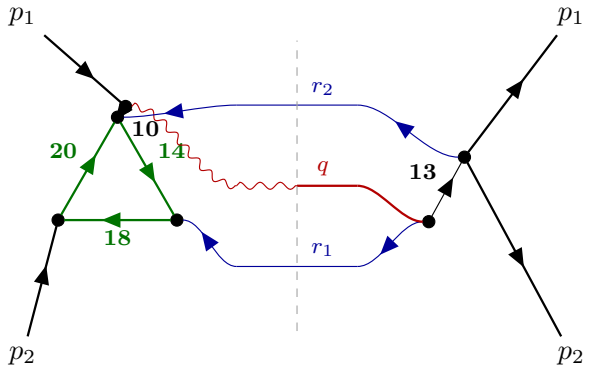
pinched: $18=(k_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

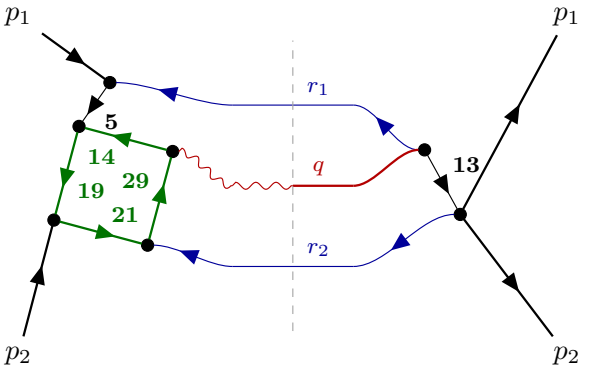
pinched: $13=(p_1+p_2+r_2)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo20 (RR family 27, pentagon q-r1-p2-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

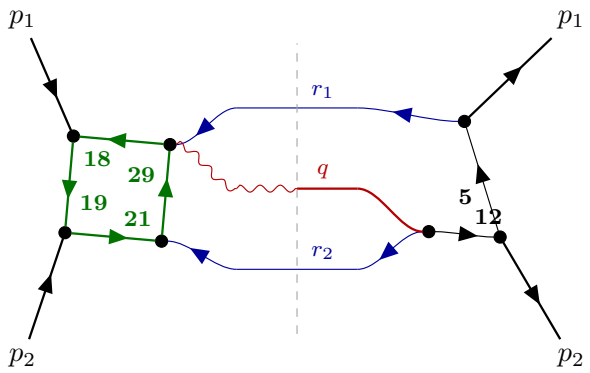
pinched: $6=(p_2+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

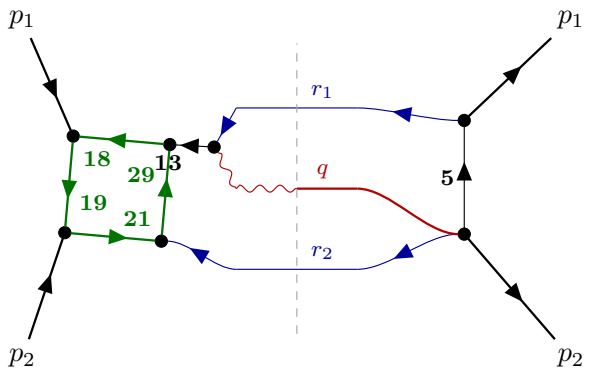
pinched: $12=(p_1+p_2+r_1)^2$, $18=(k_1+r_1)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate

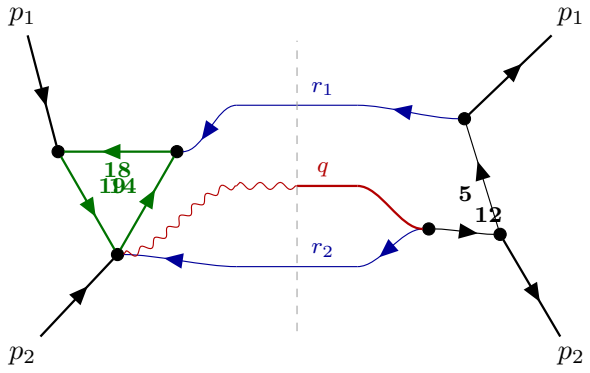
pinched: $13 = (p_1 + p_2 + r_2)^2$, $14 = (k_1)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
5	$(p_1 + r_1)^2$	conjugate

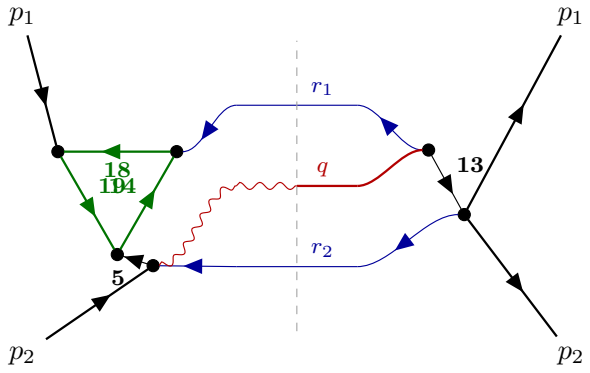
pinched: $12 = (p_1 + p_2 + r_1)^2$, $14 = (k_1)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate

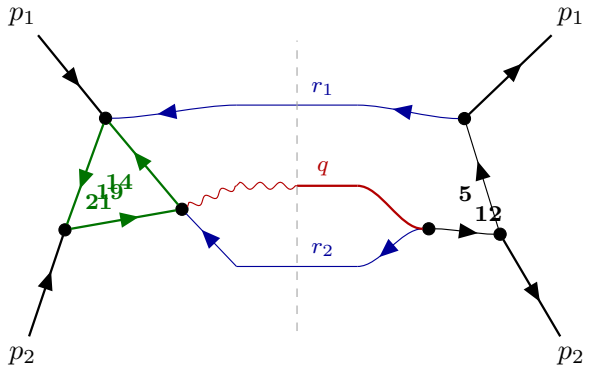
pinched: $13 = (p_1 + p_2 + r_2)^2$, $21 = (k_1 + p_1 + p_2 + r_1)^2$, $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
5	$(p_1 + r_1)^2$	amplitude (tree on the loop)
13	$(p_1 + p_2 + r_2)^2$	conjugate

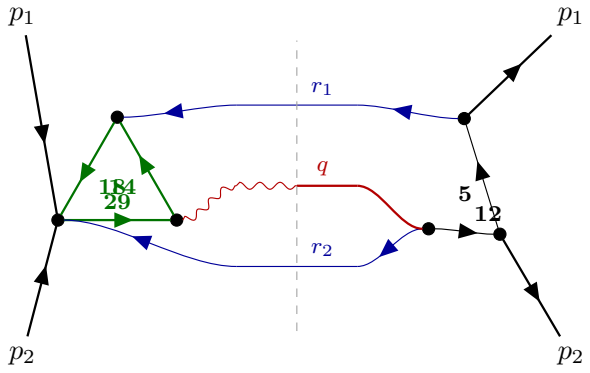
pinched: $12 = (p_1 + p_2 + r_1)^2$, $21 = (k_1 + p_1 + p_2 + r_1)^2$, $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate

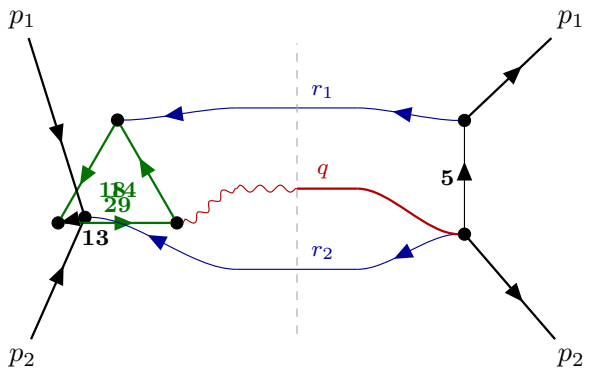
pinched: $13 = (p_1 + p_2 + r_2)^2$, $18 = (k_1 + r_1)^2$, $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate

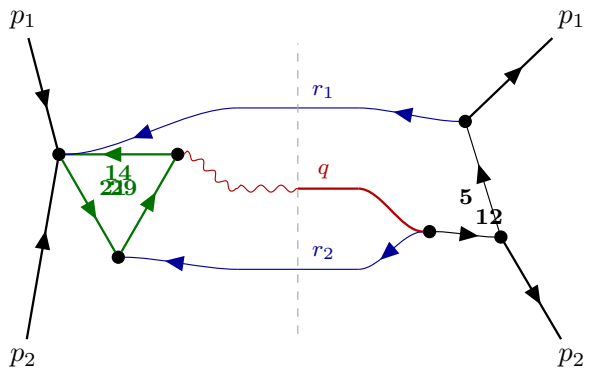
pinched: $13 = (p_1 + p_2 + r_2)^2$, $19 = (k_1 + p_1 + r_1)^2$, $21 = (k_1 + p_1 + p_2 + r_1)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

pinched: $12 = (p_1+p_2+r_1)^2$, $19 = (k_1+p_1+r_1)^2$, $21 = (k_1+p_1+p_2+r_1)^2$



topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

pinched: $13 = (p_1+p_2+r_2)^2$, $18 = (k_1+r_1)^2$, $19 = (k_1+p_1+r_1)^2$

topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

14	$(k_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

pinched: $18 = (k_1+r_1)^2$, $19 = (k_1+p_1+r_1)^2$, $21 = (k_1+p_1+p_2+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$

topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

18	$(k_1+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

pinched: $14 = (k_1)^2$, $19 = (k_1+p_1+r_1)^2$, $21 = (k_1+p_1+p_2+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$

topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

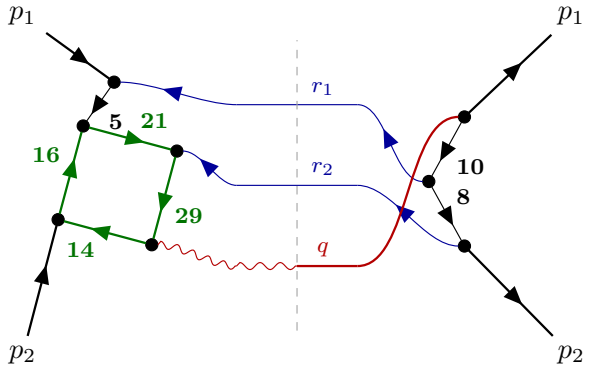
pinched: $14 = (k_1)^2$, $18 = (k_1+r_1)^2$, $19 = (k_1+p_1+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$

topo21 (RR family 23, pentagon q-r1-p1-p2-r2), sector with 4 of 8 lines

fewer than two loop lines: scaleless, zero

29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

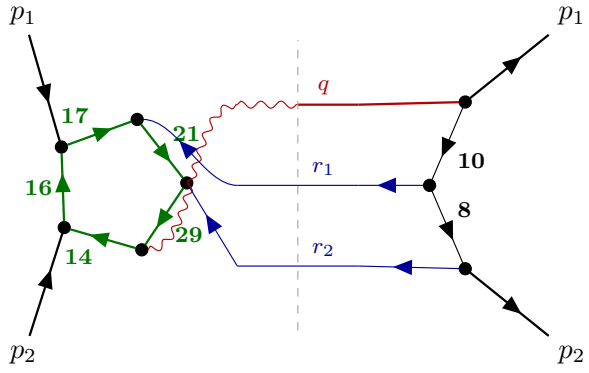
pinched: $14 = (k_1)^2$, $18 = (k_1+r_1)^2$, $19 = (k_1+p_1+r_1)^2$, $21 = (k_1+p_1+p_2+r_1)^2$



topo22 (RR family 7, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

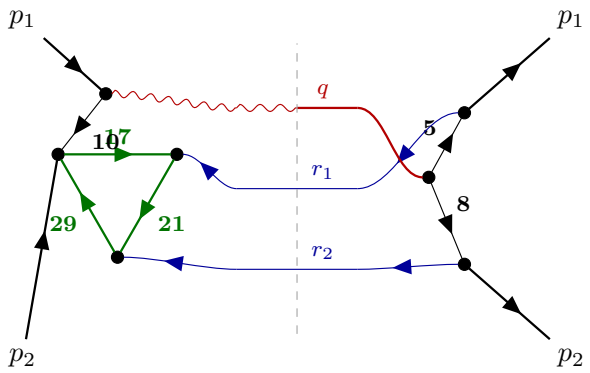
pinched: $17 = (k_1+p_1+p_2)^2$



topo22 (RR family 7, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

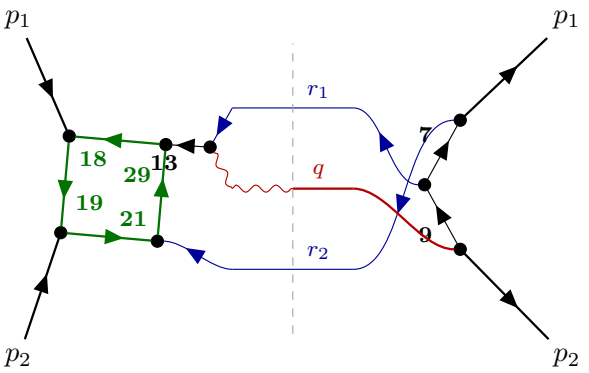
pinched: $5 = (p_1+r_1)^2$



topo22 (RR family 7, pentagon q-p2-p1-r1-r2), sector with 6 of 8 lines

17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

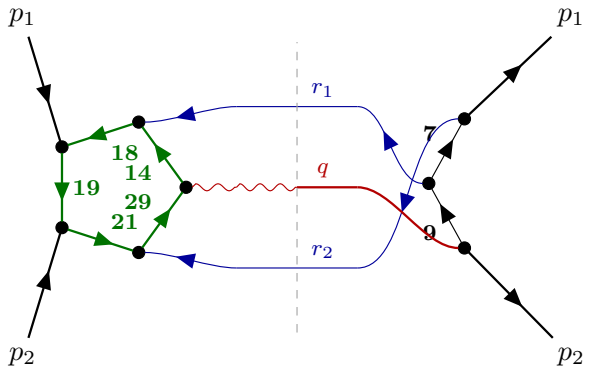
pinched: $14=(k_1)^2$, $16=(k_1+p_2)^2$



topo23 (RR family 33, pentagon q-r1-p1-p2-r2), sector with 7 of 8 lines

18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

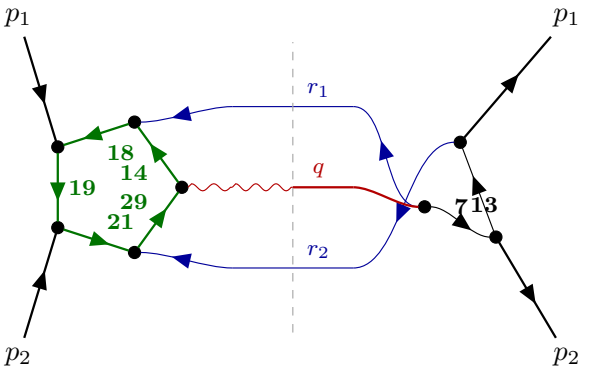
pinched: $14=(k_1)^2$



topo23 (RR family 33, pentagon q-r1-p1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

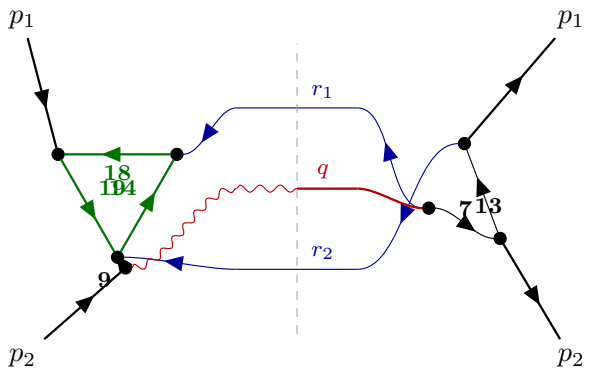
pinched: $13=(p_1+p_2+r_2)^2$



topo23 (RR family 33, pentagon q-r1-p1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

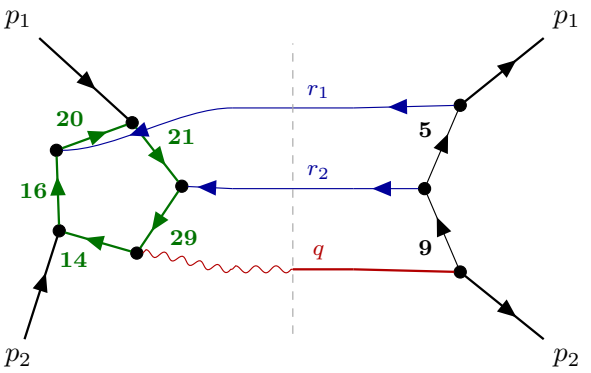
pinched: $9=(p_1+r_1+r_2)^2$



topo23 (RR family 33, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

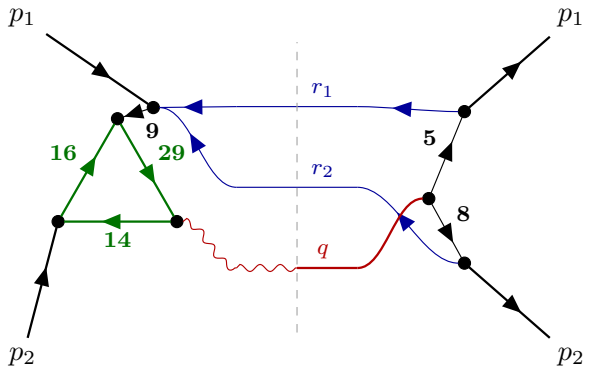
pinched: $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

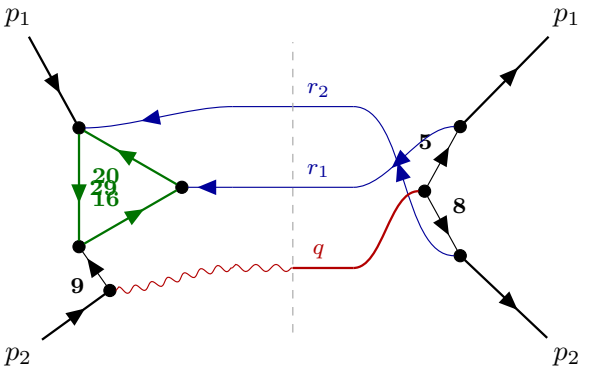
pinched: $8=(p_2+r_2)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

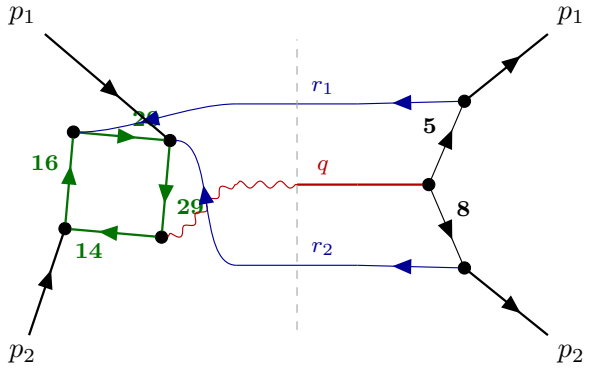
pinched: $20=(k_1+p_2+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

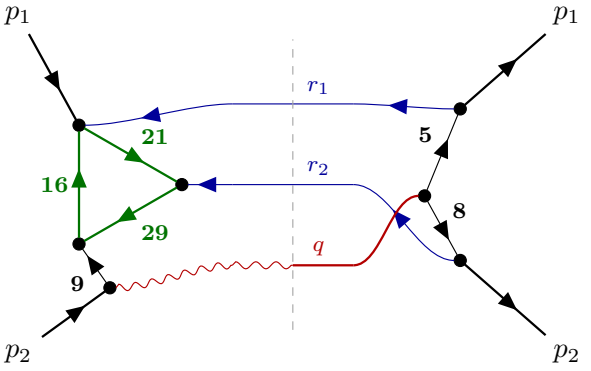
pinched: $14=(k_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

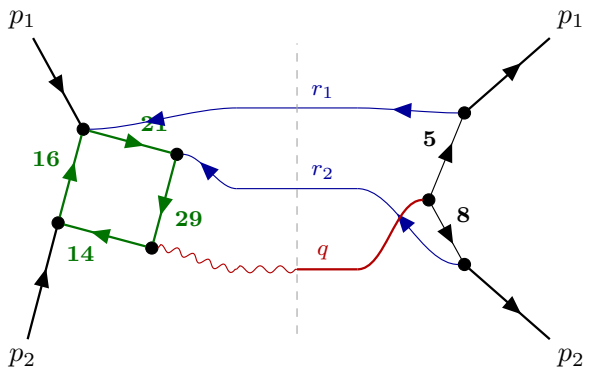
pinched: $9=(p_1+r_1+r_2)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

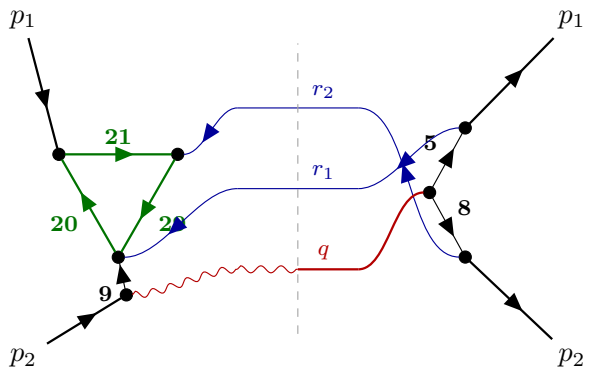
pinched: $14=(k_1)^2$, $20=(k_1+p_2+r_1)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

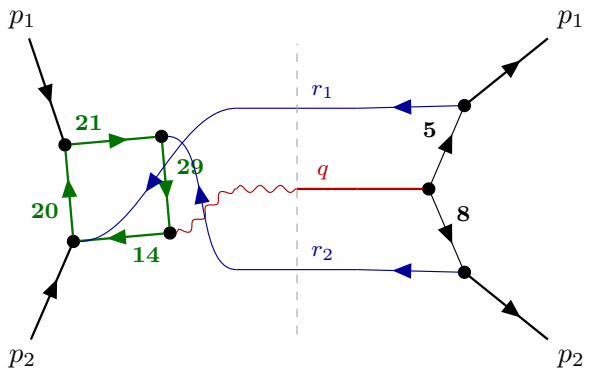
pinched: $9=(p_1+r_1+r_2)^2$, $20=(k_1+p_2+r_1)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

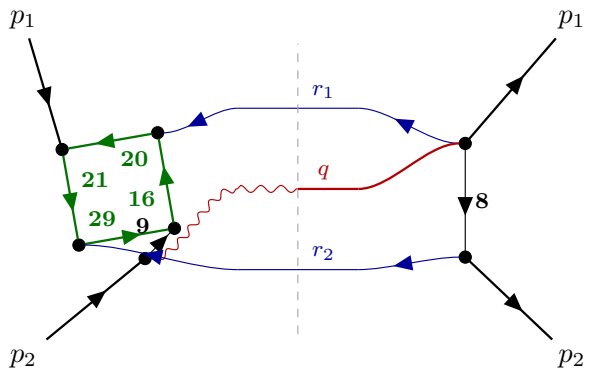
pinched: $14=(k_1)^2$, $16=(k_1+p_2)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

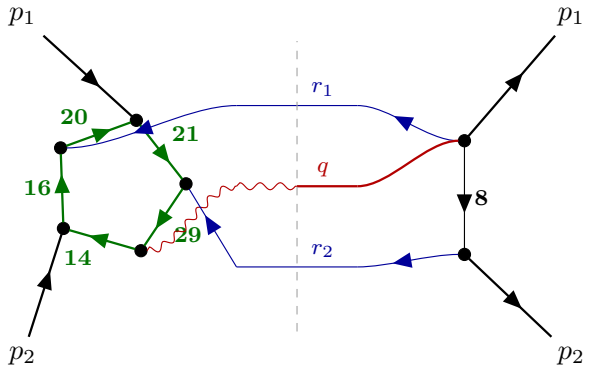
pinched: $9=(p_1+r_1+r_2)^2$, $16=(k_1+p_2)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

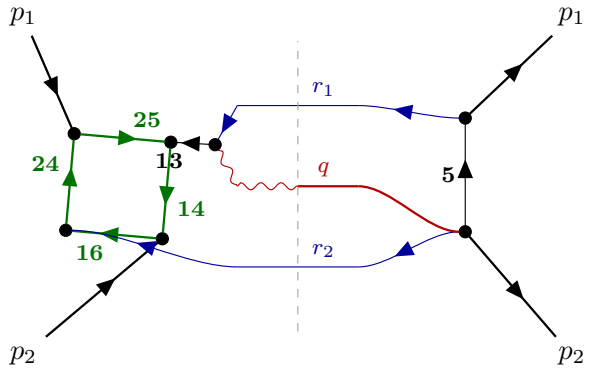
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$



topo24 (RR family 6, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate

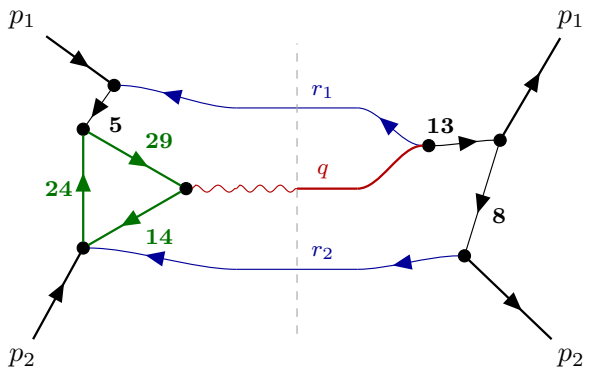
pinched: $5=(p_1+r_1)^2$, $9=(p_1+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

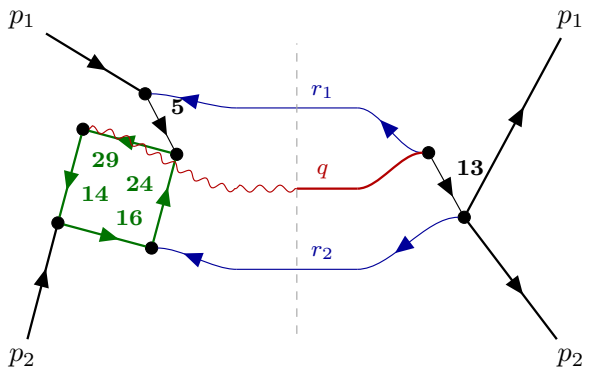
pinched: $8=(p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

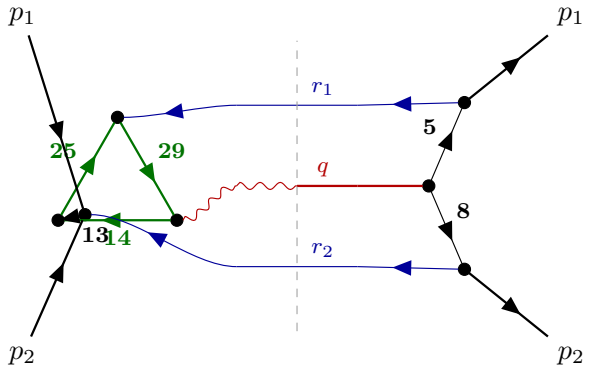
pinched: $16=(k_1+p_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

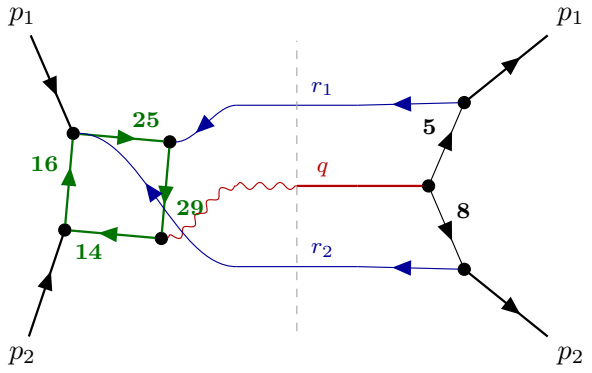
pinched: $8=(p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

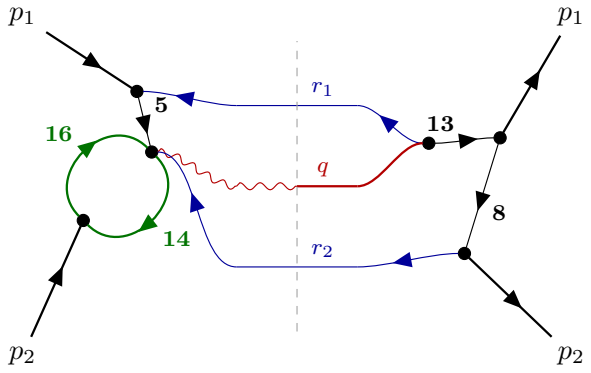
pinched: $16=(k_1+p_2)^2$, $24=(k_1+p_2+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

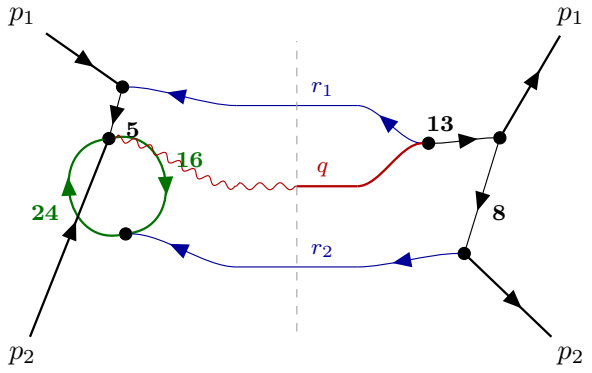
pinched: $13=(p_1+p_2+r_2)^2$, $24=(k_1+p_2+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

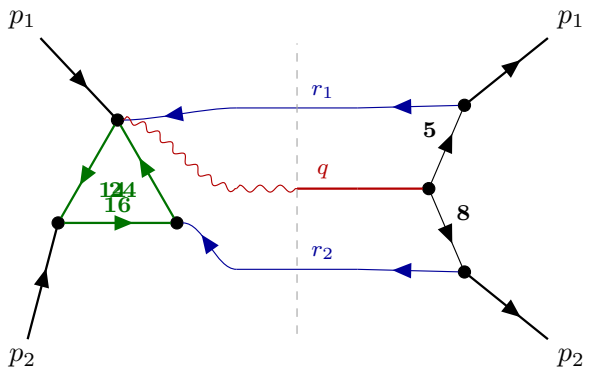
pinched: $24=(k_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

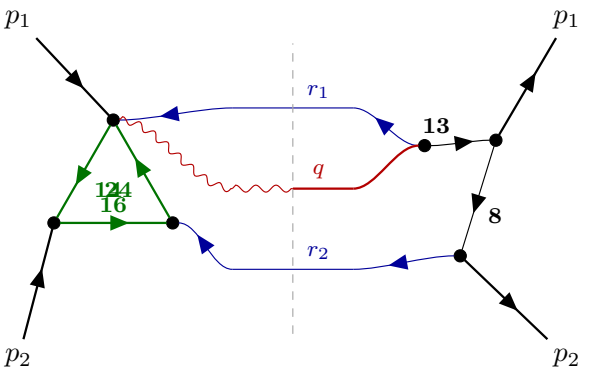
pinched: $14=(k_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

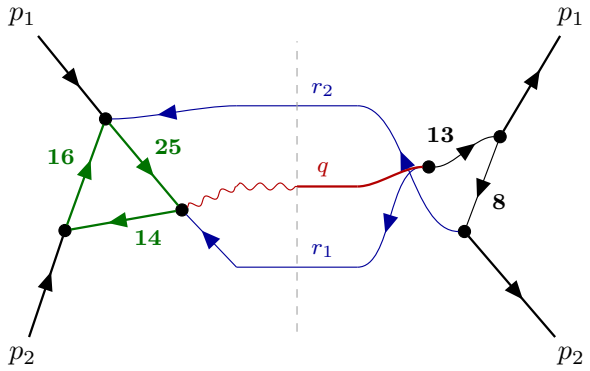
pinched: $13=(p_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

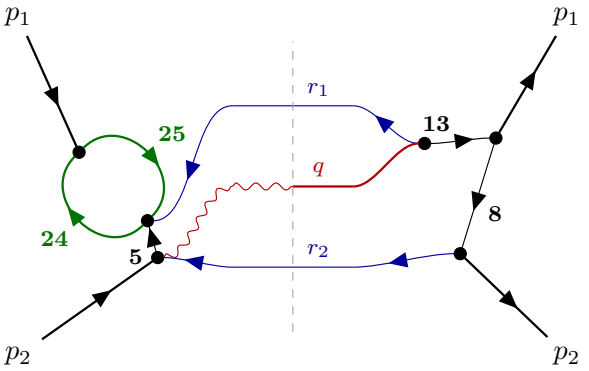
pinched: $5=(p_1+r_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

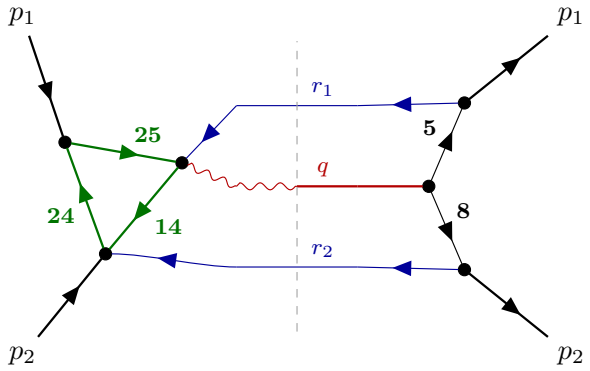
pinched: $5=(p_1+r_1)^2$, $24=(k_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

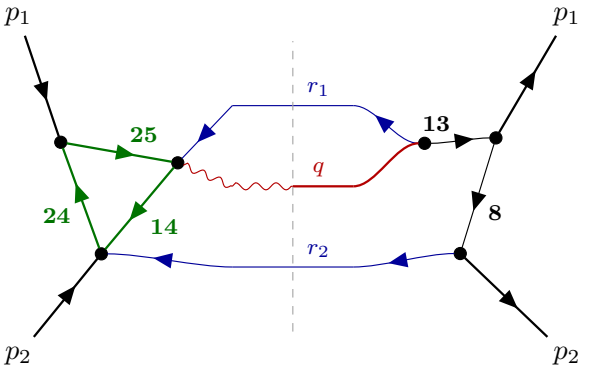
pinched: $14=(k_1)^2$, $16=(k_1+p_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

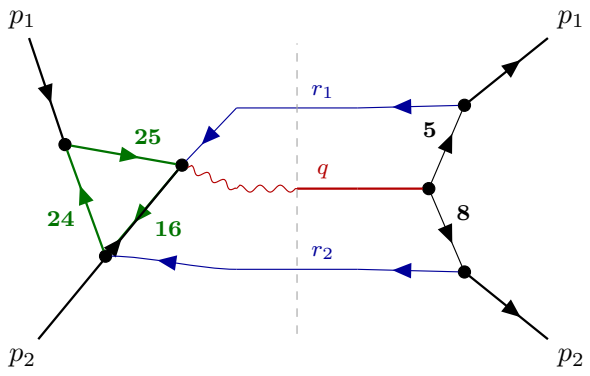
pinched: $13=(p_1+p_2+r_2)^2$, $16=(k_1+p_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

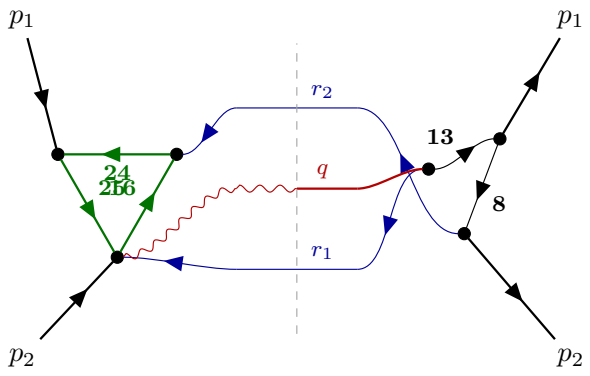
pinched: $5=(p_1+r_1)^2$, $16=(k_1+p_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

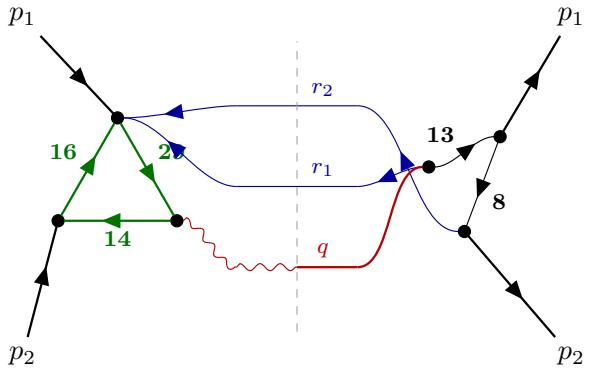
pinched: $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

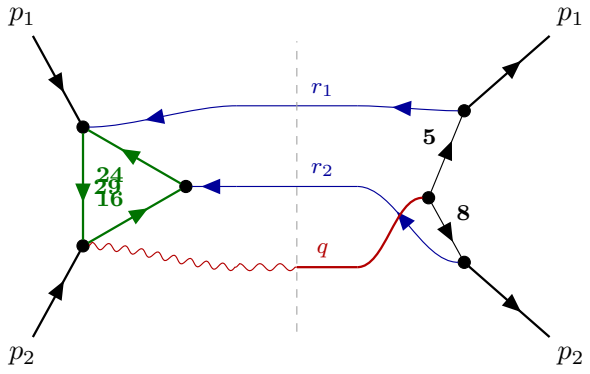
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

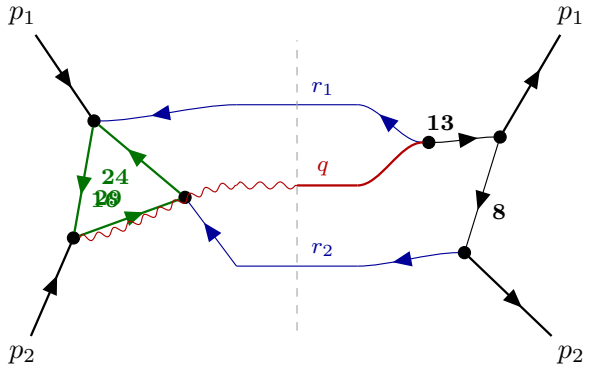
pinched: $5=(p_1+r_1)^2$, $24=(k_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

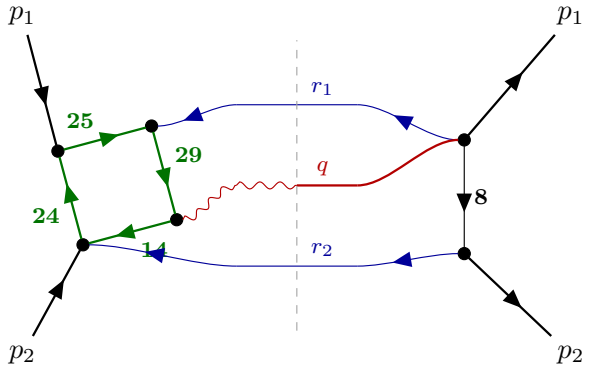
pinched: $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

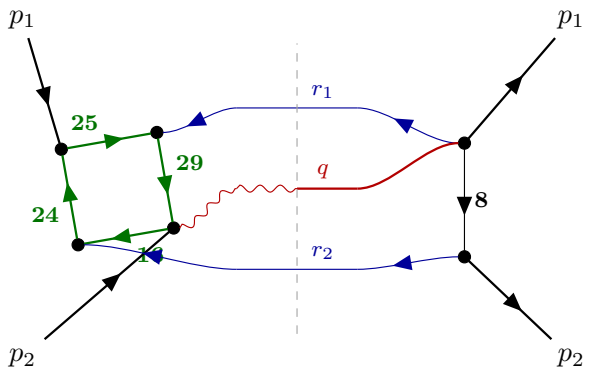
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate

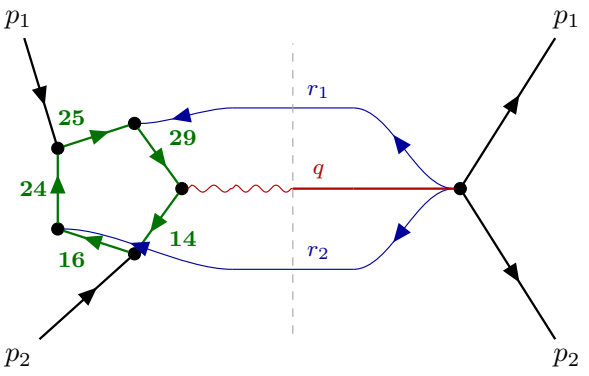
pinched: $5=(p_1+r_1)^2$, $13=(p_1+p_2+r_2)^2$, $16=(k_1+p_2)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate

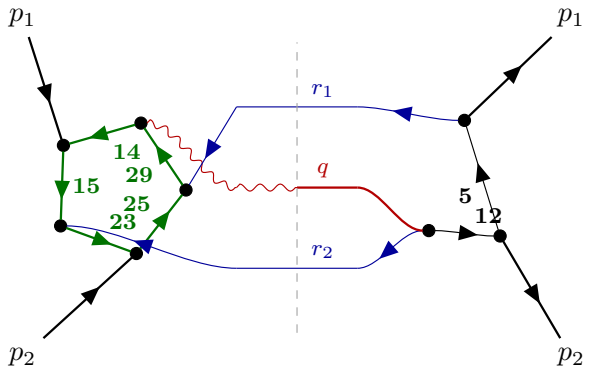
pinched: $5=(p_1+r_1)^2$, $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$



topo25 (RR family 18, pentagon q-p2-r2-p1-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop

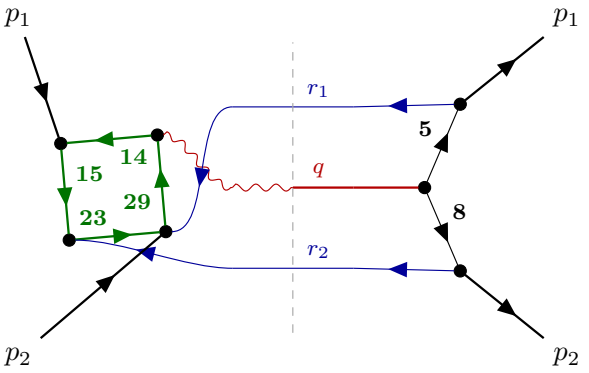
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$, $13=(p_1+p_2+r_2)^2$



topo26 (RR family 17, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

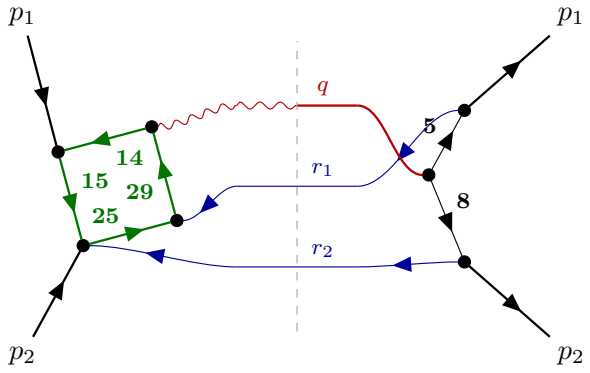
pinched: $8=(p_2+r_2)^2$



topo26 (RR family 17, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

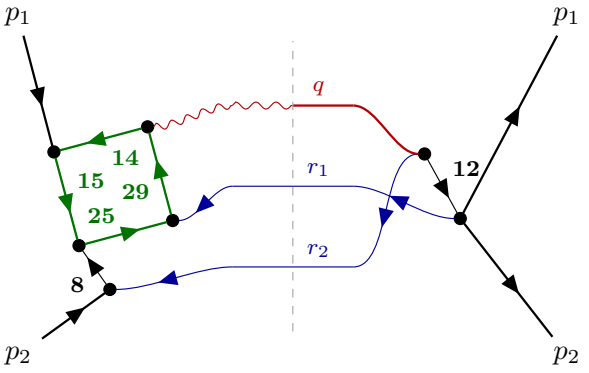
pinched: $12=(p_1+p_2+r_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo26 (RR family 17, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

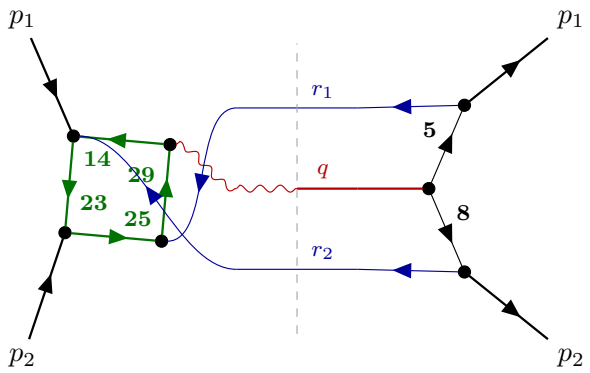
pinched: $12=(p_1+p_2+r_1)^2$, $23=(k_1+p_1+r_2)^2$



topo26 (RR family 17, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	conjugate

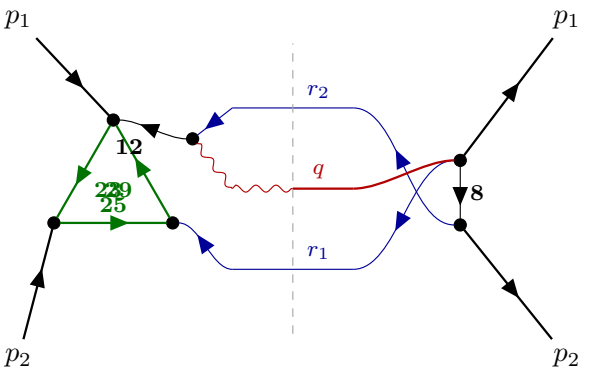
pinched: $5=(p_1+r_1)^2$, $23=(k_1+p_1+r_2)^2$



topo26 (RR family 17, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

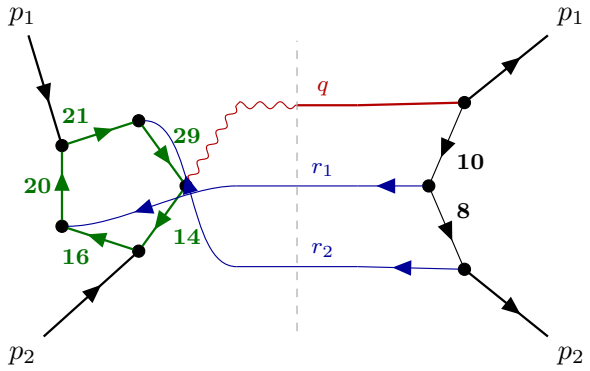
pinched: $12=(p_1+p_2+r_1)^2$, $15=(k_1+p_1)^2$



topo26 (RR family 17, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

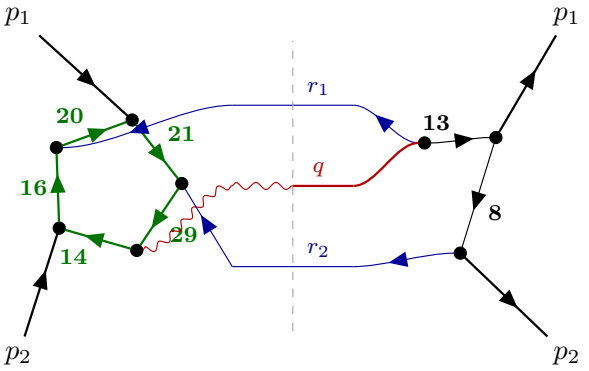
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo27 (RR family 29, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

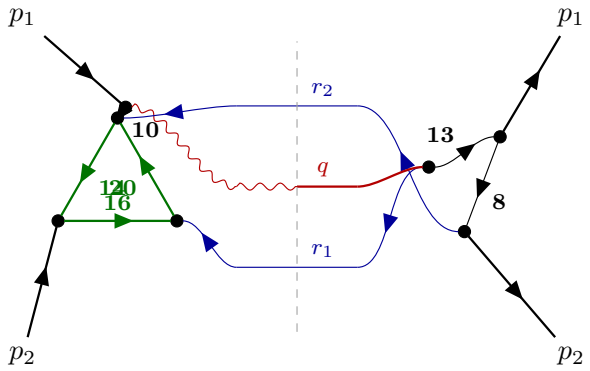
pinched: $13=(p_1+p_2+r_2)^2$



topo27 (RR family 29, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

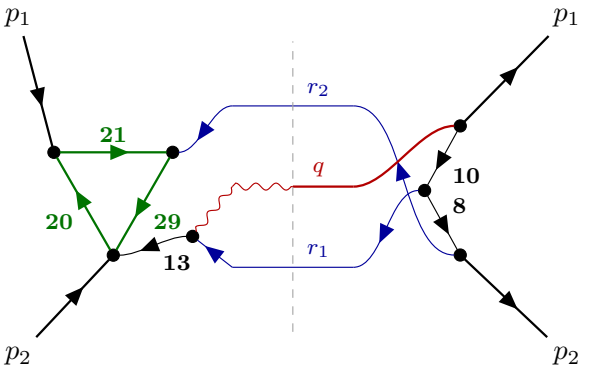
pinched: $10=(p_2+r_1+r_2)^2$



topo27 (RR family 29, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

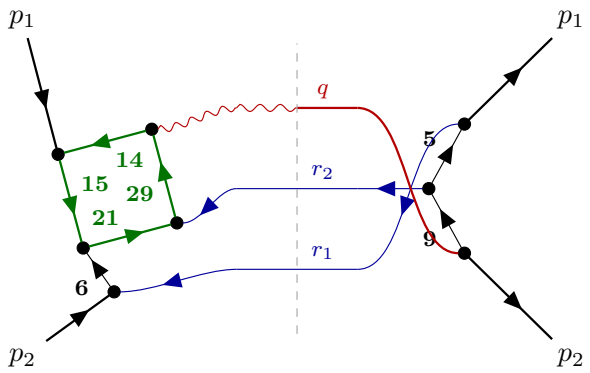
pinched: $21=(k_1+p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo27 (RR family 29, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

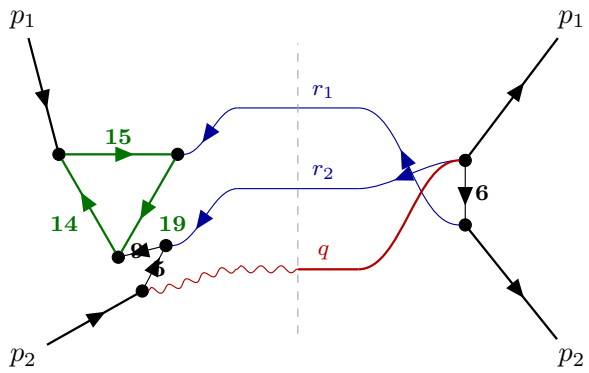
pinched: $14=(k_1)^2$, $16=(k_1+p_2)^2$



topo28 (RR family 3, pentagon q-p1-r1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

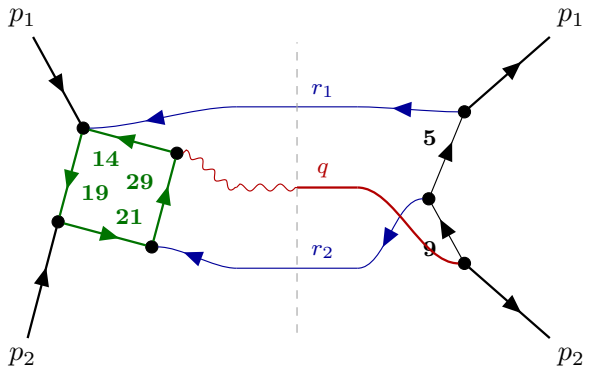
pinched: $19 = (k_1+p_1+r_1)^2$



topo28 (RR family 3, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

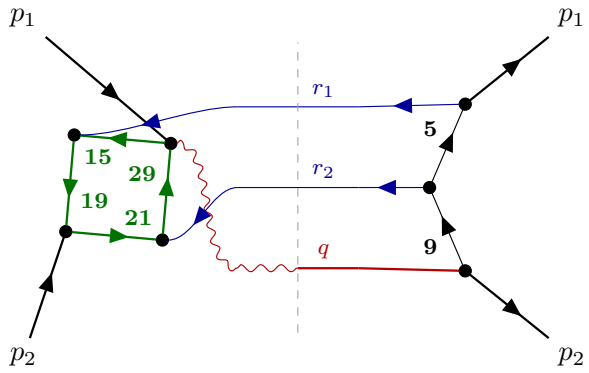
pinched: $21 = (k_1+p_1+p_2+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo28 (RR family 3, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

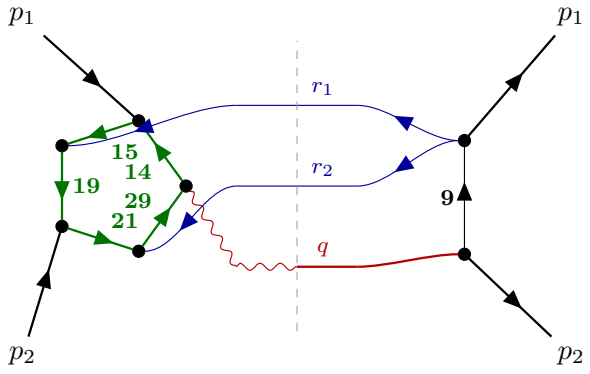
pinched: $6 = (p_2+r_1)^2$, $15 = (k_1+p_1)^2$



topo28 (RR family 3, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

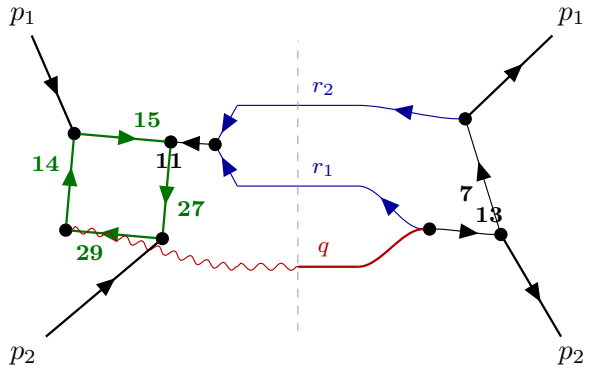
pinched: $6 = (p_2+r_1)^2$, $14 = (k_1)^2$



topo28 (RR family 3, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	conjugate

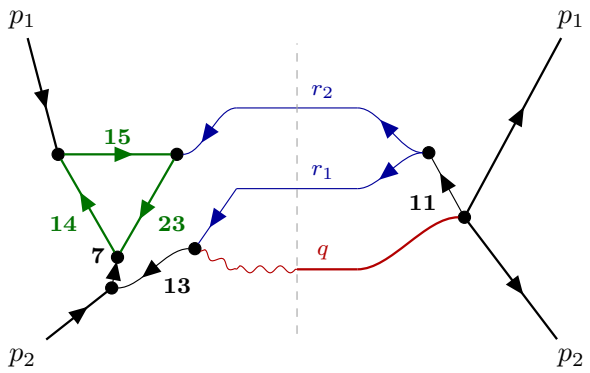
pinched: $5 = (p_1+r_1)^2$, $6 = (p_2+r_1)^2$



topo29 (RR family 30, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

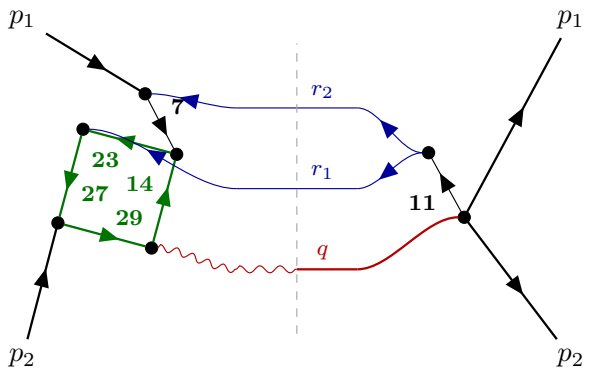
pinched: $23 = (k_1+p_1+r_2)^2$



topo29 (RR family 30, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

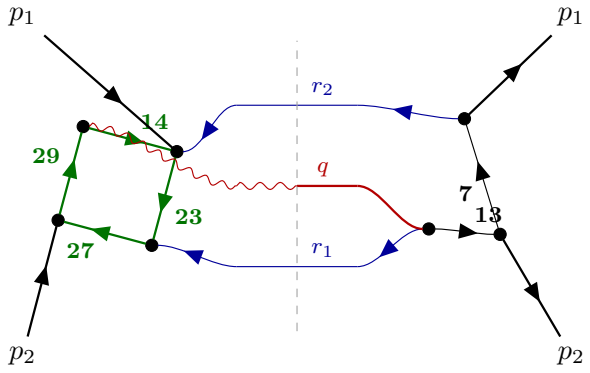
pinched: $27=(k_1+p_1+r_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo29 (RR family 30, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

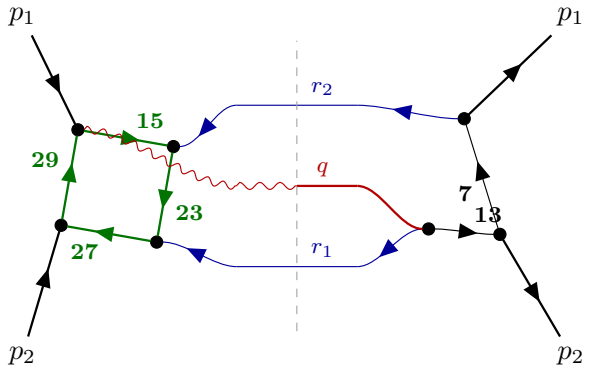
pinched: $13=(p_1+p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo29 (RR family 30, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

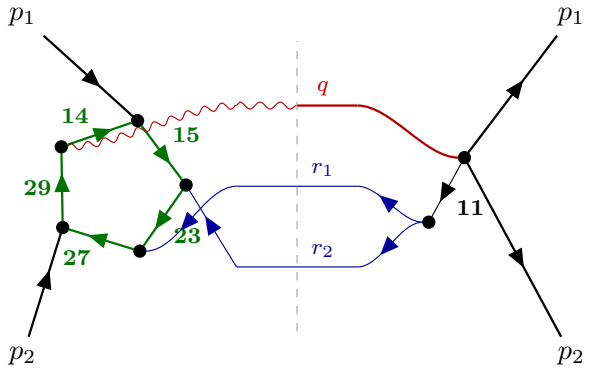
pinched: $11=(r_1+r_2)^2$, $15=(k_1+p_1)^2$



topo29 (RR family 30, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

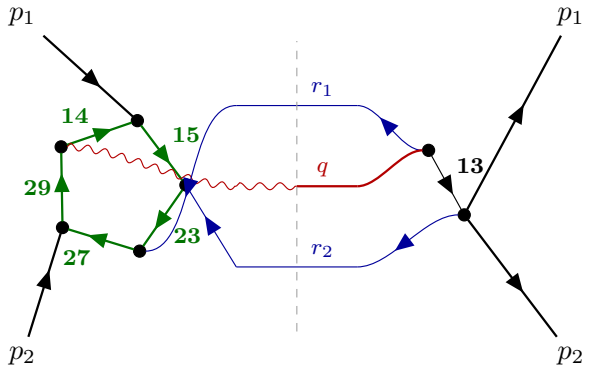
pinched: $11=(r_1+r_2)^2$, $14=(k_1)^2$



topo29 (RR family 30, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

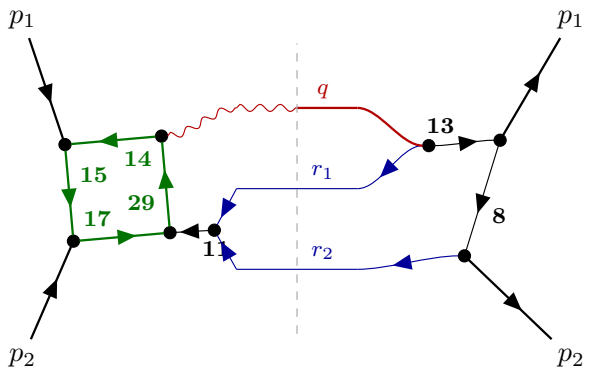
pinched: $7=(p_1+r_2)^2$, $13=(p_1+p_2+r_2)^2$



topo29 (RR family 30, pentagon q-p1-r2-r1-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	conjugate

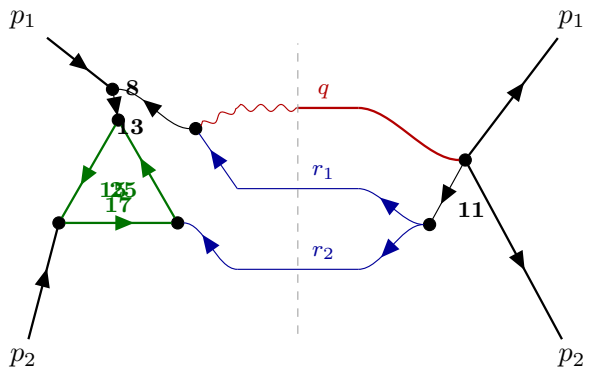
pinched: $7=(p_1+r_2)^2$, $11=(r_1+r_2)^2$



topo30 (RR family 31, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

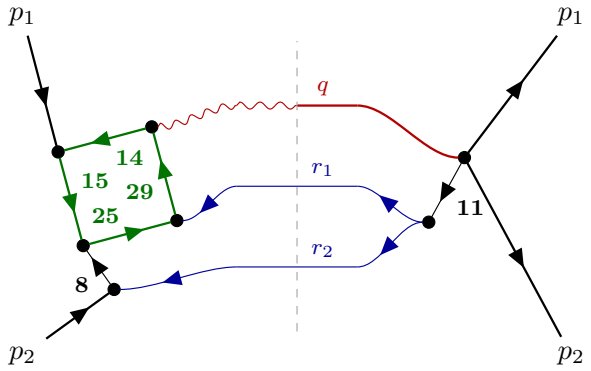
pinched: $25 = (k_1+p_1+p_2+r_2)^2$



topo30 (RR family 31, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

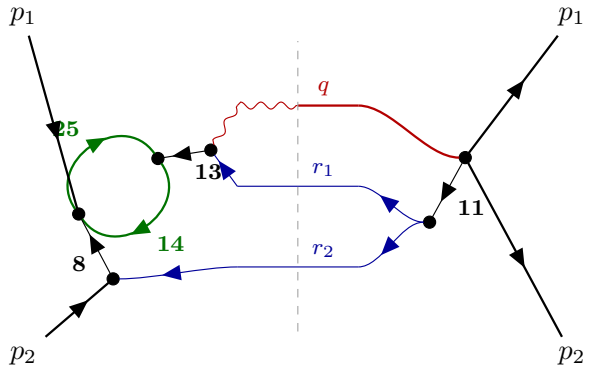
pinched: $14 = (k_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo30 (RR family 31, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

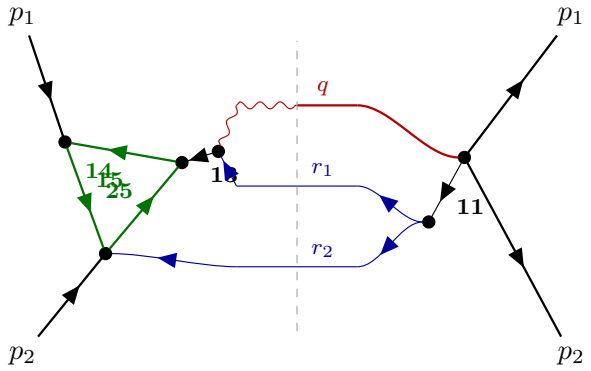
pinched: $13 = (p_1+p_2+r_2)^2$, $17 = (k_1+p_1+p_2)^2$



topo30 (RR family 31, pentagon q-p1-p2-r2-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

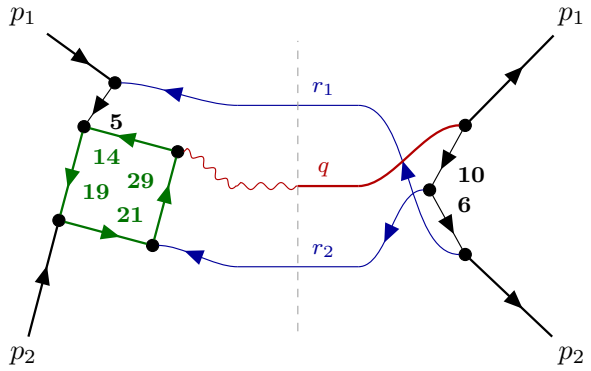
pinched: $15 = (k_1+p_1)^2$, $17 = (k_1+p_1+p_2)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo30 (RR family 31, pentagon q-p1-p2-r2-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

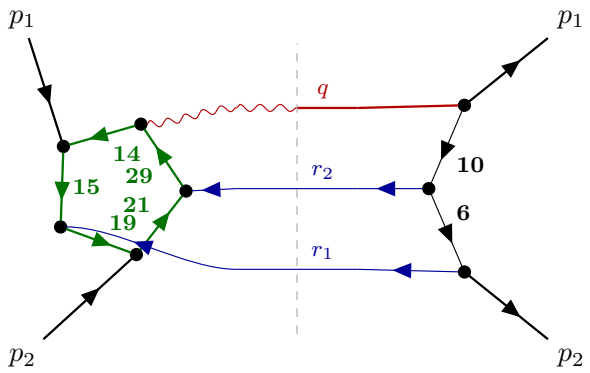
pinched: $8 = (p_2+r_2)^2$, $17 = (k_1+p_1+p_2)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo31 (RR family 4, pentagon q-p1-r1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

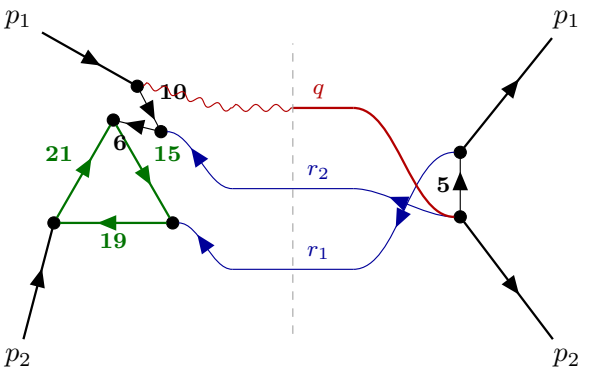
pinched: $15 = (k_1+p_1)^2$



topo31 (RR family 4, pentagon q-p1-r1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

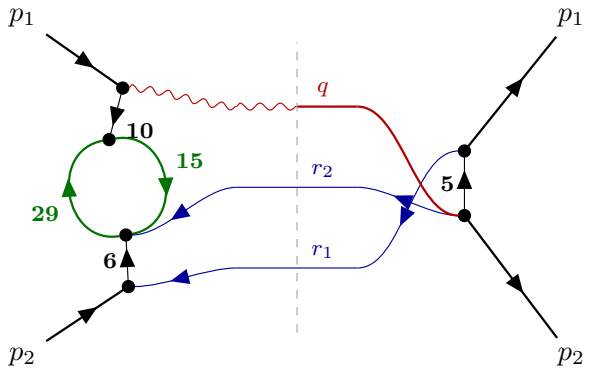
pinched: $5=(p_1+r_1)^2$



topo31 (RR family 4, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

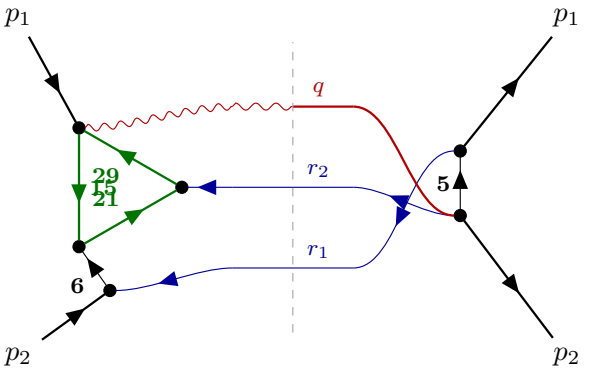
pinched: $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo31 (RR family 4, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

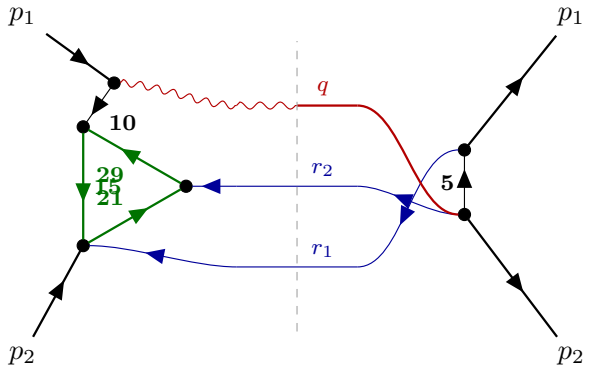
pinched: $14=(k_1)^2$, $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo31 (RR family 4, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

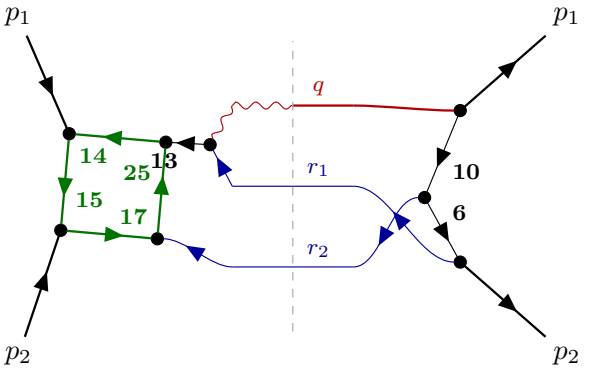
pinched: $10=(p_2+r_1+r_2)^2$, $14=(k_1)^2$, $19=(k_1+p_1+r_1)^2$



topo31 (RR family 4, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

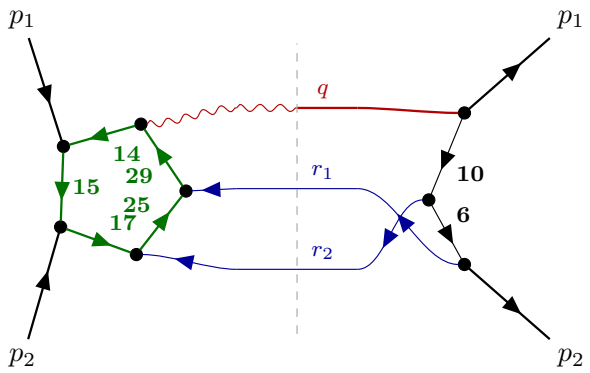
pinched: $6=(p_2+r_1)^2$, $14=(k_1)^2$, $19=(k_1+p_1+r_1)^2$



topo32 (RR family 27, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

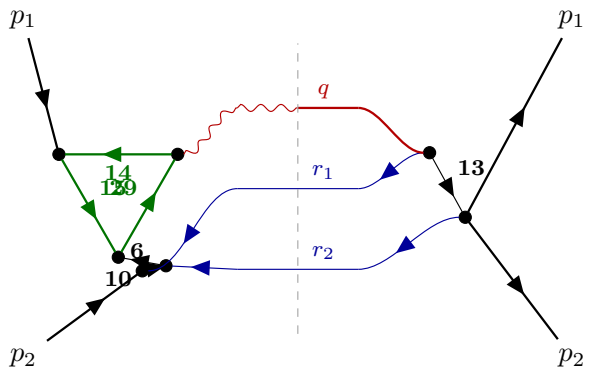
pinched: $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo32 (RR family 27, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

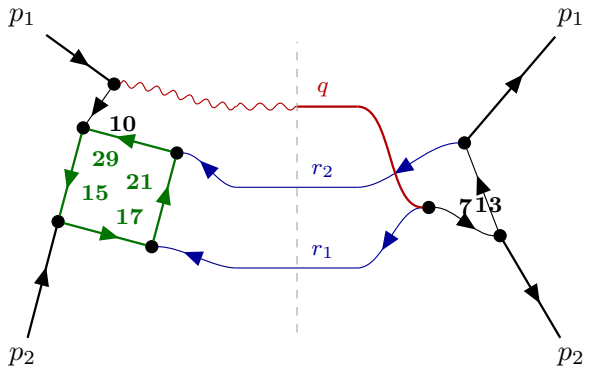
pinched: $13=(p_1+p_2+r_2)^2$



topo32 (RR family 27, pentagon q-p1-p2-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

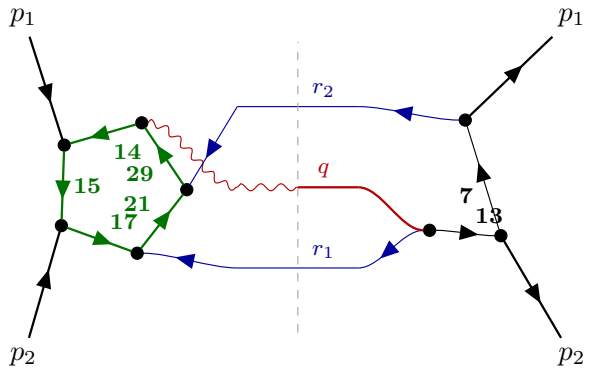
pinched: $17=(k_1+p_1+p_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo33 (RR family 28, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

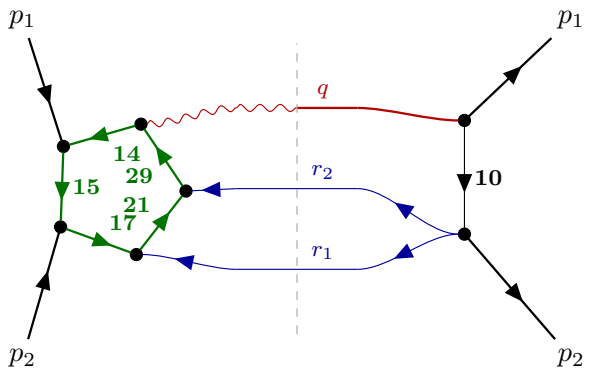
pinched: $14=(k_1)^2$



topo33 (RR family 28, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

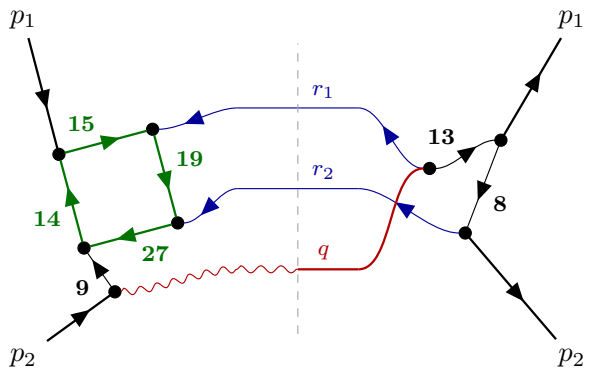
pinched: $10=(p_2+r_1+r_2)^2$



topo33 (RR family 28, pentagon q-p1-p2-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	conjugate

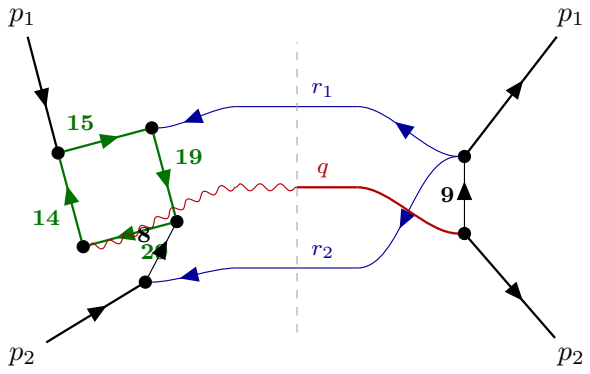
pinched: $7=(p_1+r_2)^2$, $13=(p_1+p_2+r_2)^2$



topo34 (RR family 34, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

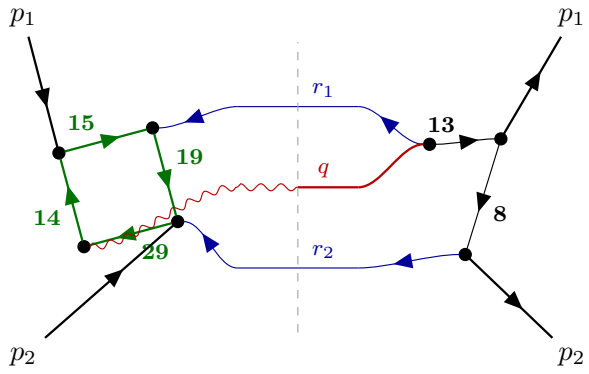
pinched: $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo34 (RR family 34, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	conjugate

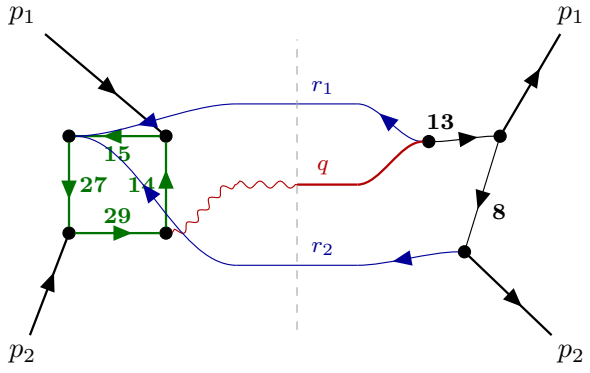
pinched: $13=(p_1+p_2+r_2)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo34 (RR family 34, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

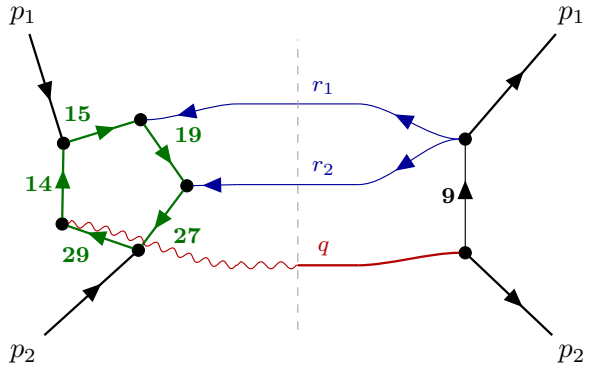
pinched: $9=(p_1+r_1+r_2)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo34 (RR family 34, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

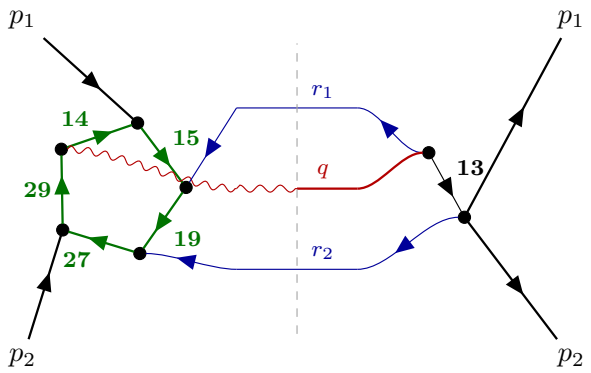
pinched: $9=(p_1+r_1+r_2)^2$, $19=(k_1+p_1+r_1)^2$



topo34 (RR family 34, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	conjugate

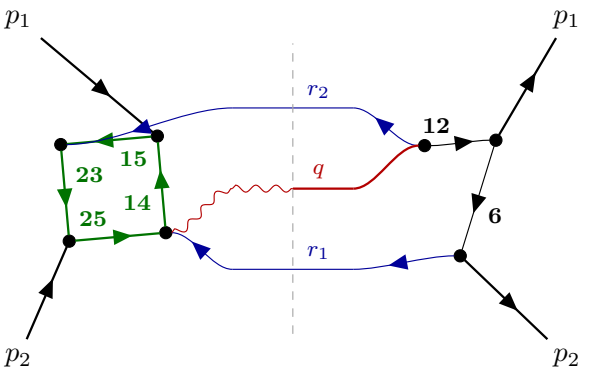
pinched: $8=(p_2+r_2)^2$, $13=(p_1+p_2+r_2)^2$



topo34 (RR family 34, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	conjugate

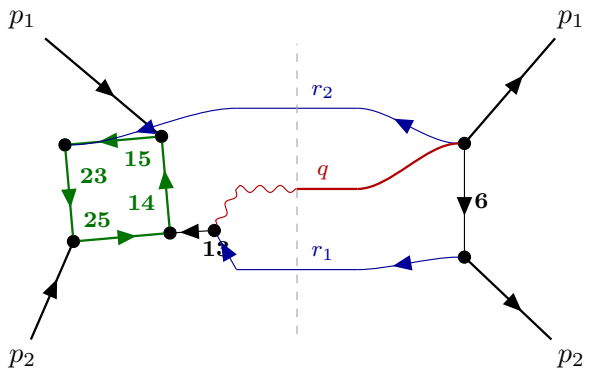
pinched: $8=(p_2+r_2)^2$, $9=(p_1+r_1+r_2)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

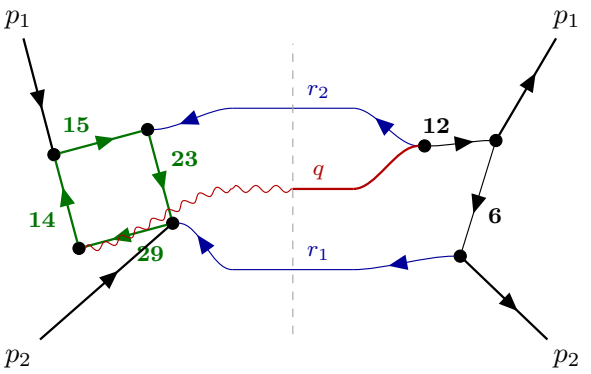
pinched: $13=(p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

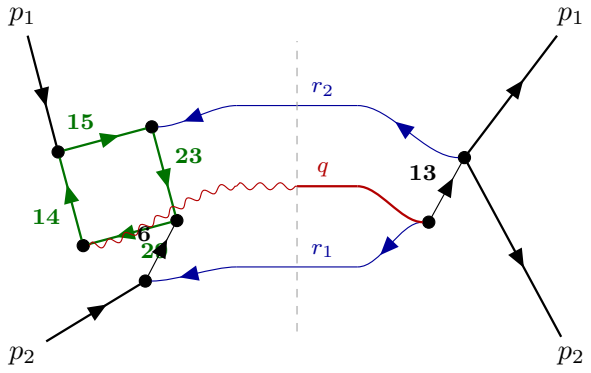
pinched: $12=(p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

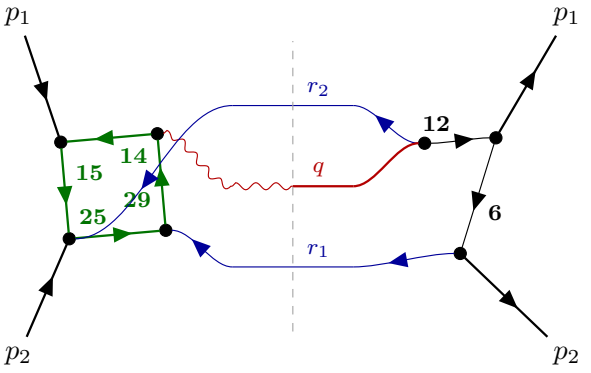
pinched: $13=(p_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

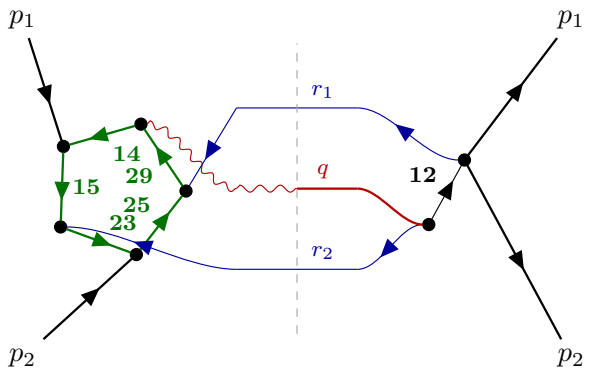
pinched: $12=(p_1+p_2+r_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

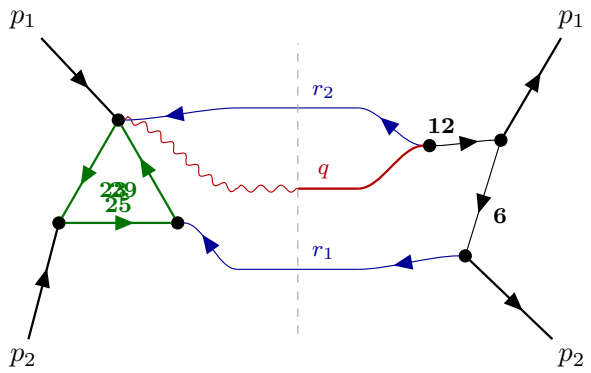
pinched: $13=(p_1+p_2+r_2)^2$, $23=(k_1+p_1+r_2)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
12	$(p_1+p_2+r_1)^2$	conjugate

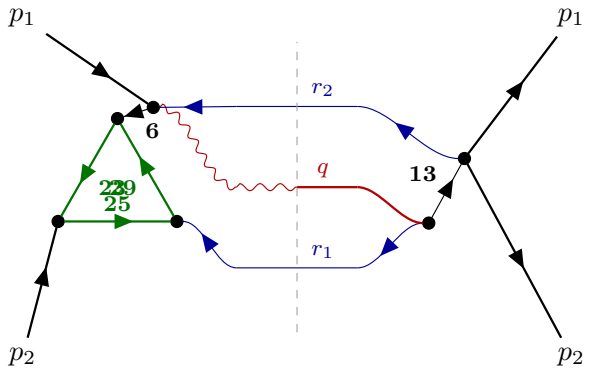
pinched: $6=(p_2+r_1)^2$, $13=(p_1+p_2+r_2)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

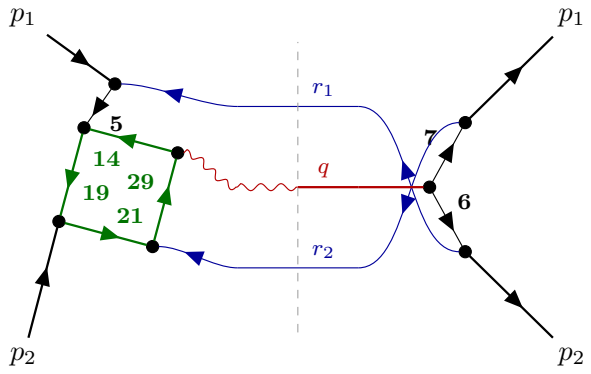
pinched: $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo35 (RR family 24, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

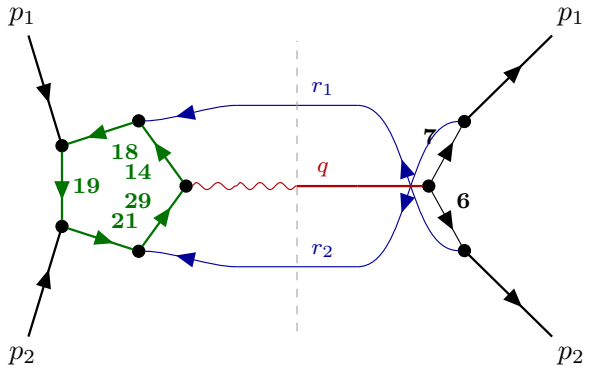
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$, $15=(k_1+p_1)^2$



topo36 (RR family 1, pentagon q-r1-p1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

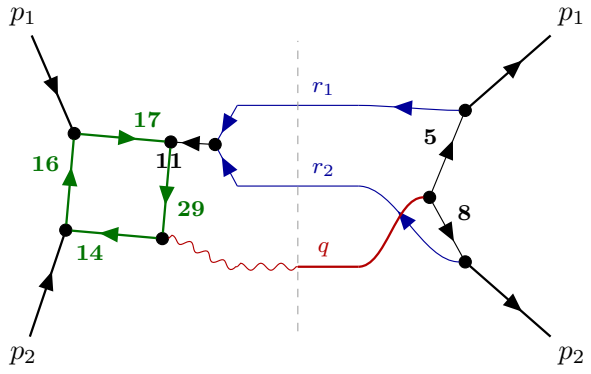
pinched: $18=(k_1+r_1)^2$



topo36 (RR family 1, pentagon q-r1-p1-p2-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

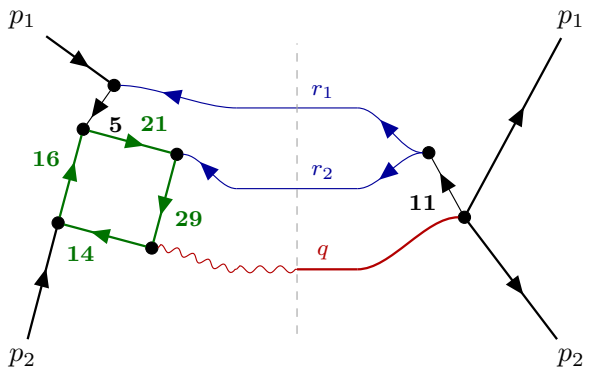
pinched: $5=(p_1+r_1)^2$



topo37 (RR family 8, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

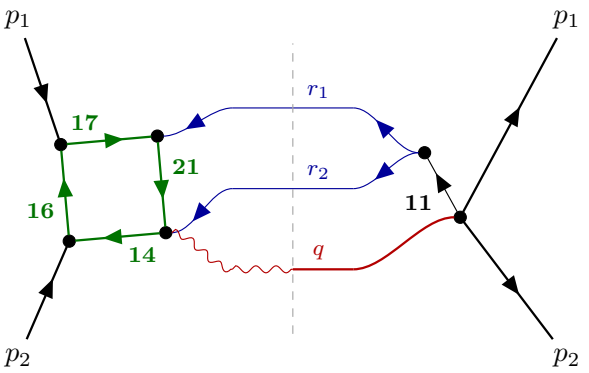
pinched: $21=(k_1+p_1+p_2+r_1)^2$



topo37 (RR family 8, pentagon q-p2-p1-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

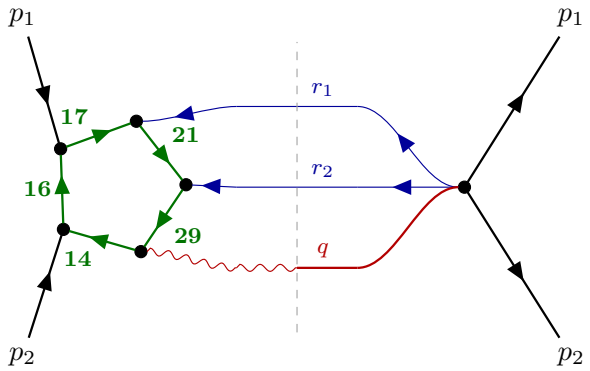
pinched: $8=(p_2+r_2)^2$, $17=(k_1+p_1+p_2)^2$



topo37 (RR family 8, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
11	$(r_1+r_2)^2$	conjugate

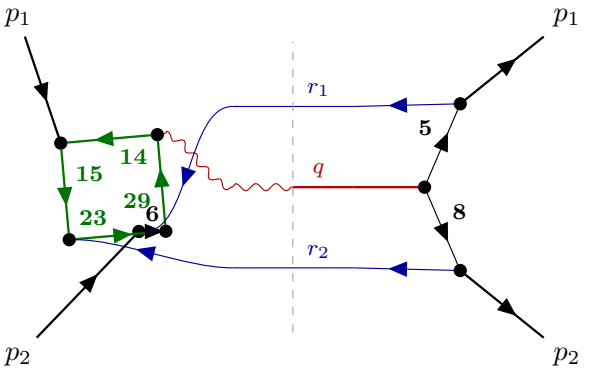
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo37 (RR family 8, pentagon q-p2-p1-r1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop

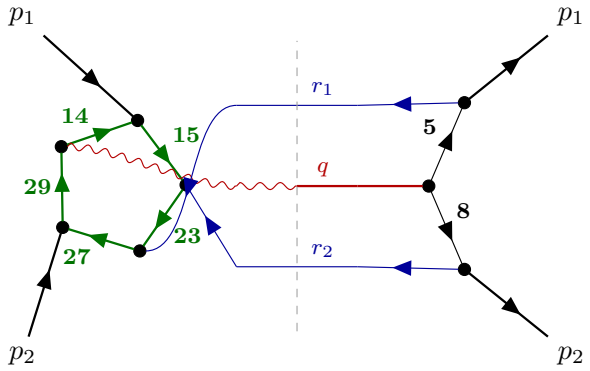
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$, $11=(r_1+r_2)^2$



topo38 (RR family 2, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

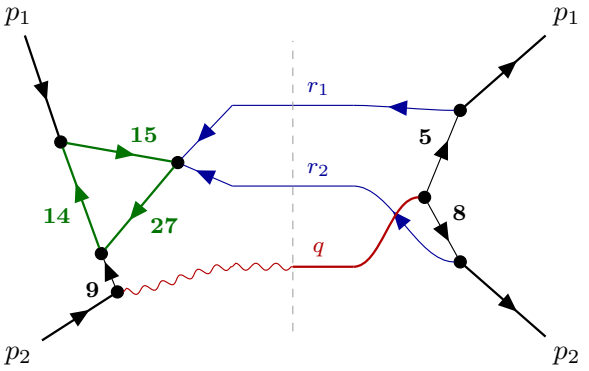
pinched: $27=(k_1+p_1+r_1+r_2)^2$



topo38 (RR family 2, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

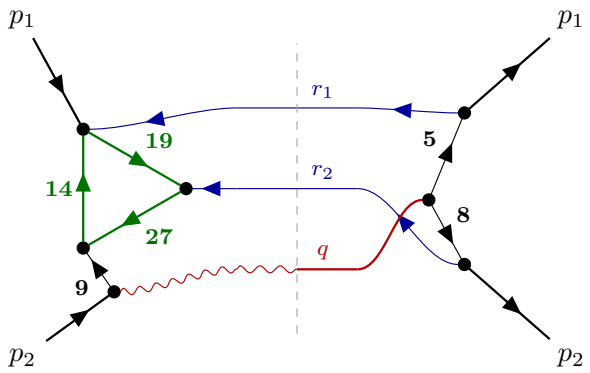
pinched: $6=(p_2+r_1)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

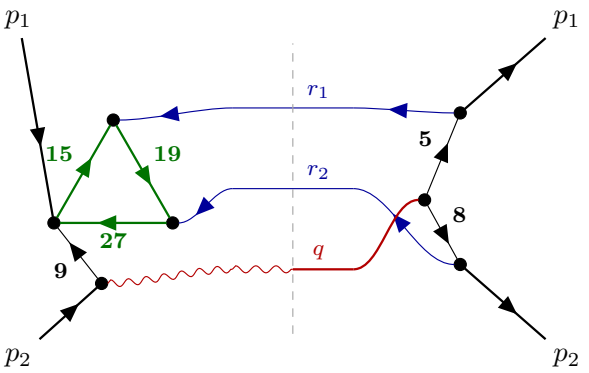
pinched: $19=(k_1+p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

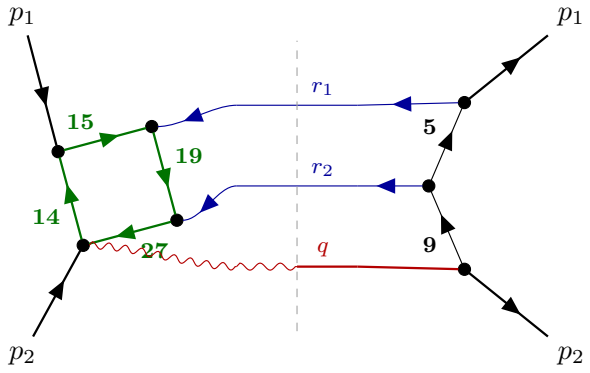
pinched: $15 = (k_1+p_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

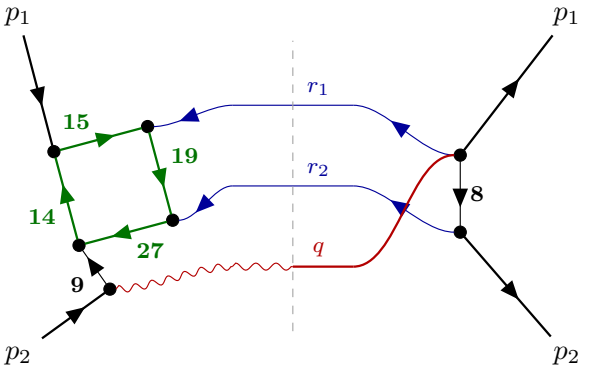
pinched: $14 = (k_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

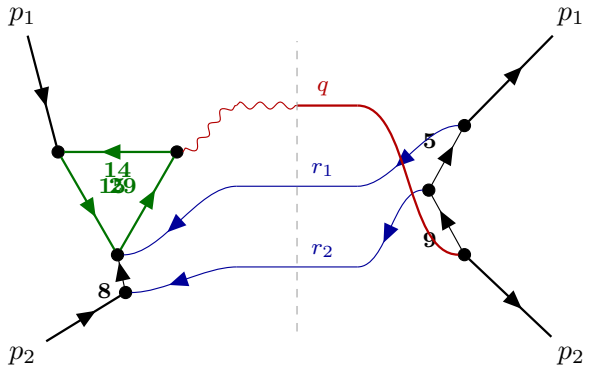
pinched: $8 = (p_2+r_2)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

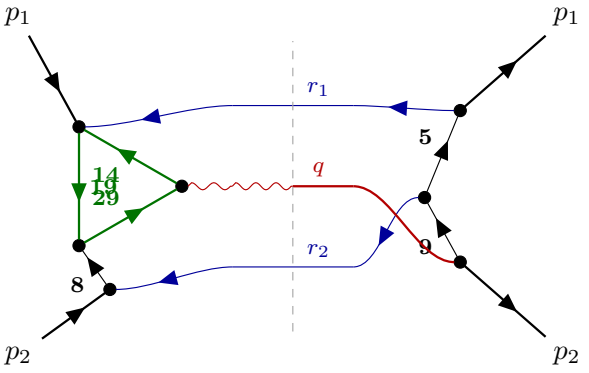
pinched: $5 = (p_1+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

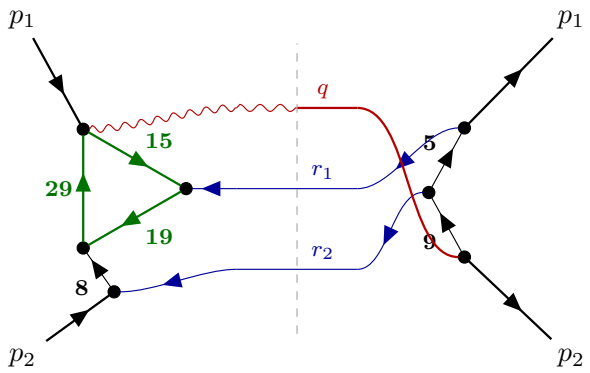
pinched: $19 = (k_1+p_1+r_1)^2$, $27 = (k_1+p_1+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

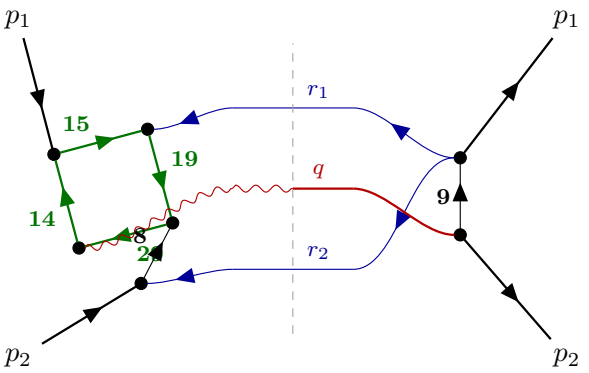
pinched: $15 = (k_1+p_1)^2$, $27 = (k_1+p_1+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

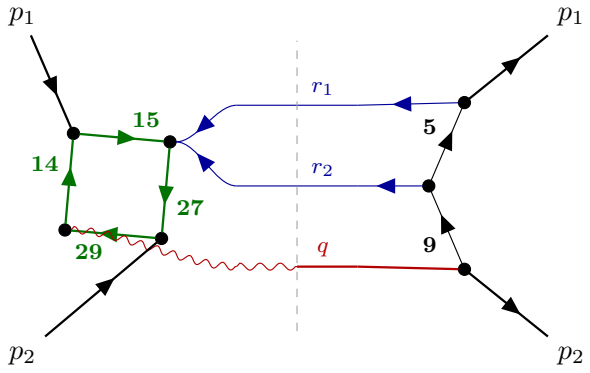
pinched: $14=(k_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	conjugate

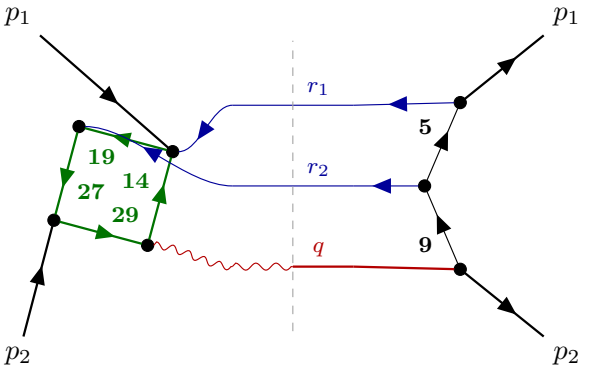
pinched: $5=(p_1+r_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

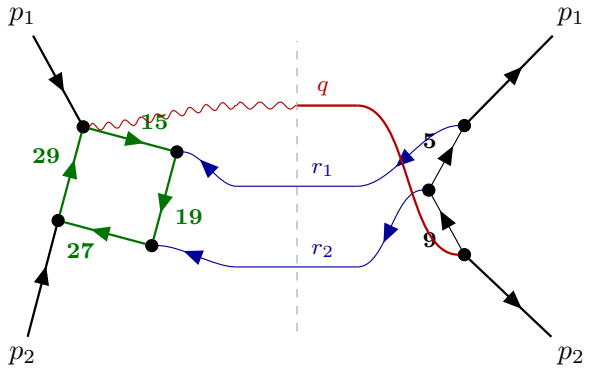
pinched: $8=(p_2+r_2)^2$, $19=(k_1+p_1+r_1)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

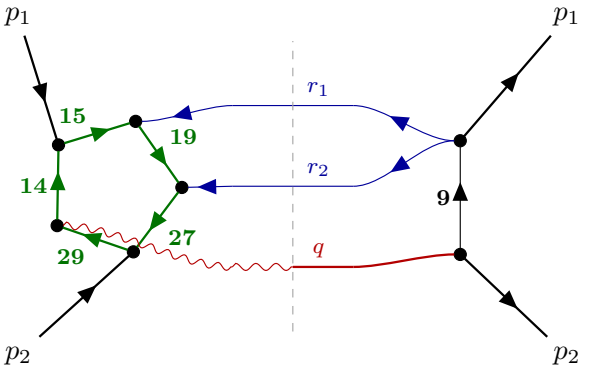
pinched: $8=(p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

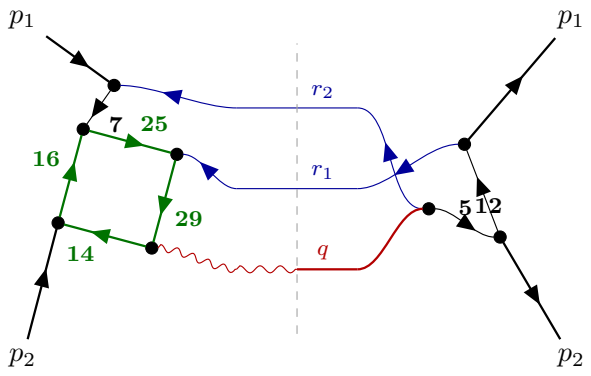
pinched: $8=(p_2+r_2)^2$, $14=(k_1)^2$



topo39 (RR family 6, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	conjugate

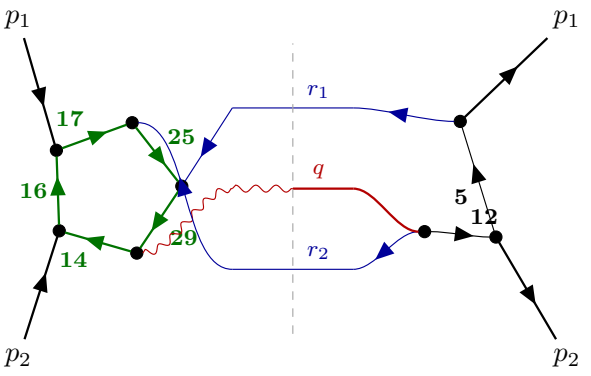
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo40 (RR family 15, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

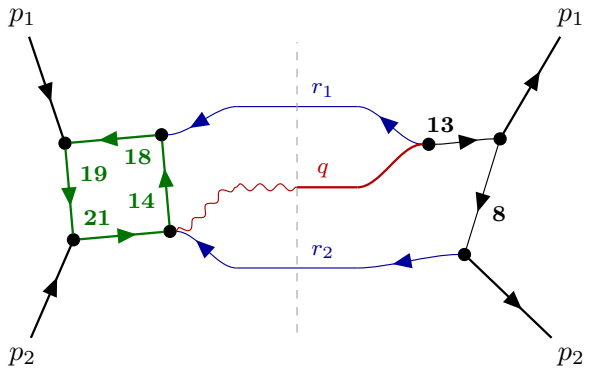
pinched: $17=(k_1+p_1+p_2)^2$



topo40 (RR family 15, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

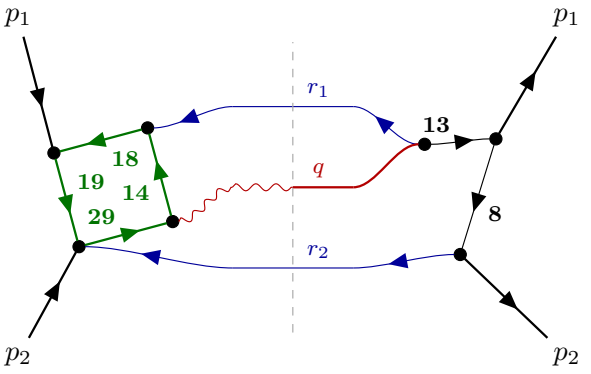
pinched: $7=(p_1+r_2)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

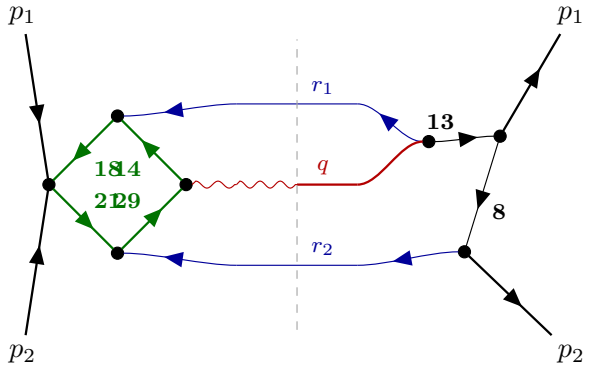
pinched: $5=(p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

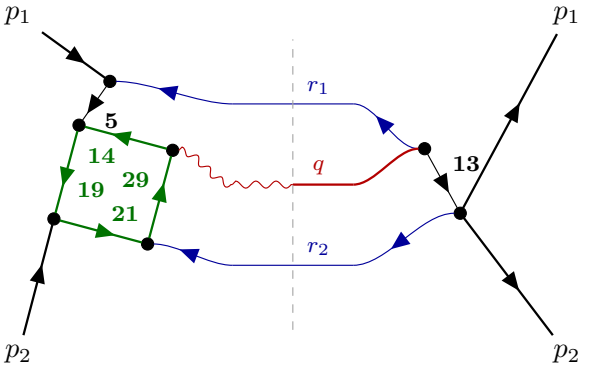
pinched: $5=(p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

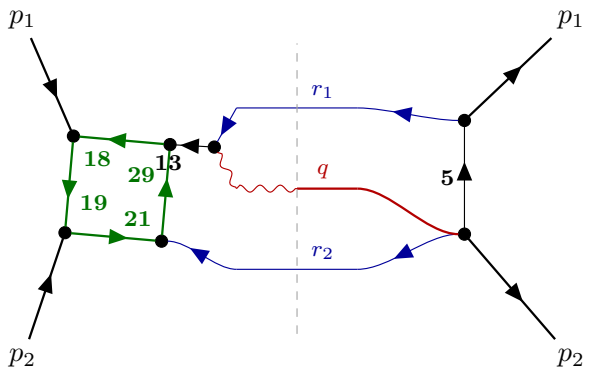
pinched: $5=(p_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

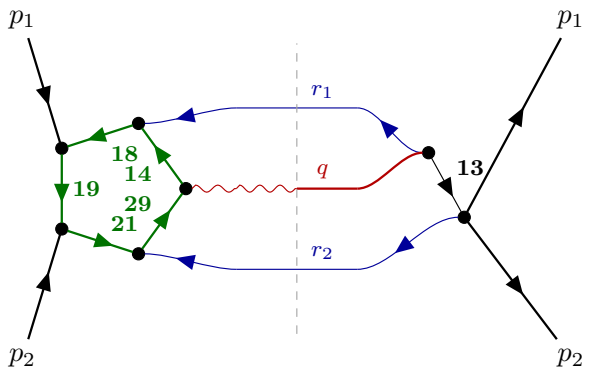
pinched: $8=(p_2+r_2)^2$, $18=(k_1+r_1)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
5	$(p_1 + r_1)^2$	conjugate

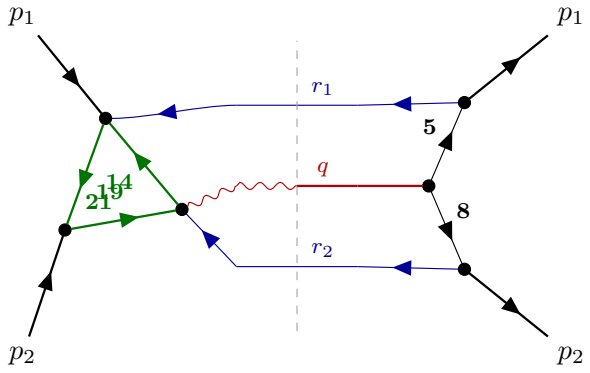
pinched: $8 = (p_2 + r_2)^2$, $14 = (k_1)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	conjugate

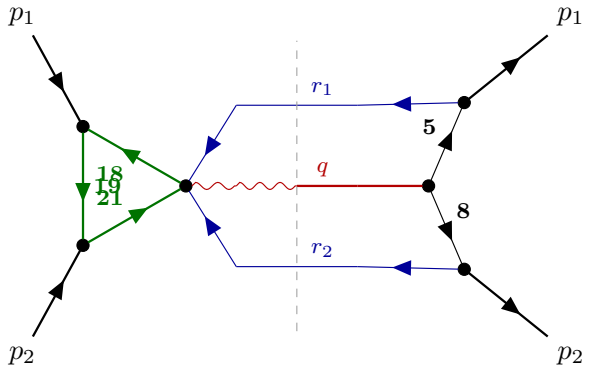
pinched: $5 = (p_1 + r_1)^2$, $8 = (p_2 + r_2)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate

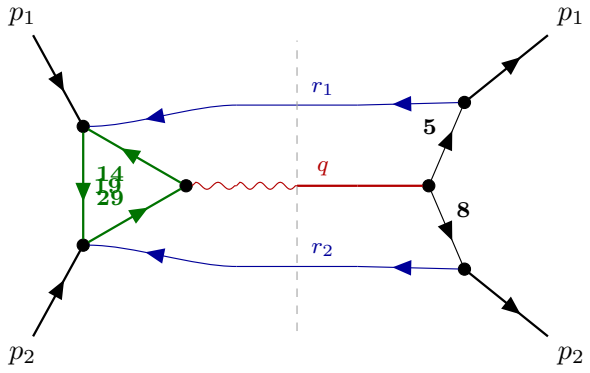
pinched: $13 = (p_1 + p_2 + r_2)^2$, $18 = (k_1 + r_1)^2$, $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate

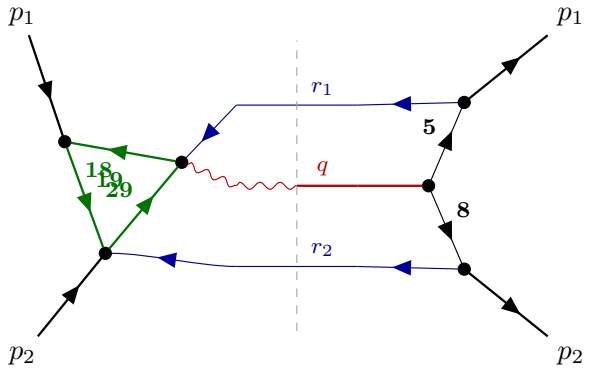
pinched: $13 = (p_1 + p_2 + r_2)^2$, $14 = (k_1)^2$, $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate

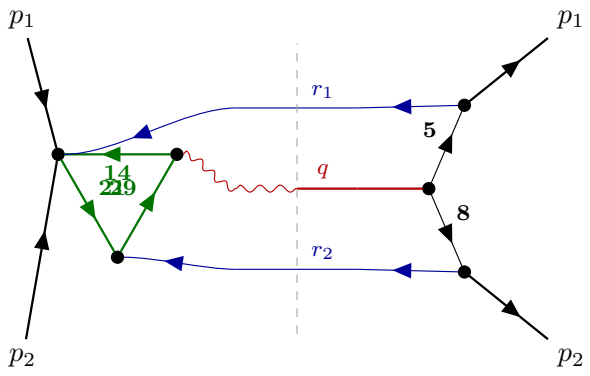
pinched: $13 = (p_1 + p_2 + r_2)^2$, $18 = (k_1 + r_1)^2$, $21 = (k_1 + p_1 + p_2 + r_1)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate

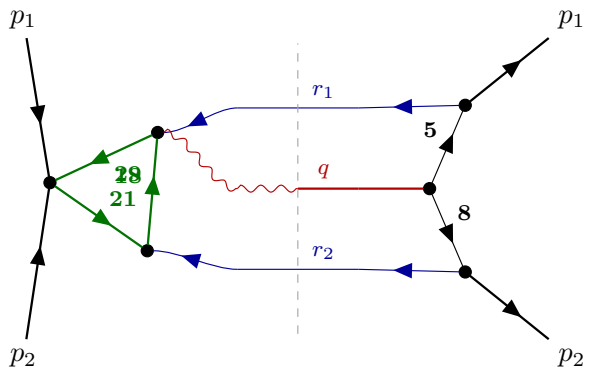
pinched: $13 = (p_1 + p_2 + r_2)^2$, $14 = (k_1)^2$, $21 = (k_1 + p_1 + p_2 + r_1)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

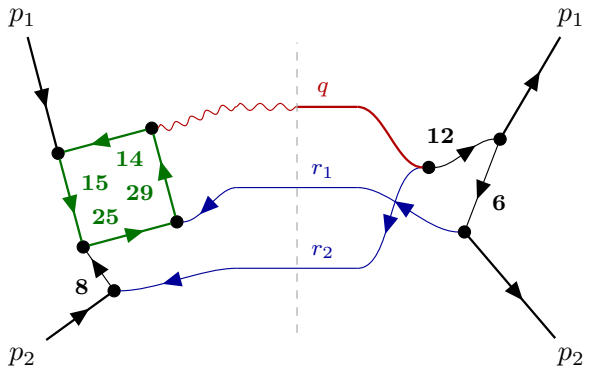
pinched: $13=(p_1+p_2+r_2)^2$, $18=(k_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$



topo41 (RR family 18, pentagon q-r1-p1-p2-r2), sector with 5 of 8 lines

18	$(k_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

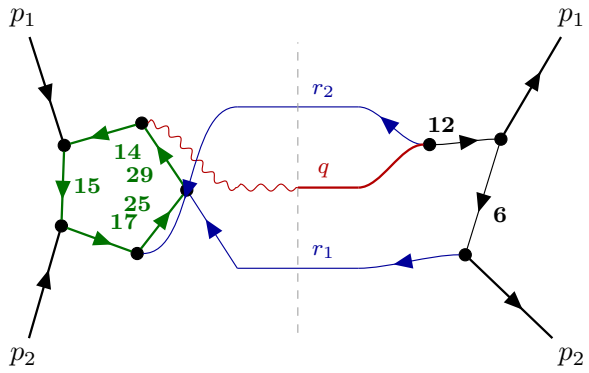
pinched: $13=(p_1+p_2+r_2)^2$, $14=(k_1)^2$, $19=(k_1+p_1+r_1)^2$



topo42 (RR family 21, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

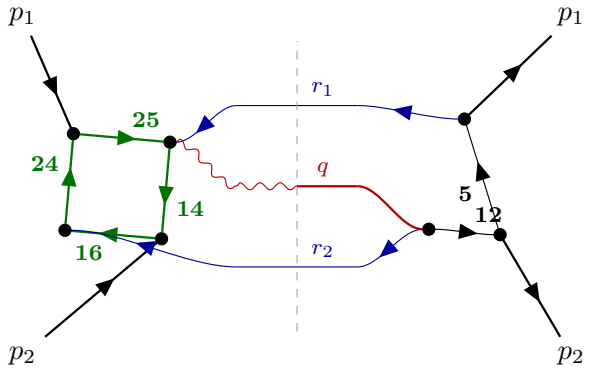
pinched: $17=(k_1+p_1+p_2)^2$



topo42 (RR family 21, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

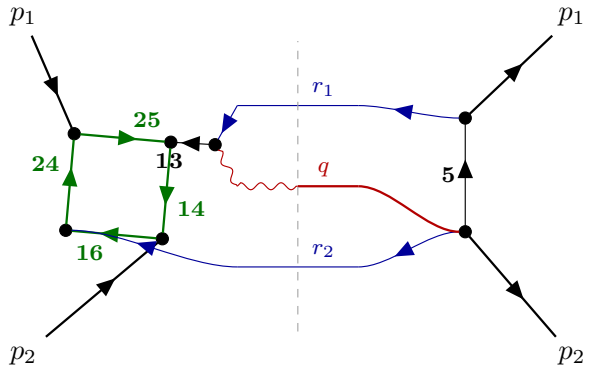
pinched: $8=(p_2+r_2)^2$



topo43 (RR family 23, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

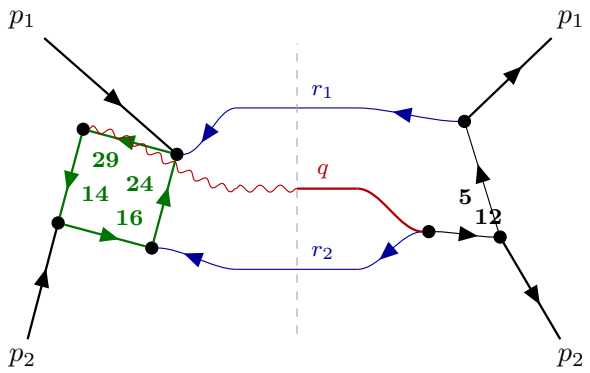
pinched: $13=(p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo43 (RR family 23, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

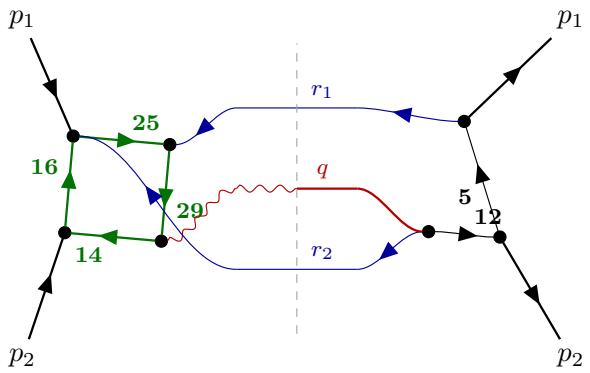
pinched: $12=(p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo43 (RR family 23, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

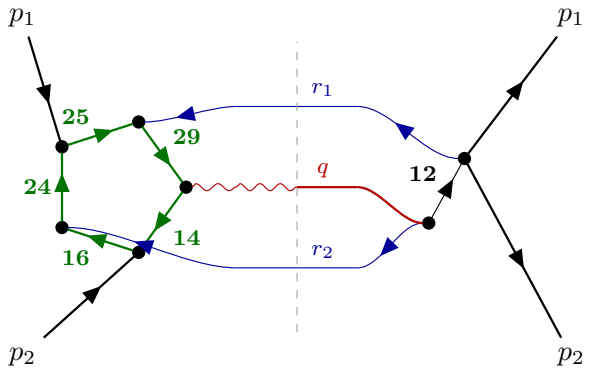
pinched: $13=(p_1+p_2+r_2)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo43 (RR family 23, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

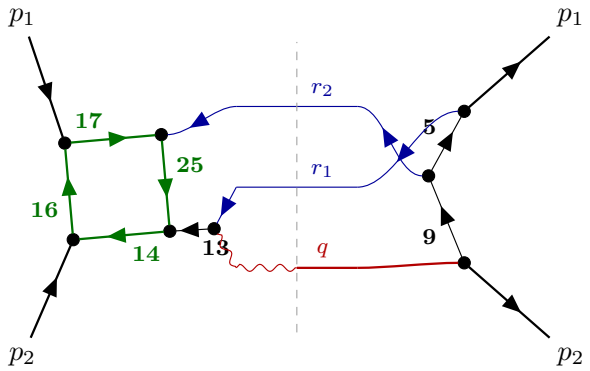
pinched: $13=(p_1+p_2+r_2)^2$, $24=(k_1+p_2+r_2)^2$



topo43 (RR family 23, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
12	$(p_1+p_2+r_1)^2$	conjugate

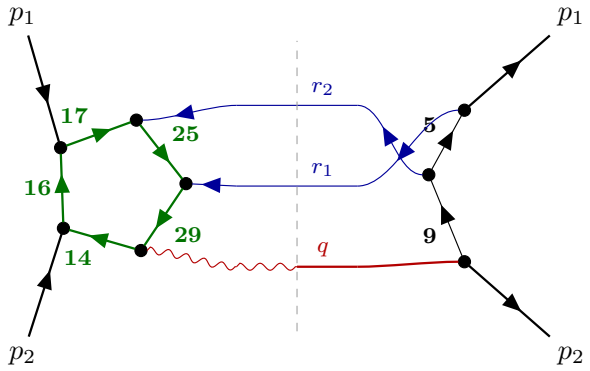
pinched: $5=(p_1+r_1)^2$, $13=(p_1+p_2+r_2)^2$



topo44 (RR family 32, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

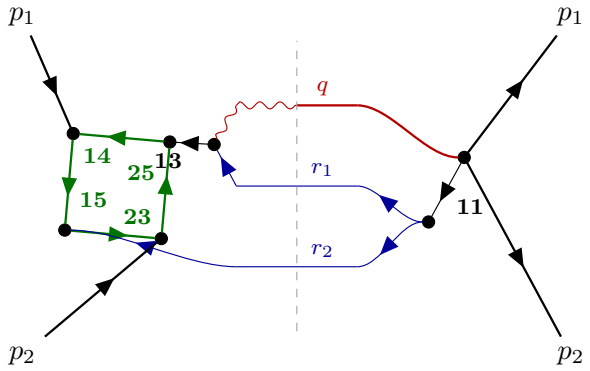
pinched: $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo44 (RR family 32, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

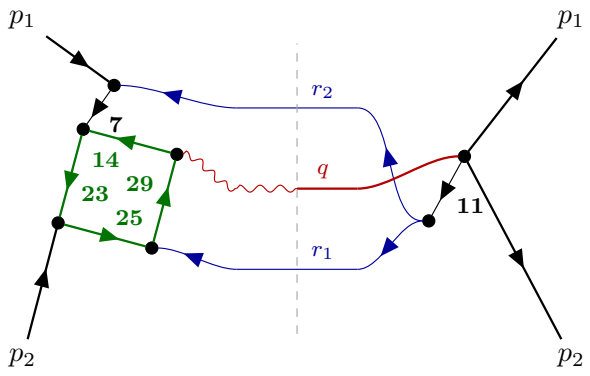
pinched: $13=(p_1+p_2+r_2)^2$



topo45 (RR family 30, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

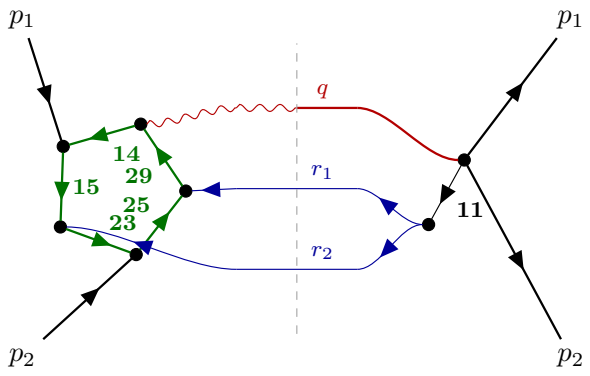
pinched: $7=(p_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo45 (RR family 30, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

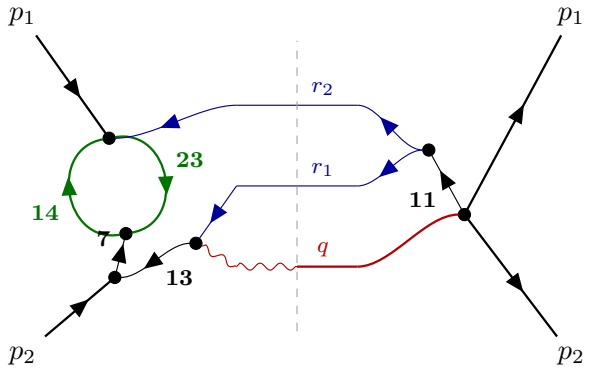
pinched: $13=(p_1+p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo45 (RR family 30, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

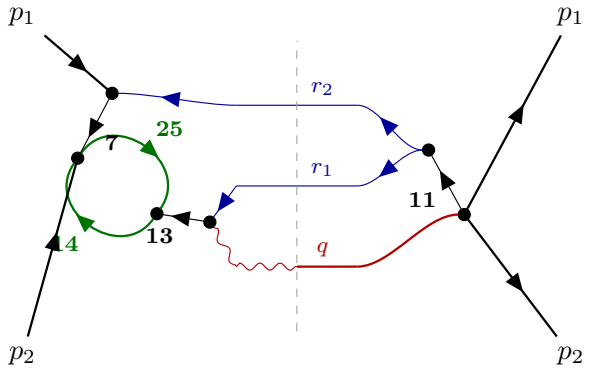
pinched: $7=(p_1+r_2)^2$, $13=(p_1+p_2+r_2)^2$



topo45 (RR family 30, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

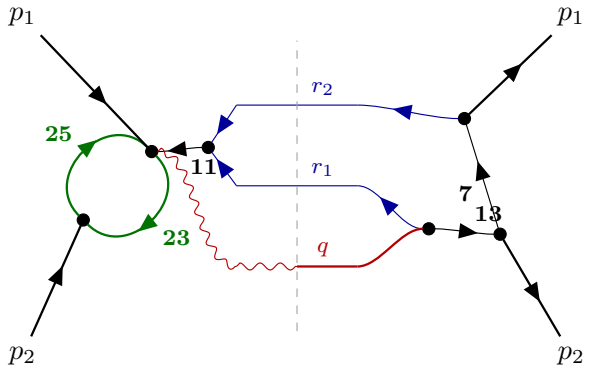
pinched: $15=(k_1+p_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo45 (RR family 30, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

14	$(k_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

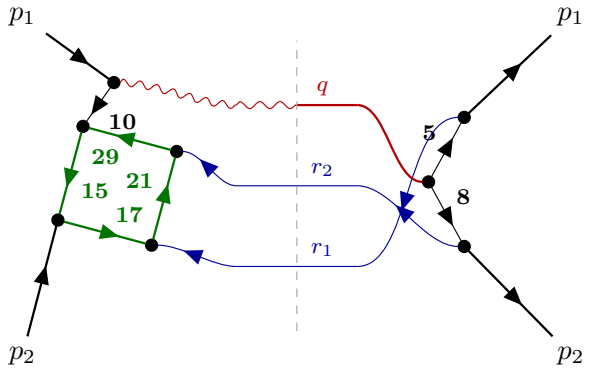
pinched: $15=(k_1+p_1)^2$, $23=(k_1+p_1+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo45 (RR family 30, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

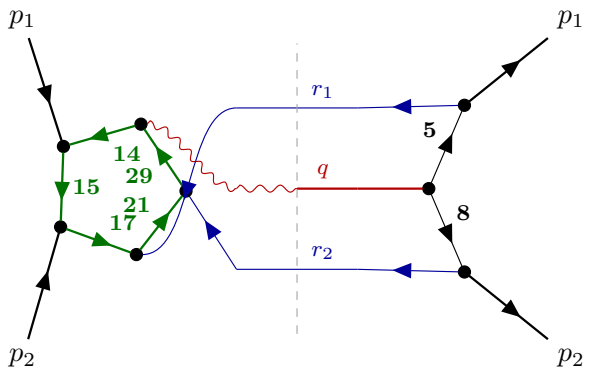
pinched: $14=(k_1)^2$, $15=(k_1+p_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo46 (RR family 7, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

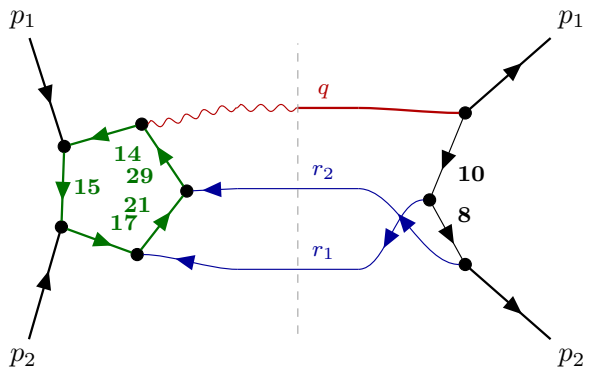
pinched: $14=(k_1)^2$



topo46 (RR family 7, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

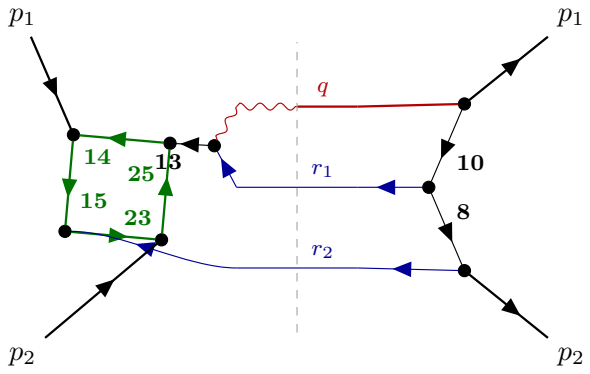
pinched: $10 = (p_2+r_1+r_2)^2$



topo46 (RR family 7, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

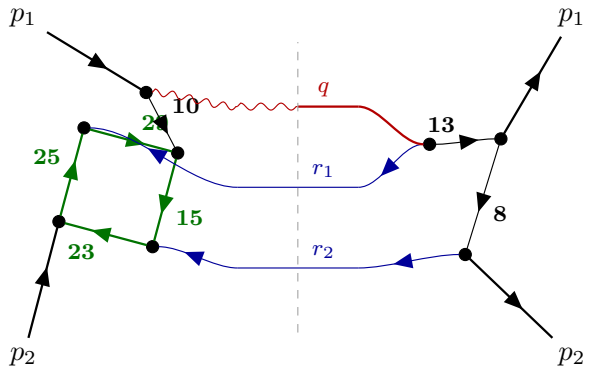
pinched: $5 = (p_1+r_1)^2$



topo47 (RR family 29, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

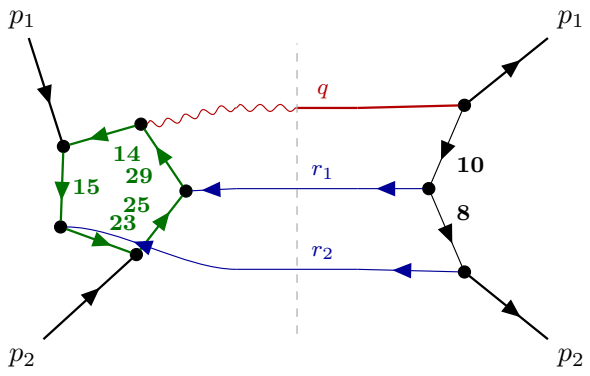
pinched: $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo47 (RR family 29, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

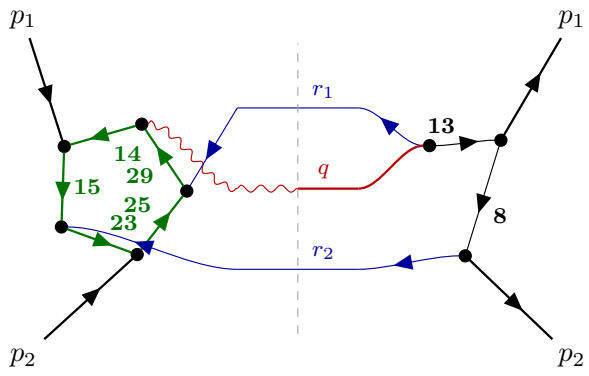
pinched: $14 = (k_1)^2$



topo47 (RR family 29, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

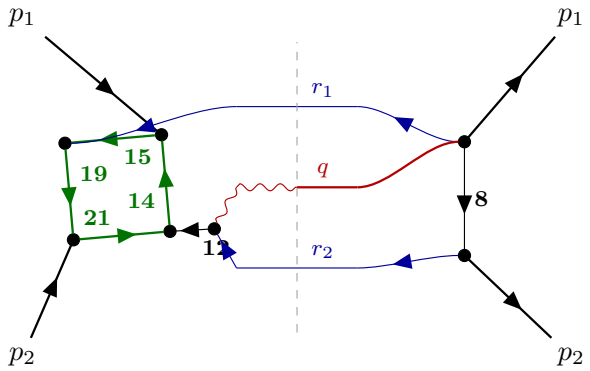
pinched: $13 = (p_1+p_2+r_2)^2$



topo47 (RR family 29, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

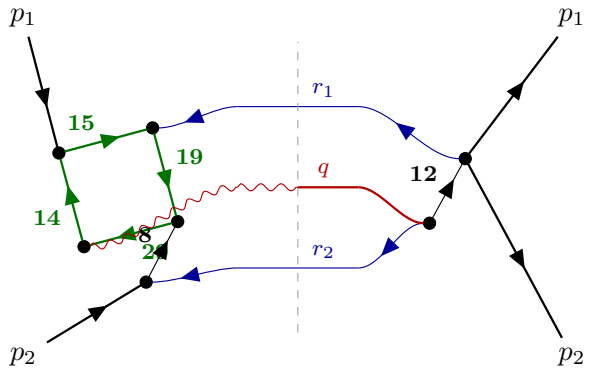
pinched: $10 = (p_2+r_1+r_2)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

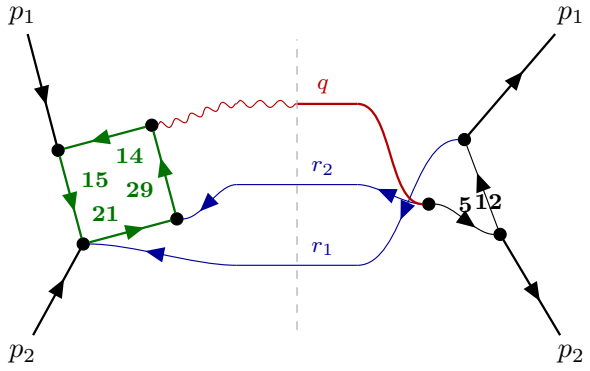
pinched: $5 = (p_1+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
12	$(p_1+p_2+r_1)^2$	conjugate

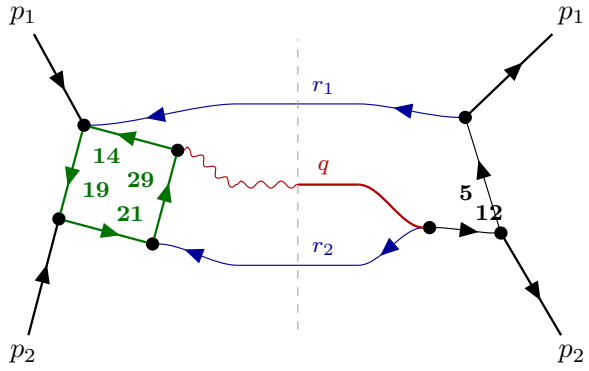
pinched: $5 = (p_1+r_1)^2$, $21 = (k_1+p_1+p_2+r_1)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

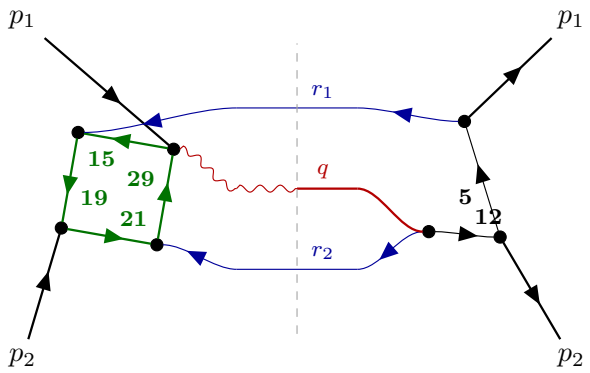
pinched: $8 = (p_2+r_2)^2$, $19 = (k_1+p_1+r_1)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

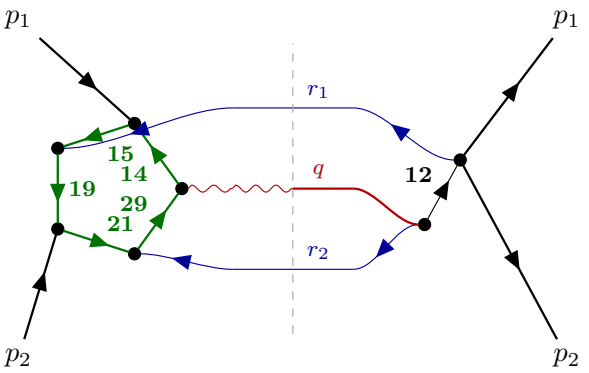
pinched: $8 = (p_2+r_2)^2$, $15 = (k_1+p_1)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

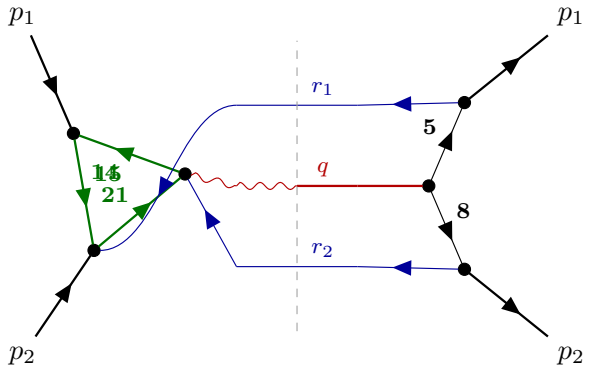
pinched: $8=(p_2+r_2)^2$, $14=(k_1)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
12	$(p_1+p_2+r_1)^2$	conjugate

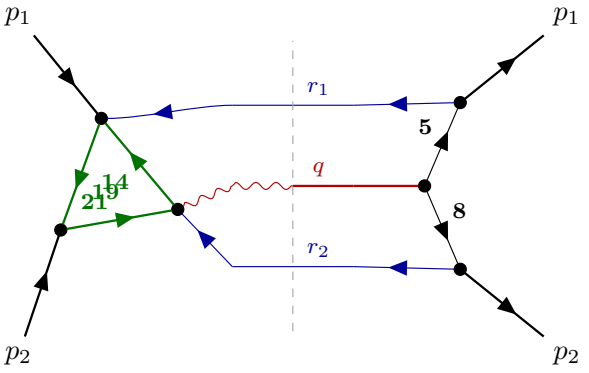
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

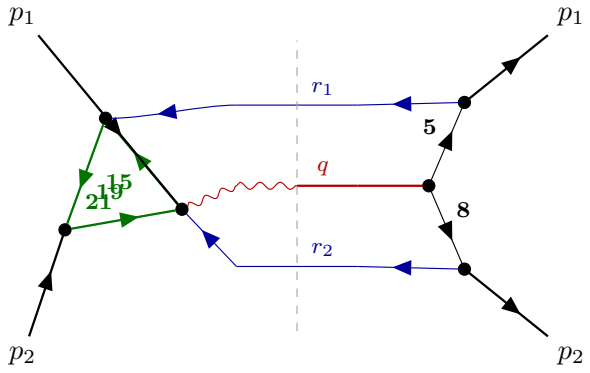
pinched: $12=(p_1+p_2+r_1)^2$, $19=(k_1+p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

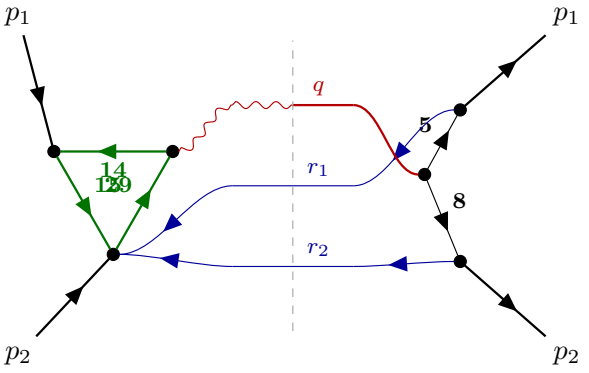
pinched: $12=(p_1+p_2+r_1)^2$, $15=(k_1+p_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

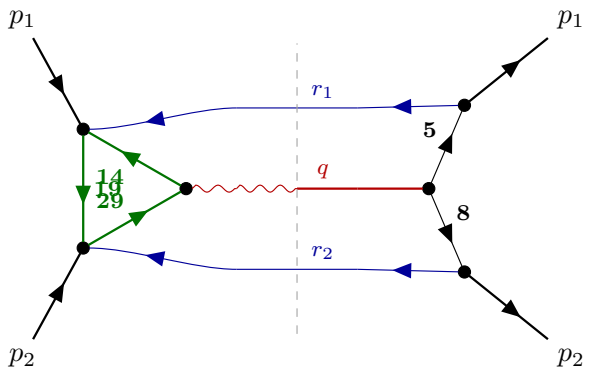
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

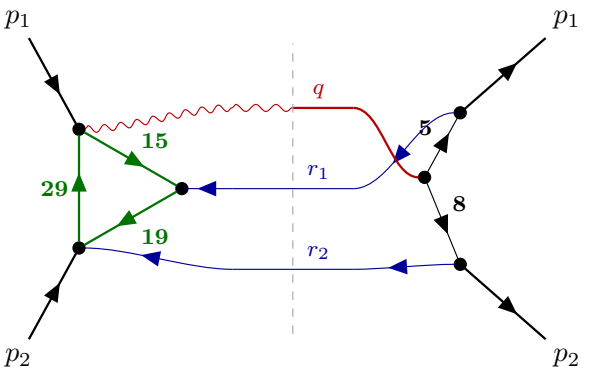
pinched: $12=(p_1+p_2+r_1)^2$, $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

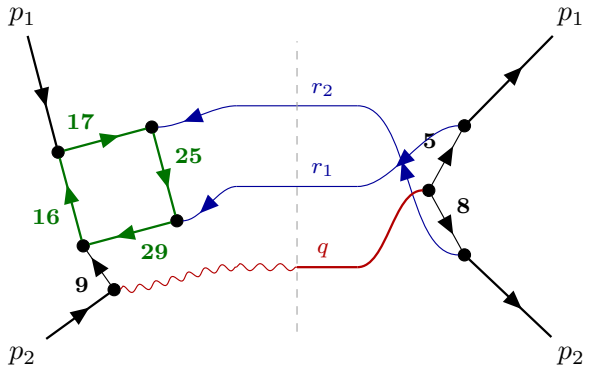
pinched: $12=(p_1+p_2+r_1)^2$, $15=(k_1+p_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo48 (RR family 17, pentagon q-p1-r1-p2-r2), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

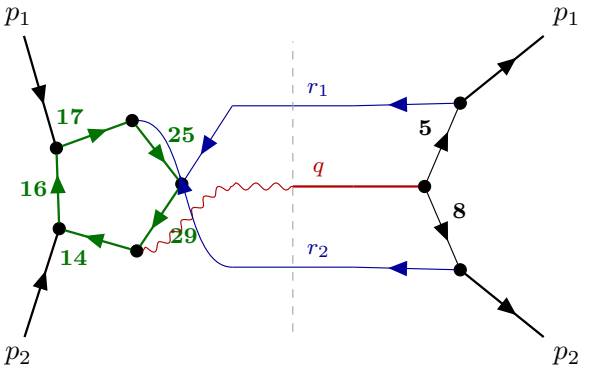
pinched: $12=(p_1+p_2+r_1)^2$, $14=(k_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo49 (RR family 6, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

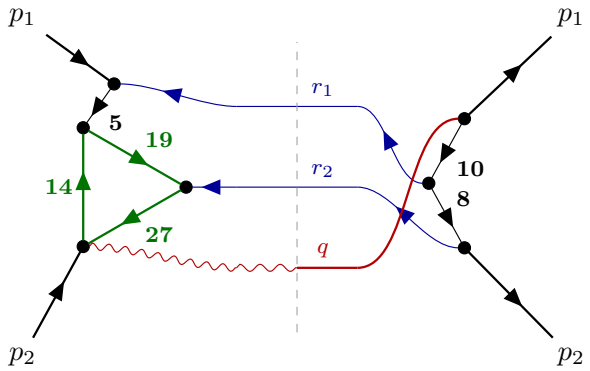
pinched: $14=(k_1)^2$



topo49 (RR family 6, pentagon q-p2-p1-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

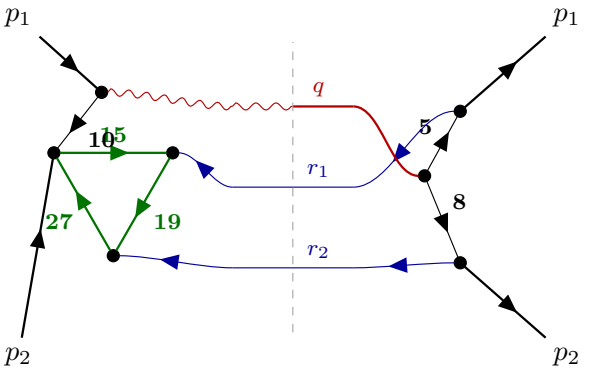
pinched: $9=(p_1+r_1+r_2)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

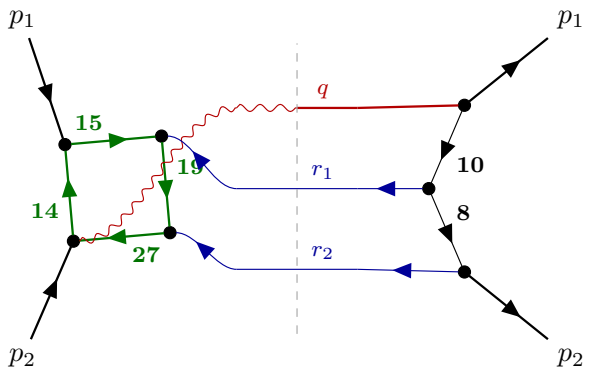
pinched: $15=(k_1+p_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

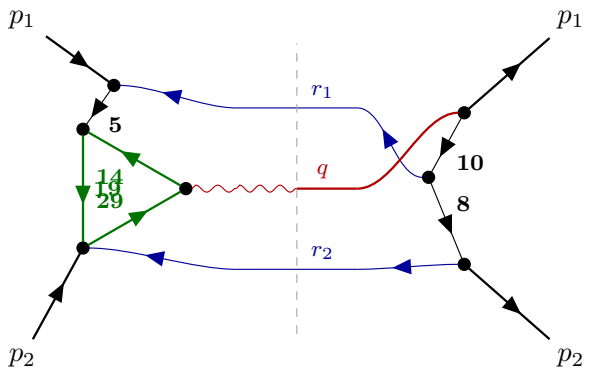
pinched: $14=(k_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

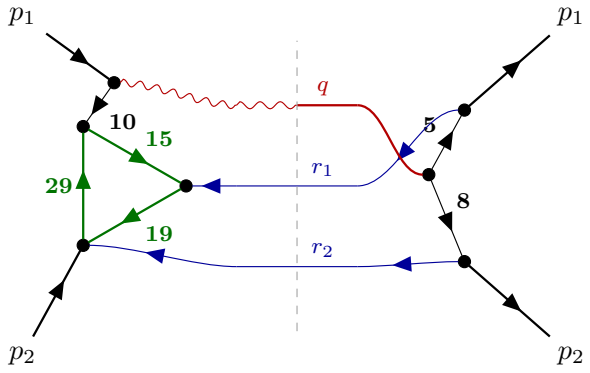
pinched: $5=(p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

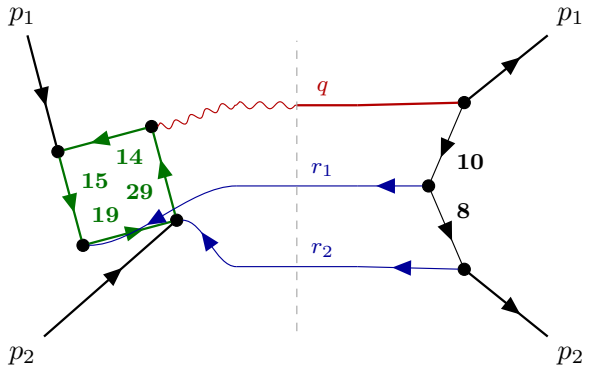
pinched: $15=(k_1+p_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

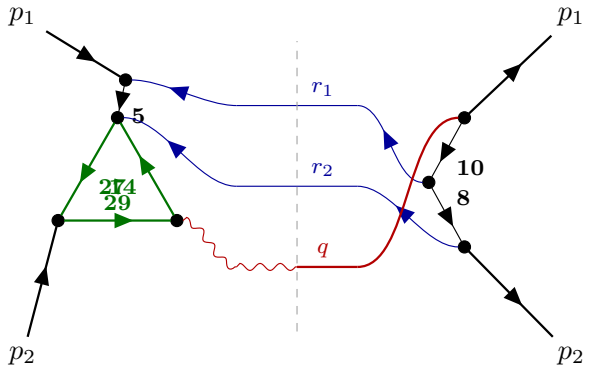
pinched: $14=(k_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

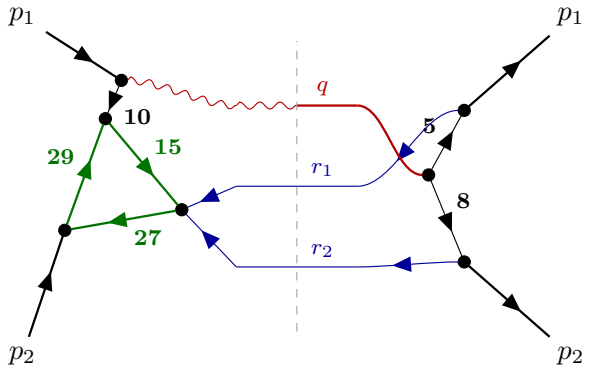
pinched: $5=(p_1+r_1)^2$, $27=(k_1+p_1+r_1+r_2)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

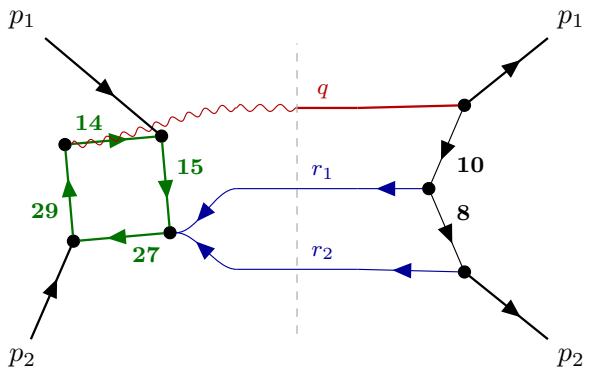
pinched: $15=(k_1+p_1)^2$, $19=(k_1+p_1+r_1)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

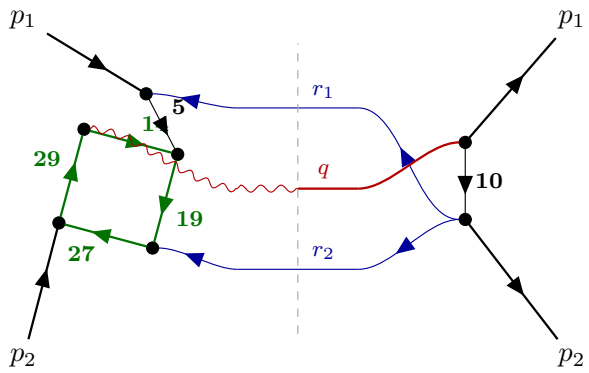
pinched: $14=(k_1)^2$, $19=(k_1+p_1+r_1)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

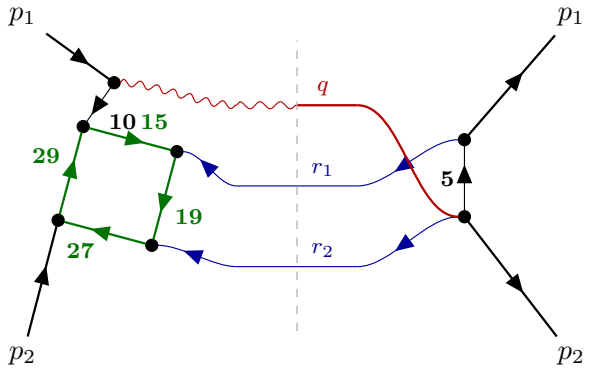
pinched: $5=(p_1+r_1)^2$, $19=(k_1+p_1+r_1)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	conjugate

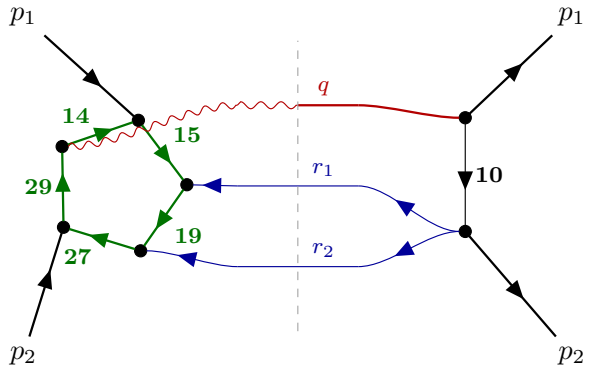
pinched: $8=(p_2+r_2)^2$, $15=(k_1+p_1)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

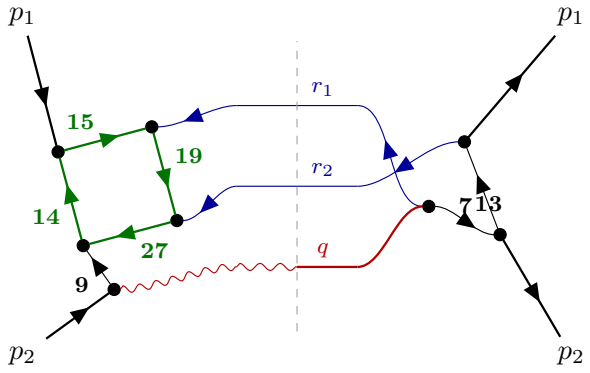
pinched: $8=(p_2+r_2)^2$, $14=(k_1)^2$



topo50 (RR family 7, pentagon q-p1-r1-r2-p2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	conjugate

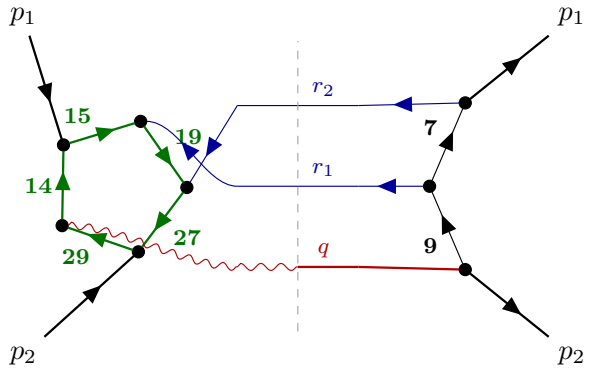
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo51 (RR family 33, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

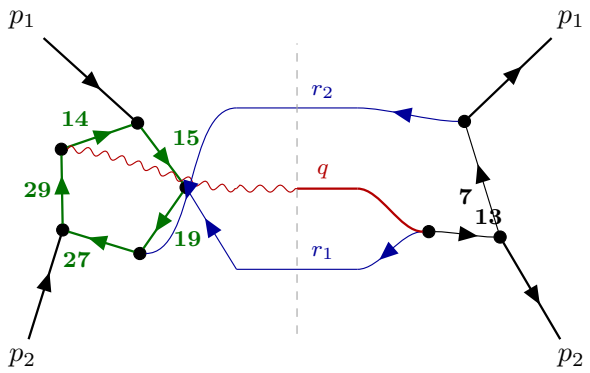
pinched: $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo51 (RR family 33, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

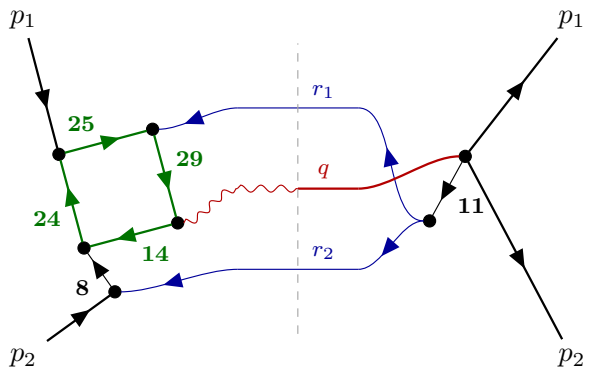
pinched: $13=(p_1+p_2+r_2)^2$



topo51 (RR family 33, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

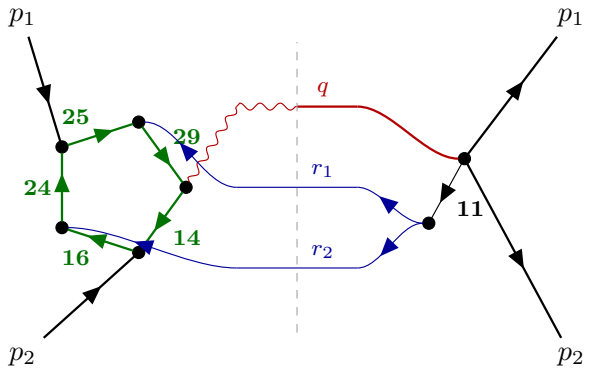
pinched: $9=(p_1+r_1+r_2)^2$



topo52 (RR family 8, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

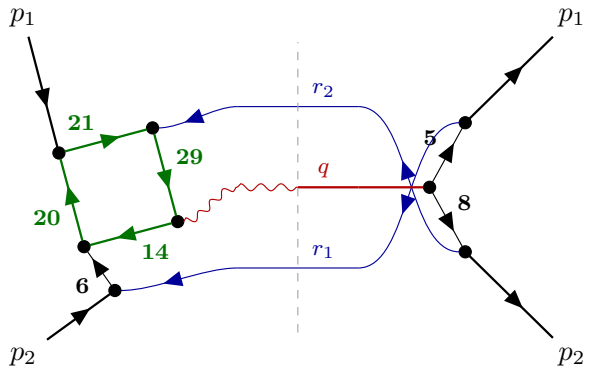
pinched: $5=(p_1+r_1)^2$, $16=(k_1+p_2)^2$



topo52 (RR family 8, pentagon q-p2-r2-p1-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

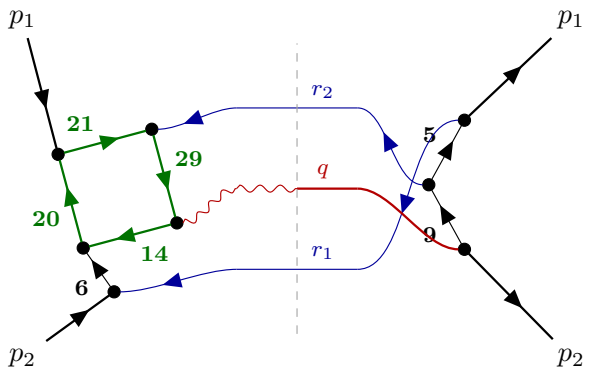
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo53 (RR family 2, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

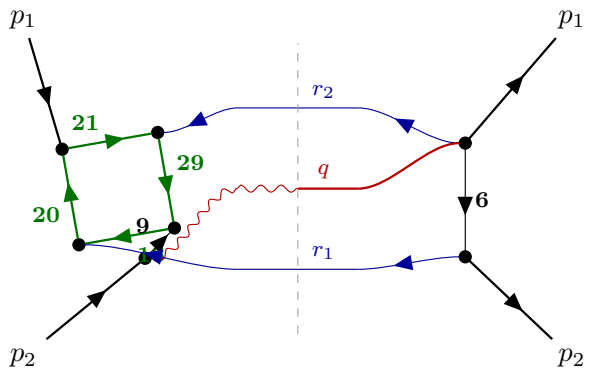
pinched: $16=(k_1+p_2)^2$



topo54 (RR family 3, pentagon q-p2-r1-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

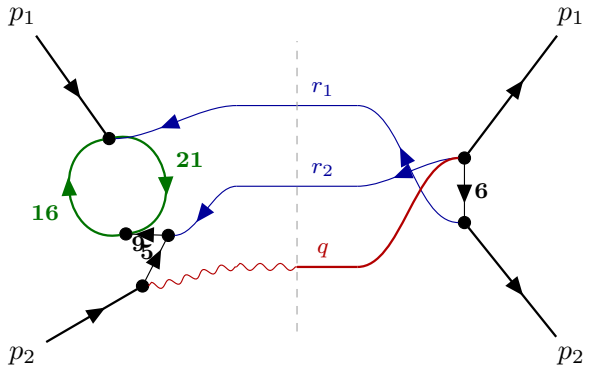
pinched: $16 = (k_1+p_2)^2$



topo54 (RR family 3, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

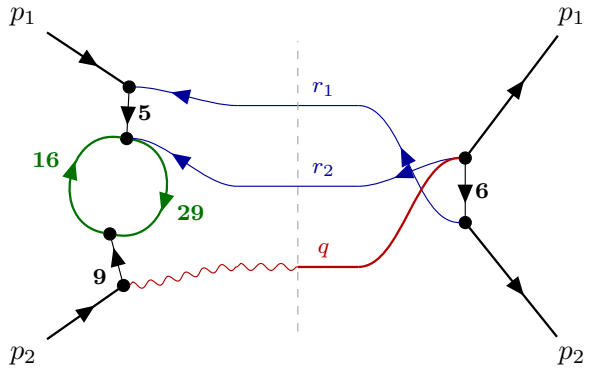
pinched: $5 = (p_1+r_1)^2$, $14 = (k_1)^2$



topo54 (RR family 3, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

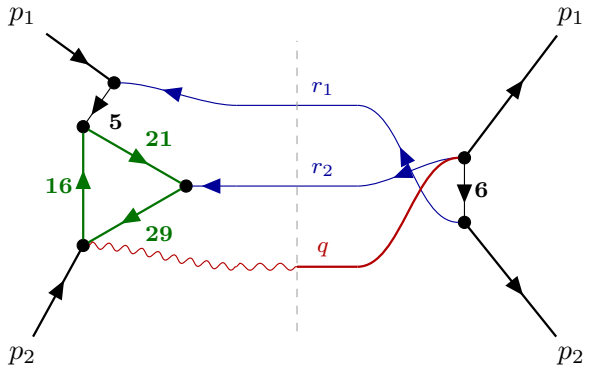
pinched: $14 = (k_1)^2$, $20 = (k_1+p_2+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo54 (RR family 3, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

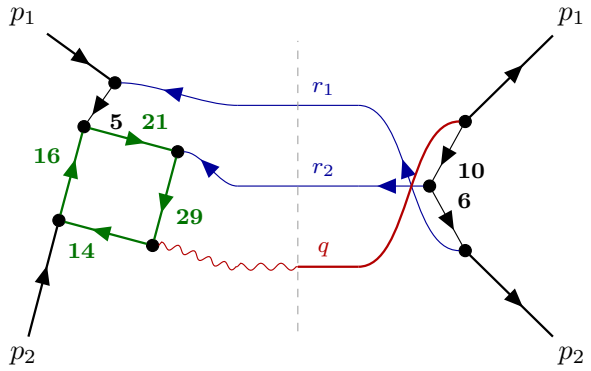
pinched: $14 = (k_1)^2$, $20 = (k_1+p_2+r_1)^2$, $21 = (k_1+p_1+p_2+r_1)^2$



topo54 (RR family 3, pentagon q-p2-r1-p1-r2), sector with 5 of 8 lines

16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate

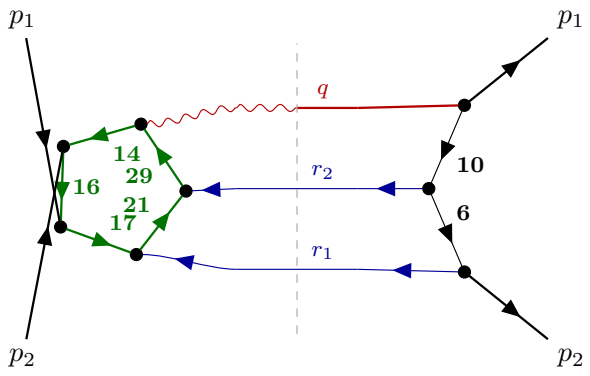
pinched: $9 = (p_1+r_1+r_2)^2$, $14 = (k_1)^2$, $20 = (k_1+p_2+r_1)^2$



topo55 (RR family 4, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

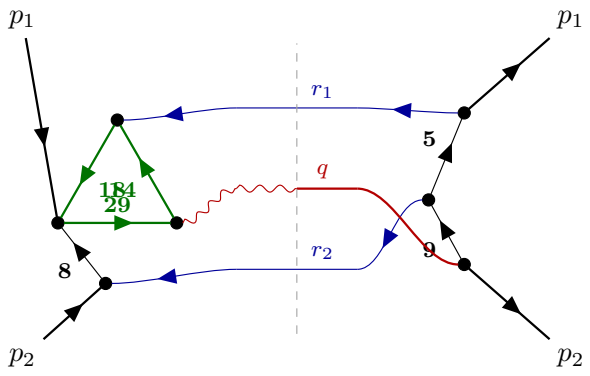
pinched: $17 = (k_1+p_1+p_2)^2$



topo55 (RR family 4, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

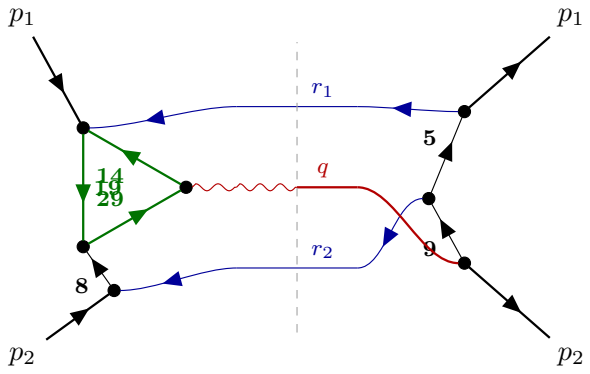
pinched: $5=(p_1+r_1)^2$



topo56 (RR family 6, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

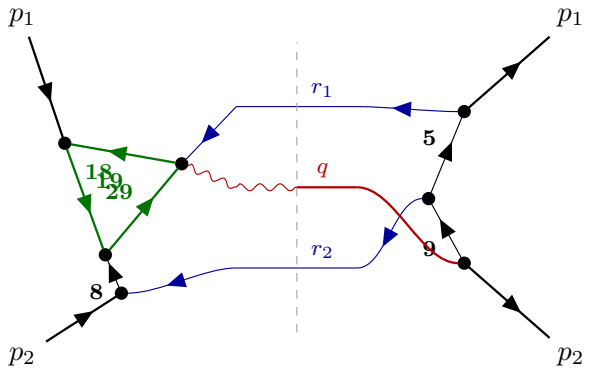
pinched: $19=(k_1+p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo56 (RR family 6, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

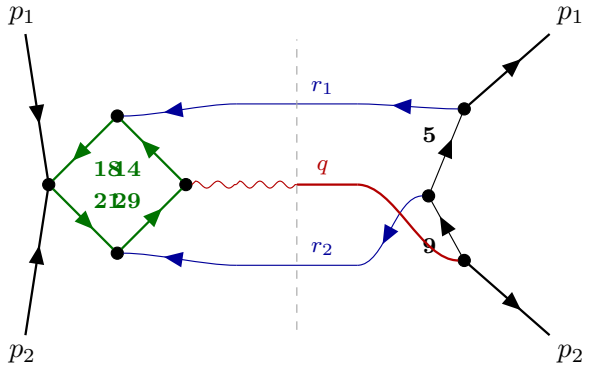
pinched: $18=(k_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo56 (RR family 6, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
9	$(p_1+r_1+r_2)^2$	conjugate

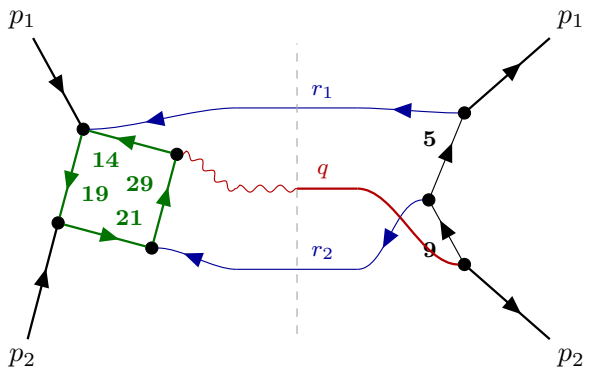
pinched: $5=(p_1+r_1)^2$, $21=(k_1+p_1+p_2+r_1)^2$



topo56 (RR family 6, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

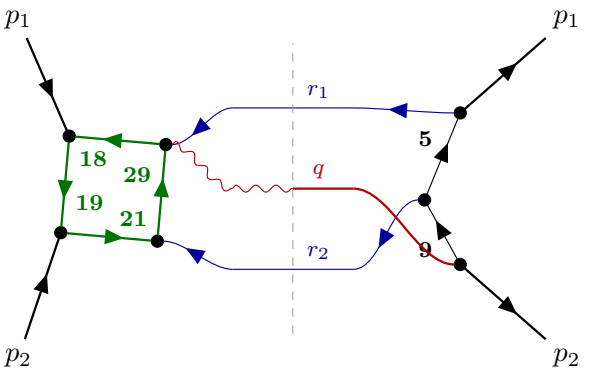
pinched: $8=(p_2+r_2)^2$, $19=(k_1+p_1+r_1)^2$



topo56 (RR family 6, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

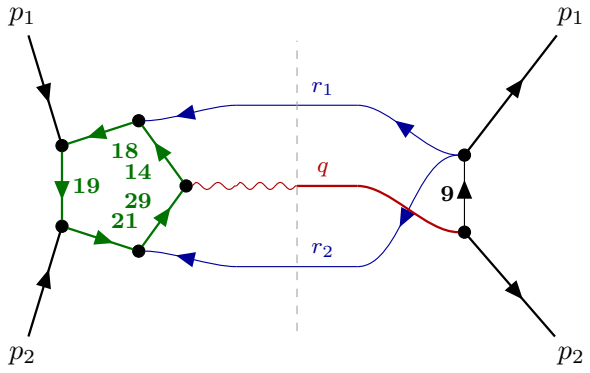
pinched: $8=(p_2+r_2)^2$, $18=(k_1+r_1)^2$



topo56 (RR family 6, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

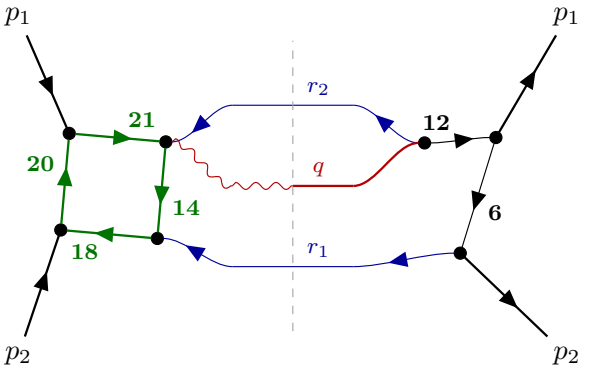
pinched: $8=(p_2+r_2)^2$, $14=(k_1)^2$



topo56 (RR family 6, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	conjugate

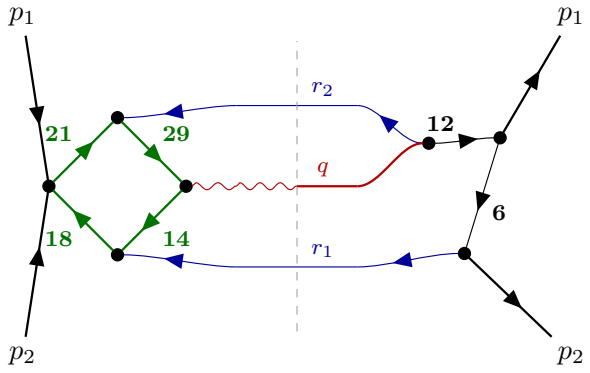
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

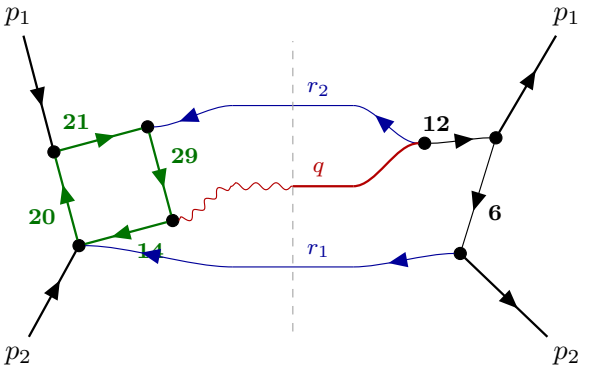
pinched: $13=(p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

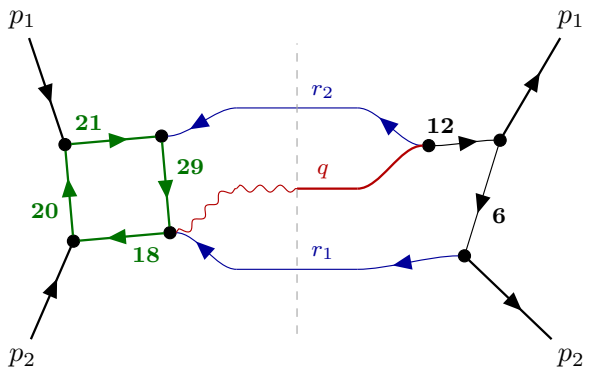
pinched: $13=(p_1+p_2+r_2)^2$, $20=(k_1+p_2+r_1)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

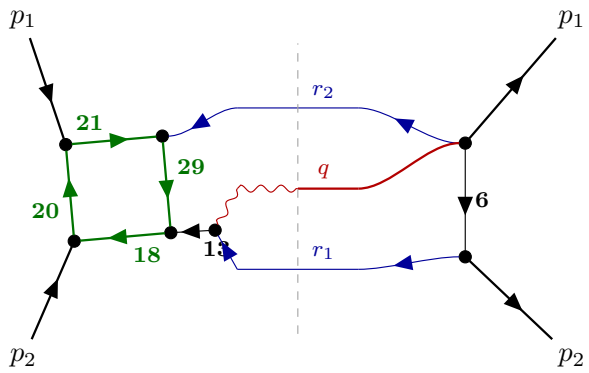
pinched: $13=(p_1+p_2+r_2)^2$, $18=(k_1+r_1)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

18	$(k_1 + r_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
6	$(p_2 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate

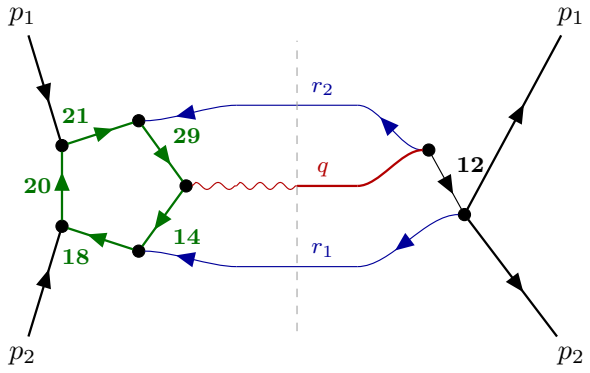
pinched: $13 = (p_1 + p_2 + r_2)^2$, $14 = (k_1)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

18	$(k_1 + r_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
6	$(p_2 + r_1)^2$	conjugate

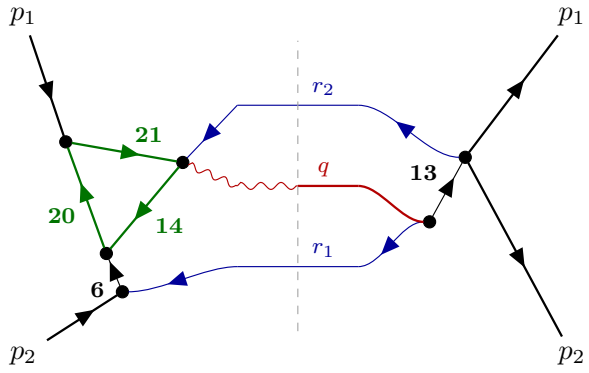
pinched: $12 = (p_1 + p_2 + r_1)^2$, $14 = (k_1)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
12	$(p_1 + p_2 + r_1)^2$	conjugate

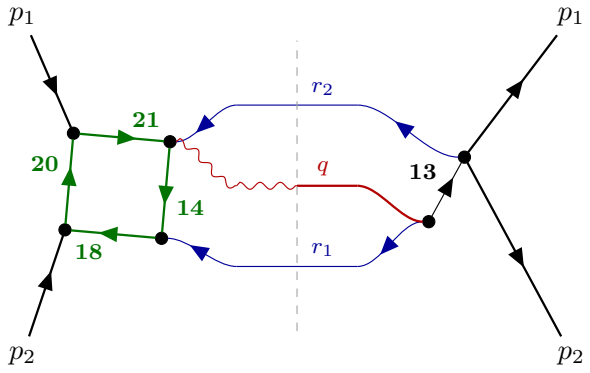
pinched: $6 = (p_2 + r_1)^2$, $13 = (p_1 + p_2 + r_2)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
6	$(p_2 + r_1)^2$	amplitude (tree on the loop)
13	$(p_1 + p_2 + r_2)^2$	conjugate

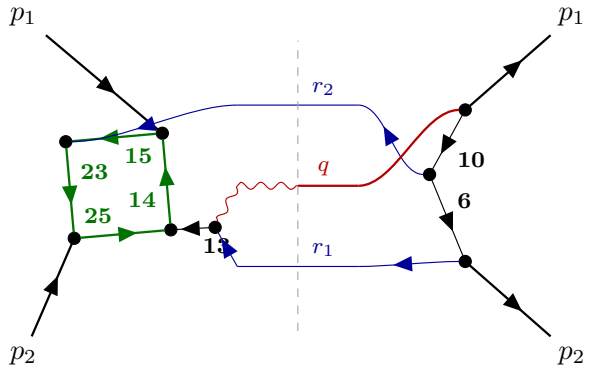
pinched: $12 = (p_1 + p_2 + r_1)^2$, $18 = (k_1 + r_1)^2$, $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo57 (RR family 24, pentagon q-r1-p2-p1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	conjugate

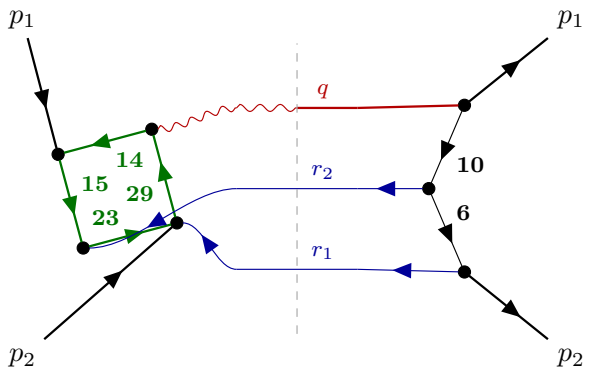
pinched: $6 = (p_2 + r_1)^2$, $12 = (p_1 + p_2 + r_1)^2$, $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
23	$(k_1 + p_1 + r_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
6	$(p_2 + r_1)^2$	conjugate
10	$(p_2 + r_1 + r_2)^2$	conjugate

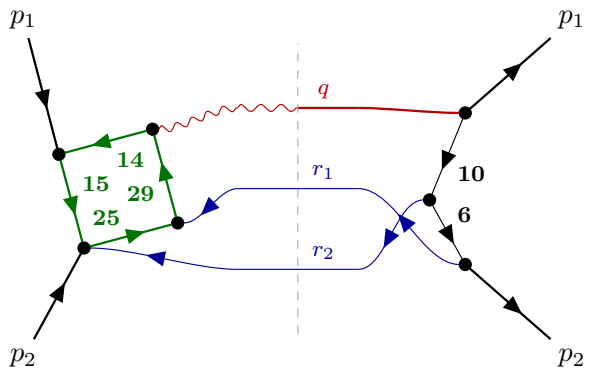
pinched: $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

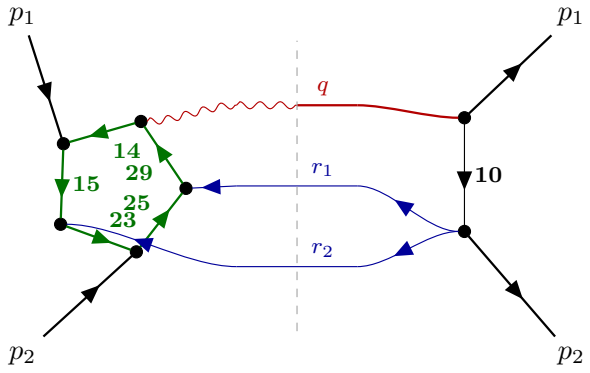
pinched: $13 = (p_1+p_2+r_2)^2$, $25 = (k_1+p_1+p_2+r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

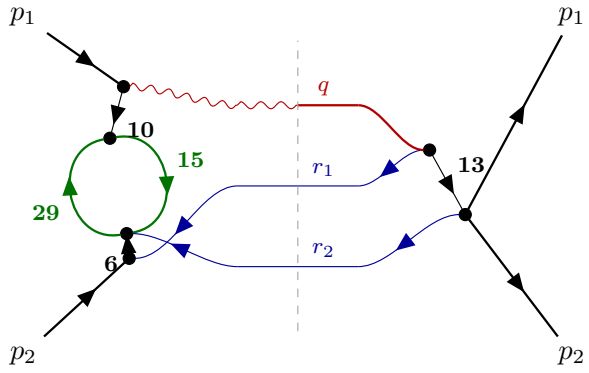
pinched: $13 = (p_1+p_2+r_2)^2$, $23 = (k_1+p_1+r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	conjugate

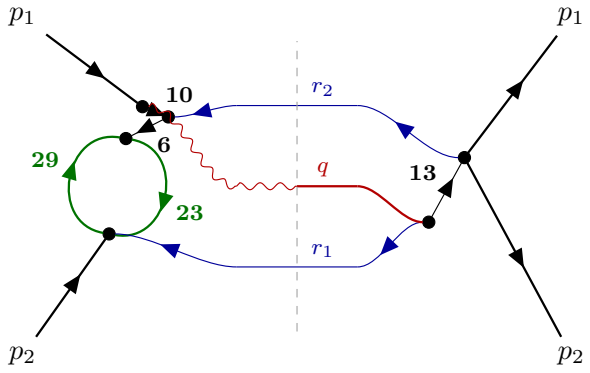
pinched: $6 = (p_2+r_1)^2$, $13 = (p_1+p_2+r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

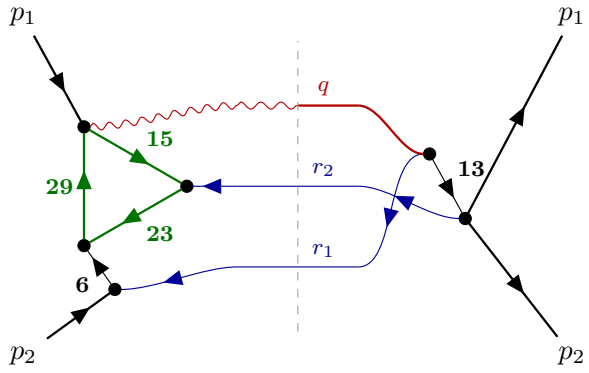
pinched: $14 = (k_1)^2$, $23 = (k_1+p_1+r_2)^2$, $25 = (k_1+p_1+p_2+r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

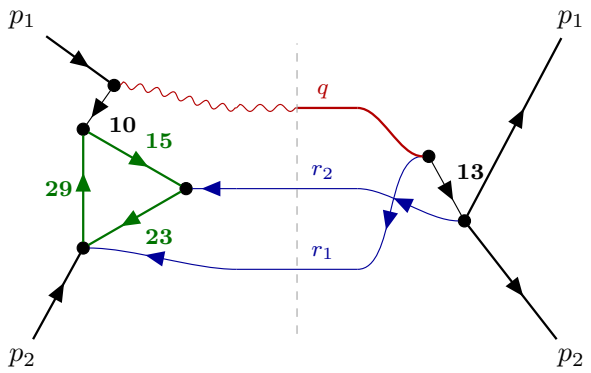
pinched: $14 = (k_1)^2$, $15 = (k_1+p_1)^2$, $25 = (k_1+p_1+p_2+r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

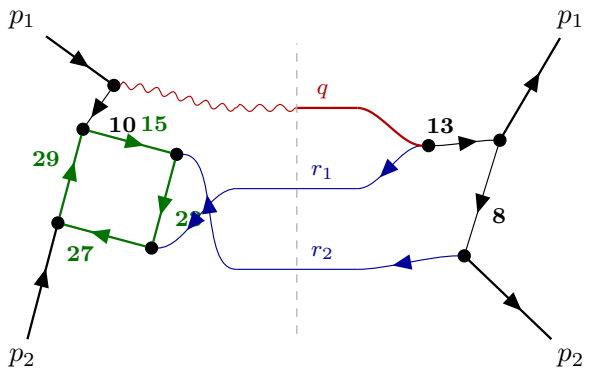
pinched: $10 = (p_2+r_1+r_2)^2$, $14 = (k_1)^2$, $25 = (k_1+p_1+p_2+r_2)^2$



topo58 (RR family 27, pentagon q-p1-r2-p2-r1), sector with 5 of 8 lines

15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
13	$(p_1+p_2+r_2)^2$	conjugate

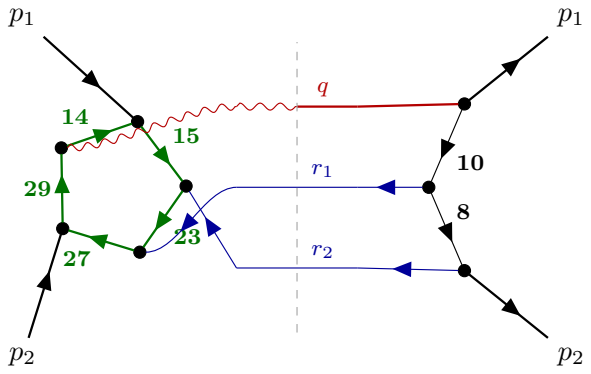
pinched: $6=(p_2+r_1)^2$, $14=(k_1)^2$, $25=(k_1+p_1+p_2+r_2)^2$



topo59 (RR family 29, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

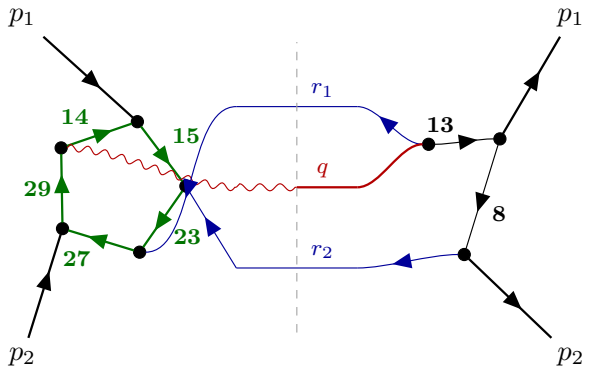
pinched: $14=(k_1)^2$



topo59 (RR family 29, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

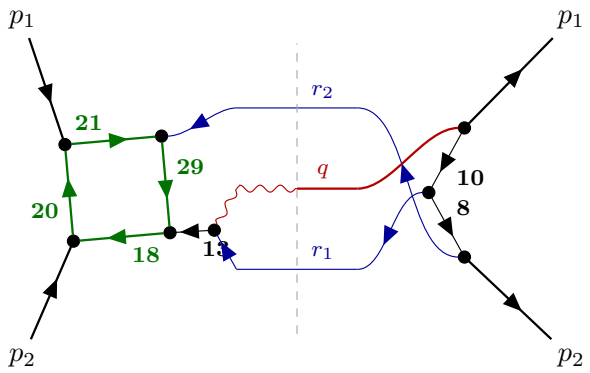
pinched: $13=(p_1+p_2+r_2)^2$



topo59 (RR family 29, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

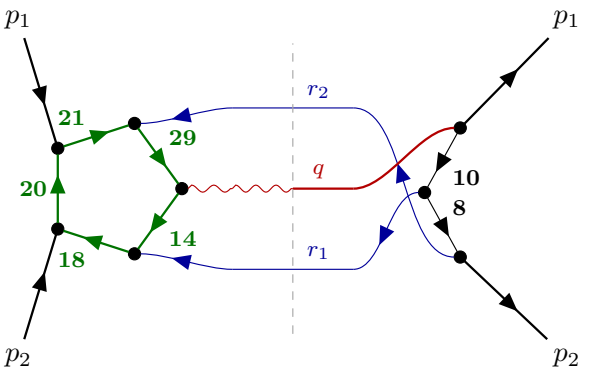
pinched: $10=(p_2+r_1+r_2)^2$



topo60 (RR family 29, pentagon q-r1-p2-p1-r2), sector with 7 of 8 lines

18	$(k_1 + r_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
8	$(p_2 + r_2)^2$	conjugate
10	$(p_2 + r_1 + r_2)^2$	conjugate

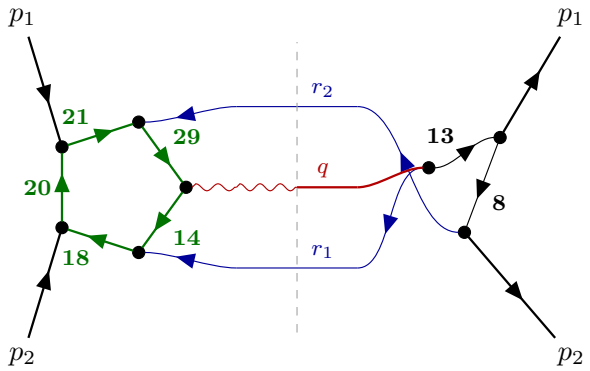
pinched: $14 = (k_1)^2$



topo60 (RR family 29, pentagon q-r1-p2-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
8	$(p_2 + r_2)^2$	conjugate
10	$(p_2 + r_1 + r_2)^2$	conjugate

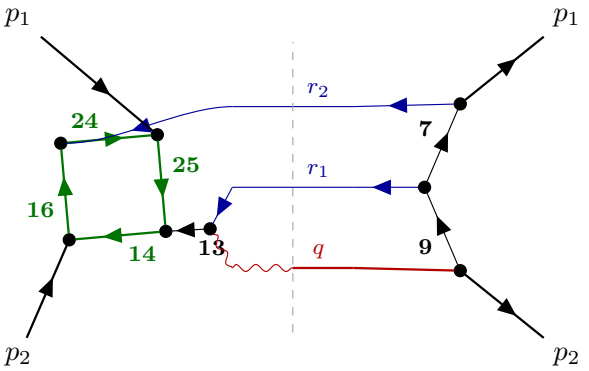
pinched: $13 = (p_1 + p_2 + r_2)^2$



topo60 (RR family 29, pentagon q-r1-p2-p1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
8	$(p_2 + r_2)^2$	conjugate

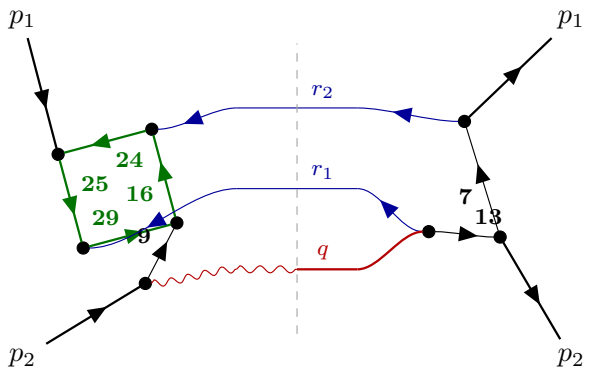
pinched: $10 = (p_2 + r_1 + r_2)^2$



topo61 (RR family 33, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
24	$(k_1 + p_2 + r_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
13	$(p_1 + p_2 + r_2)^2$	amplitude (tree on the loop)
7	$(p_1 + r_2)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate

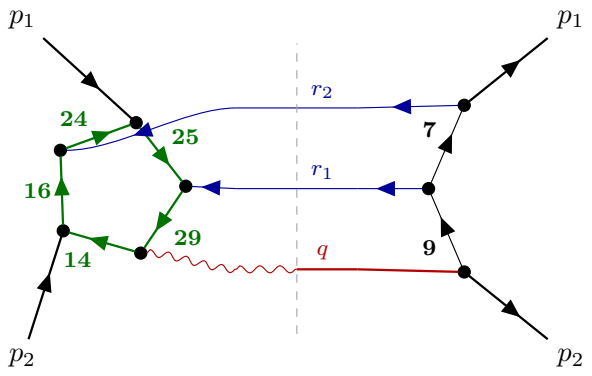
pinched: $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$



topo61 (RR family 33, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

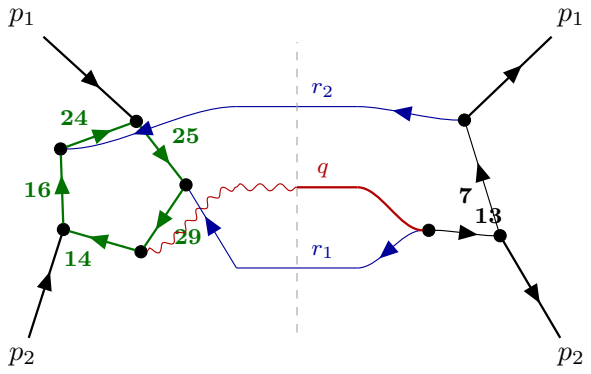
pinched: $14 = (k_1)^2$



topo61 (RR family 33, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

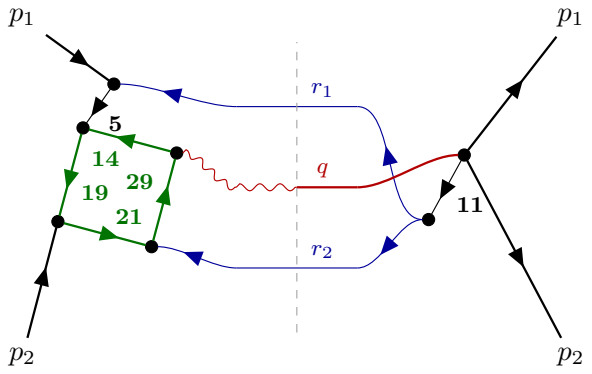
pinched: $13 = (p_1+p_2+r_2)^2$



topo61 (RR family 33, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

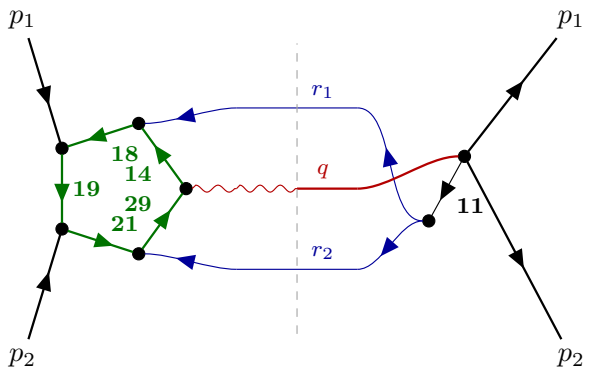
pinched: $9 = (p_1+r_1+r_2)^2$



topo62 (RR family 8, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
11	$(r_1+r_2)^2$	conjugate

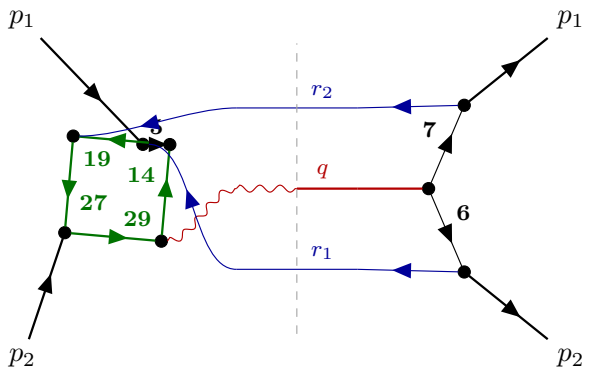
pinched: $8 = (p_2+r_2)^2$, $18 = (k_1+r_1)^2$



topo62 (RR family 8, pentagon q-r1-p1-p2-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

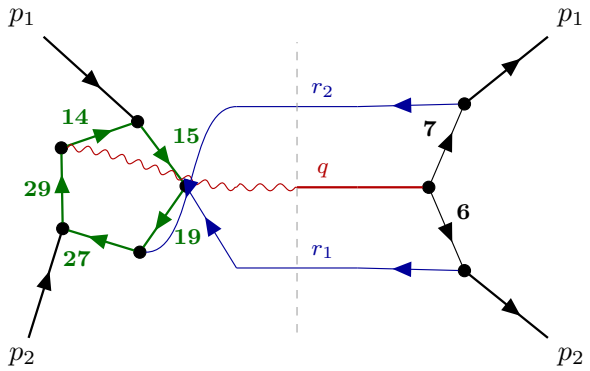
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$



topo63 (RR family 1, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

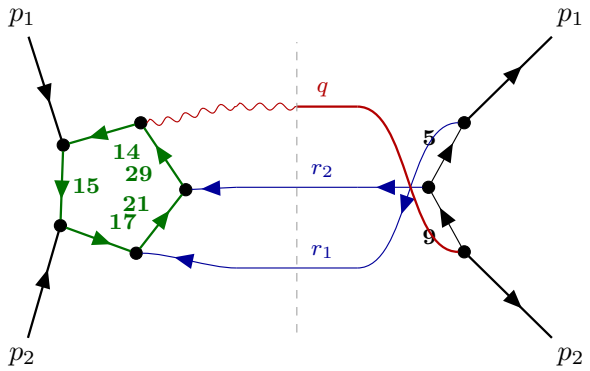
pinched: $15=(k_1+p_1)^2$



topo63 (RR family 1, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate

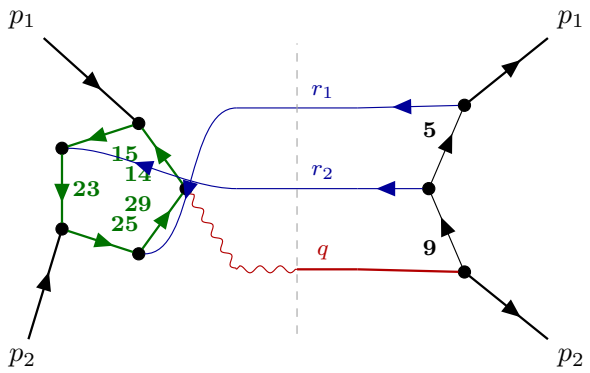
pinched: $5=(p_1+r_1)^2$



topo64 (RR family 3, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

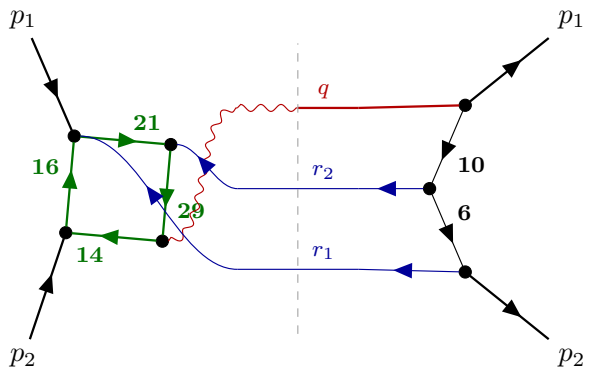
pinched: $6=(p_2+r_1)^2$



topo65 (RR family 3, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

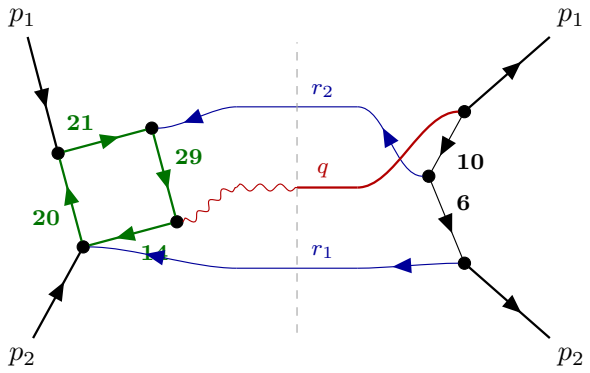
pinched: $6=(p_2+r_1)^2$



topo66 (RR family 4, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

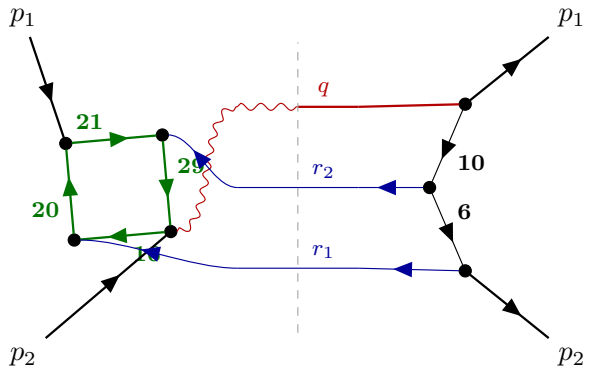
pinched: $5=(p_1+r_1)^2$, $20=(k_1+p_2+r_1)^2$



topo66 (RR family 4, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

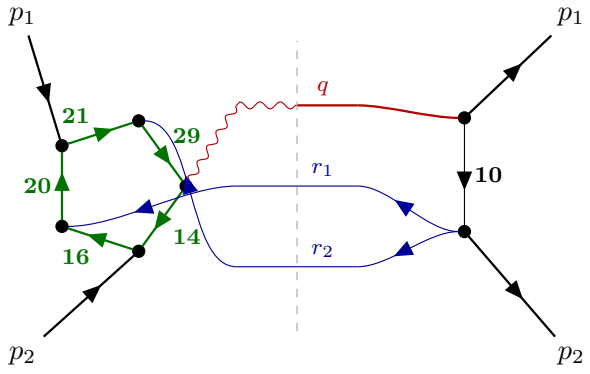
pinched: $5=(p_1+r_1)^2$, $16=(k_1+p_2)^2$



topo66 (RR family 4, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate

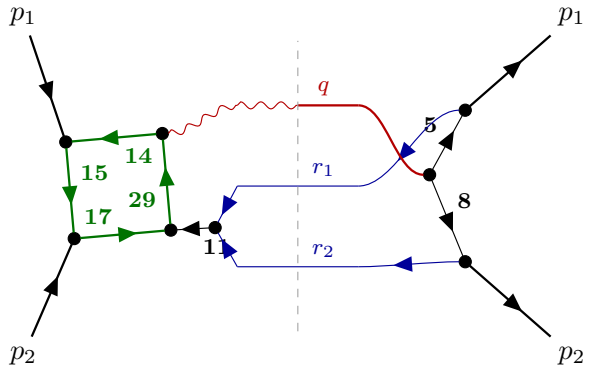
pinched: $5=(p_1+r_1)^2$, $14=(k_1)^2$



topo66 (RR family 4, pentagon q-p2-r1-p1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	conjugate

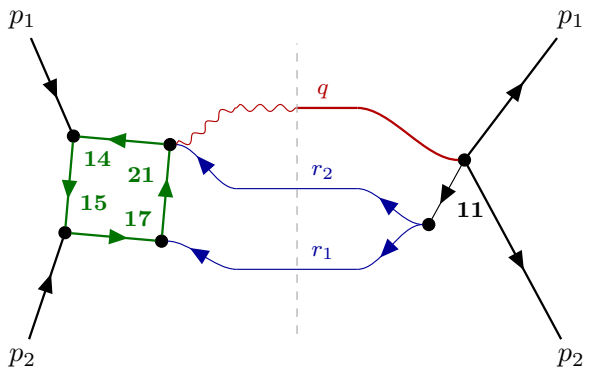
pinched: $5=(p_1+r_1)^2$, $6=(p_2+r_1)^2$



topo67 (RR family 8, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

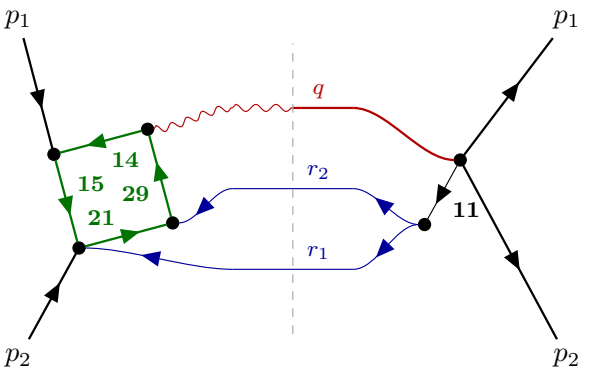
pinched: $21=(k_1+p_1+p_2+r_1)^2$



topo67 (RR family 8, pentagon q-p1-p2-r1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
11	$(r_1+r_2)^2$	conjugate

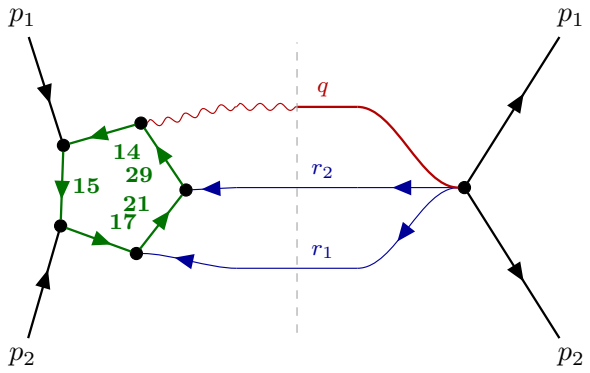
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo67 (RR family 8, pentagon q-p1-p2-r1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	conjugate

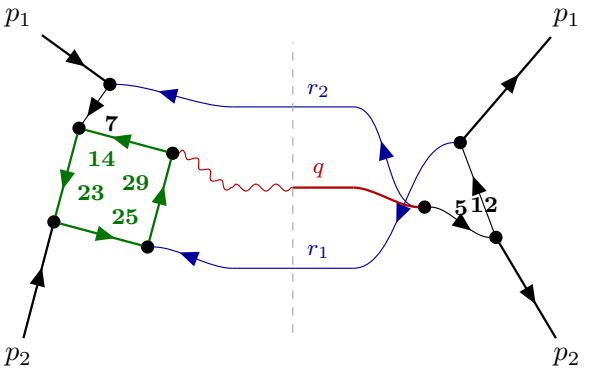
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$, $17=(k_1+p_1+p_2)^2$



topo67 (RR family 8, pentagon q-p1-p2-r1-r2), sector with 5 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop

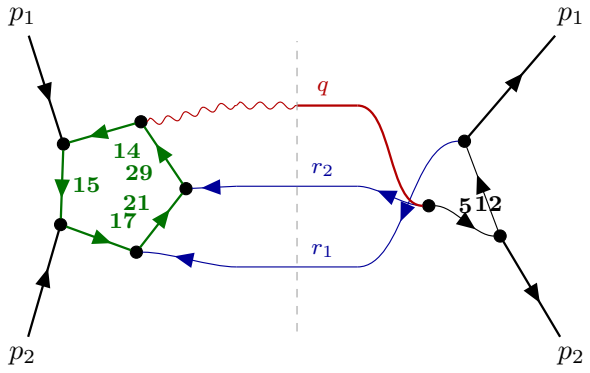
pinched: $5=(p_1+r_1)^2$, $8=(p_2+r_2)^2$, $11=(r_1+r_2)^2$



topo68 (RR family 15, pentagon q-p1-r2-p2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

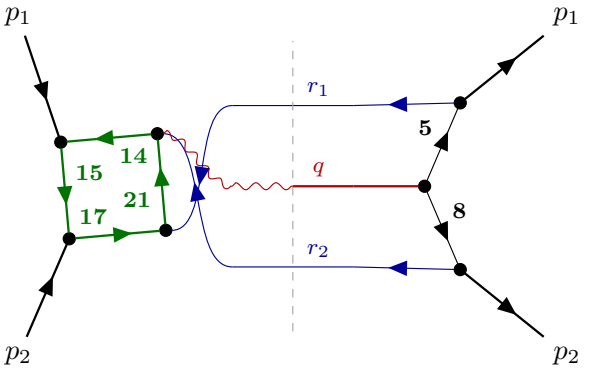
pinched: $15=(k_1+p_1)^2$



topo69 (RR family 17, pentagon q-p1-p2-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

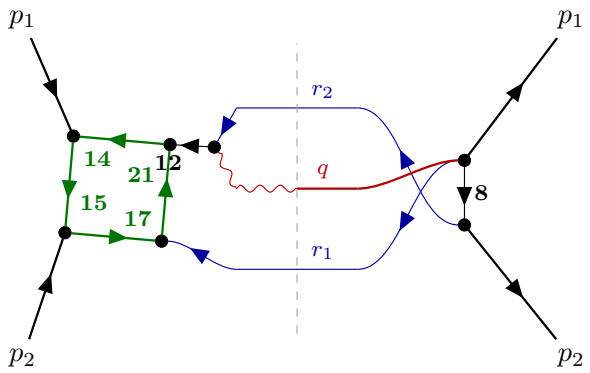
pinched: $8=(p_2+r_2)^2$



topo69 (RR family 17, pentagon q-p1-p2-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

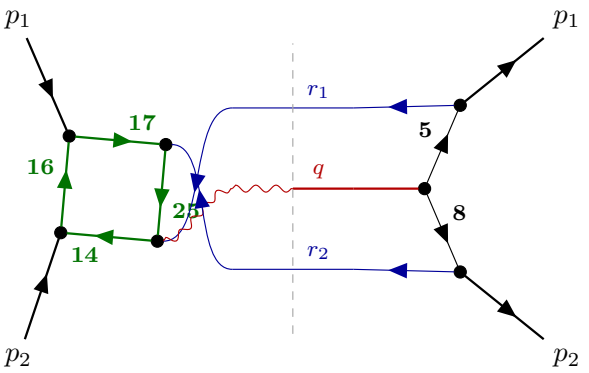
pinched: $12=(p_1+p_2+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo69 (RR family 17, pentagon q-p1-p2-r1-r2), sector with 6 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
12	$(p_1+p_2+r_1)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate

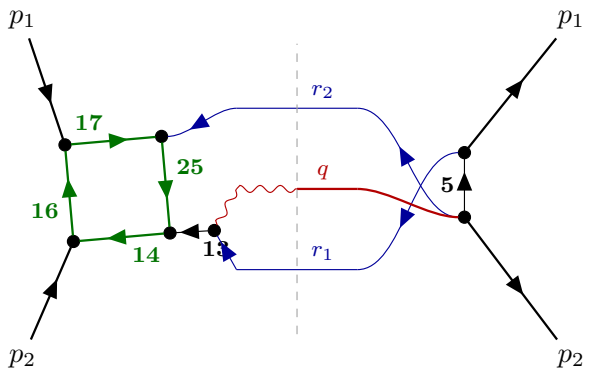
pinched: $5=(p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo70 (RR family 18, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate

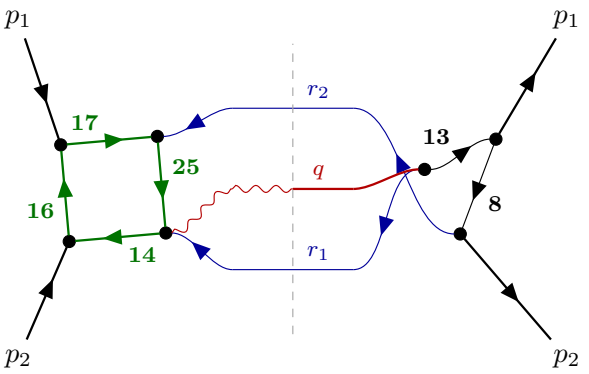
pinched: $13=(p_1+p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo70 (RR family 18, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

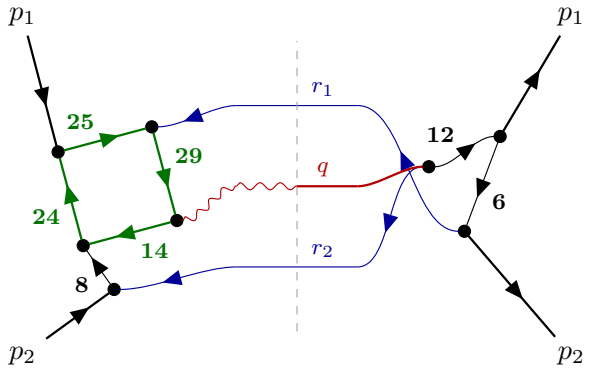
pinched: $8=(p_2+r_2)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo70 (RR family 18, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

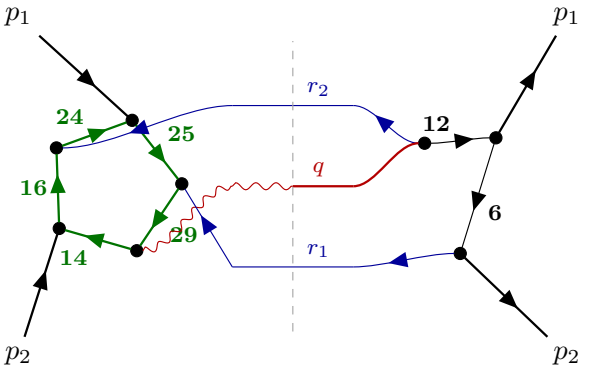
pinched: $5=(p_1+r_1)^2$, $29=(k_1+p_1+p_2+r_1+r_2)^2$



topo71 (RR family 21, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	amplitude (tree on the loop)
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

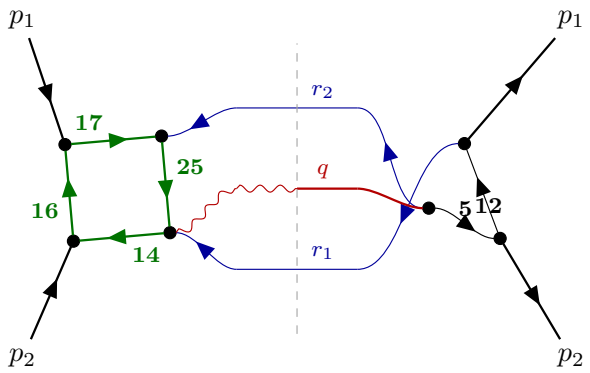
pinched: $16=(k_1+p_2)^2$



topo71 (RR family 21, pentagon q-p2-r2-p1-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

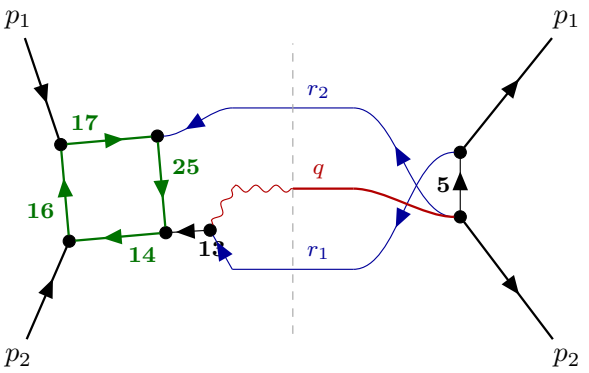
pinched: $8=(p_2+r_2)^2$



topo72 (RR family 23, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate

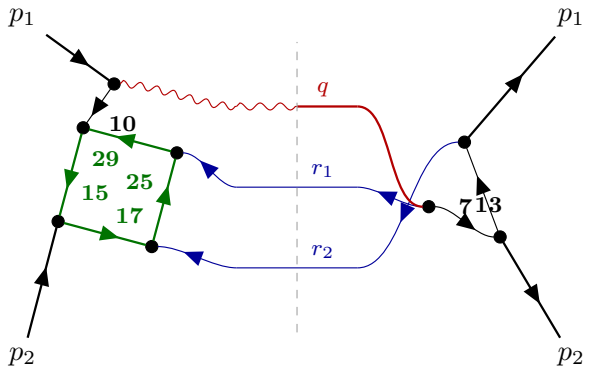
pinched: $13 = (p_1+p_2+r_2)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo72 (RR family 23, pentagon q-p2-p1-r2-r1), sector with 6 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate

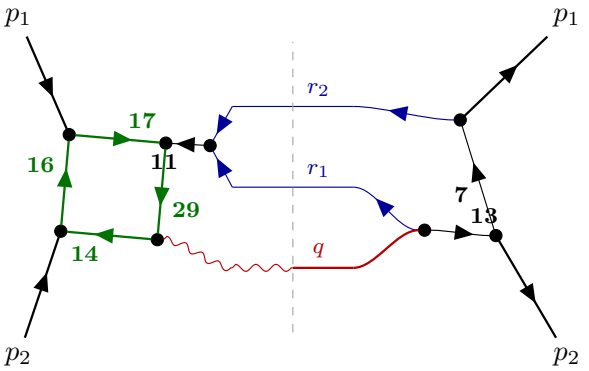
pinched: $12 = (p_1+p_2+r_1)^2$, $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo73 (RR family 28, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
10	$(p_2+r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

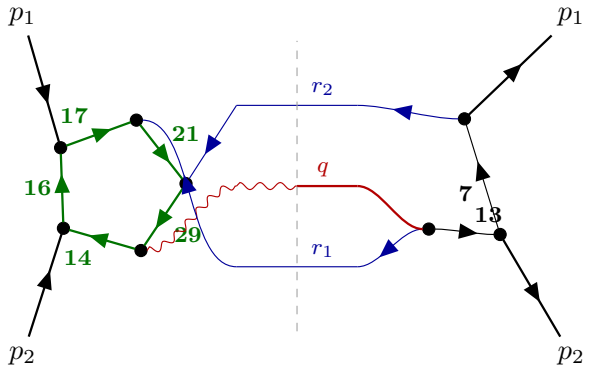
pinched: $14 = (k_1)^2$



topo74 (RR family 30, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

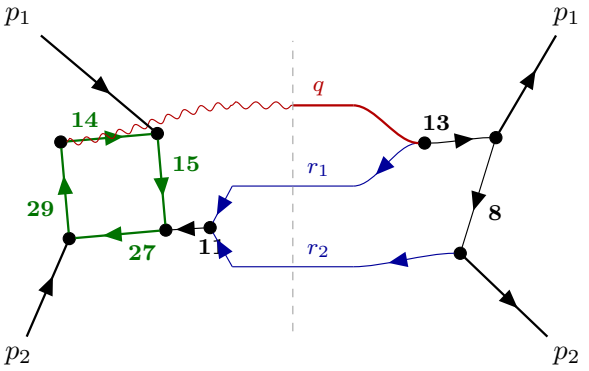
pinched: $21 = (k_1+p_1+p_2+r_1)^2$



topo74 (RR family 30, pentagon q-p2-p1-r1-r2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

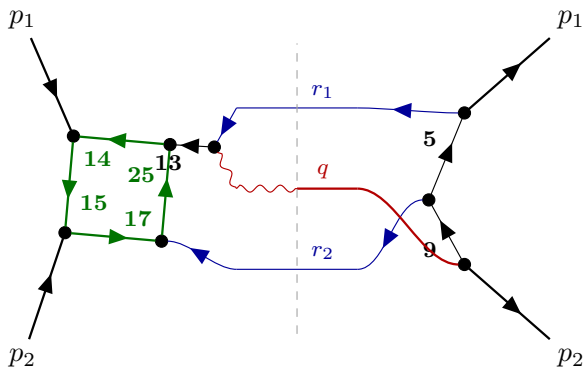
pinched: $11 = (r_1+r_2)^2$



topo75 (RR family 31, pentagon q-p1-r1-r2-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
11	$(r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

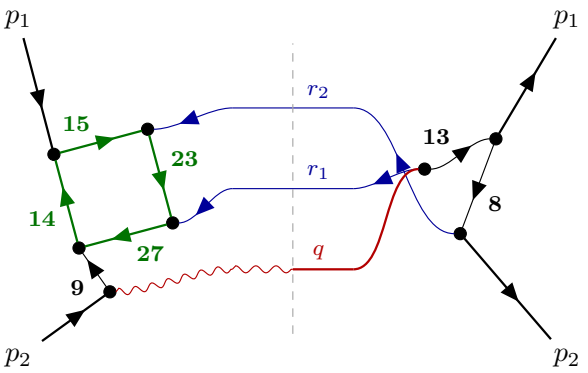
pinched: $19 = (k_1+p_1+r_1)^2$



topo76 (RR family 32, pentagon q-p1-p2-r2-r1), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
13	$(p_1+p_2+r_2)^2$	amplitude (tree on the loop)
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate

pinched: $29 = (k_1+p_1+p_2+r_1+r_2)^2$



topo77 (RR family 34, pentagon q-p1-r2-r1-p2), sector with 7 of 8 lines

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
9	$(p_1+r_1+r_2)^2$	amplitude (tree on the loop)
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate

pinched: $29 = (k_1+p_1+p_2+r_1+r_2)^2$